

COURSE DESCRIPTIONS

Undergraduate Programs

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CC102 - Programming Fundamentals

Credit Hours: 4(3+1) **Prerequisites:** None

Course Category: Computing Core Course

Description

This course furnishes students with a comprehensive foundation in fundamental concepts essential to any standard programming language. It covers key programming principles in structured ensuring students grasp these concepts. Moreover, students will acquire hands-on experience utilizing the C++ programming language, along with insights into basic understanding about the object-oriented program development.

Textbook(s)

- Modern C++ Programming Cookbook, Marius Bancila, 2020, ISBN: 978-1800208988

- A Tour of C++ (3rd Edition), Bjarne Stroustrup, 2022, ISBN: 978-0136816485

- Starting out with Programming Logic & Degins, 4th Edition, Tony Gaddis, Pearson (2018), ISBN:978-0134801155

Other supplemental materials

- Object Oriented Programming in C++ by Robert Lafore, Sams, (2001), ISBN: 978-0672323089

- C++ online tutorial - <https://www.w3schools.com/cpp/>

- C++ Online Compiler - <https://www.programiz.com/cpp-programming/online-compiler/>

Course Learning Objectives

#	Course Learning Outcomes (CLOs)	BT/ PLO #
CLO 1	Understand fundamental programming principles and apply programming logic and techniques.	C2 1,2,3
CLO 2	Apply coding basics and demonstrate proficiency in manipulating data types, variables, constants, and expressions.	C3 1,2
CLO 3	Master structured programming with C++, applying various programming aspects for effective implementation.	C3 3,4,5, 6, 8
CLO 4	Understand the basics of Object-Oriented Programming.	C4 1,2,3, 5

Week wise Lecture Schedule

W #	Lecture Topics
1	Introduction to problem solving, a brief review of Von-Neumann architecture
2	Introduction to programming, role of compiler and linker,

W #	Lecture Topics
3	introduction to algorithms and transformation into programs
4	Basic data types and variables, input/output constructs,
5	Arithmetic, comparison and logical operators
6	Conditional statements and execution flow for conditional statements
7	Repetitive statements and execution flow for repetitive statements
8	Arrays and their memory organization, multi-dimensional Arrays
9	Introduction to modular programming, function definition and calling
10	Stack rolling and unrolling, vectors
11	string and string operations
12	Pointers/references,
13	Static and dynamic memory allocation
14	File I/O operations
15	Structures. Basics about Object Oriented Programming
16	Sample Project (Case Study)

CC111- Object Oriented Programming

Credit Hours: 4(3+1) **Prerequisites:** CC102

Course Category: Computing Core Course

Description

The course aims to focus on object-oriented concepts, analysis and software development. The basic concept of OOP is covered in this course.

Textbook(s)

- Paul Deitel, Harvey Deitel. Java How to Program, Early Objects, 11th Edition, Pearson, (2017), ISBN: 978-0134743356

- Nick Samoylov. Learn Java 17 Programming: Learn the fundamentals of Java Programming with this updated guide with the latest features, 2nd Edition, Packet Publishing, (2022), ISBN: 978-1803241432

Other supplemental materials

- Online Tutorial –

https://www.w3schools.com/cpp/cpp_oop.asp

- Online Compiler - <https://ideone.com/xA0en6>

Course Learning Objectives

#	Course Learning Outcomes (CLOs)	BT/ PLO #
CLO 1	Understand principles of object-oriented paradigm	C2 1, 10
CLO 2	Identify the objects & their relationships to build object-oriented solution	C3 2,6
CLO 3	Model a solution for a given problem using object-oriented principles	C3 3,4, 8, 10
CLO 4	Examine an object-oriented solution	C4 5,6, 8

Week wise Lecture Schedule

W #	Lecture Topics
1	Introduction to object-oriented design, history and advantages of object-oriented design
2	Introduction to object-oriented programming concepts, Basic UML, Primitive vs Reference Types, Memory Models, Initializing Objects, Encapsulation, Constructors and Method overloading, Default and no argument constructors.
3	Variable declaration, data types, input output operations
4	Objects and classes, Access modifiers, Data encapsulation, constructors, destructors
5	Static data members & functions, const vs non-const functions
6	Identification of classes and their relationships, memory models, Composition, aggregation

W #	Lecture Topics
7	Inheritance, class hierarchies, super classes and subclasses
8	inheritance, multiple inheritance, constructors in subclasses, Function overloading, Operator overloading
9	Exception Handling
10	Polymorphism – abstract and concrete classes
11	Polymorphism - constructor overloading, interfaces. Case Study
12	Strings and string operations
13	Graphical User Interfaces, 2D graphics
14	Databases with GUI
15	File Handling and working with data
16	Collections

CC131- Digital Logic Design**Credit Hours:** 3(2+1) **Prerequisites:** None**Course Category:** Core Computing Course**Description**

The course introduces the concept of digital logic, gates and the digital circuits. Further, it focuses on the design and analysis combinational and sequential circuits. It also serves to familiarize the students with the logic design of basic computer hardware components.

Textbook(s)

- Thomas Floyd. Digital Fundamentals 11th Edition, Pearson, (2014), ISBN: 978-0132737968
- Fundamentals of Digital Logic with Verilog Design, Stephen Brown & Zvonko Vranesic, 2025 (4th ed.), ISBN: 978-1265120412

Other supplemental materials

- Draw Gates online -
<https://app.creately.com/d/JGzTzOr0LGa/edit>

Course Learning Objectives

#	Course Learning Outcomes (CLOs)	BT/ PLO #
CLO 1	Acquire knowledge related to the concepts, tools and techniques for the design of digital electronic circuits	C2 1,2, 10
CLO 2	Demonstrate the skills to design and analyze both combinational and sequential circuits using a variety of techniques	C3 3,4, 5, 10
CLO 3	Apply the acquired knowledge to simulate and implement small-scale digital circuits	C3 4,5
CLO 4	Understand the relationship between abstract logic characterizations and practical electrical implementations.	C2 3

Week wise Lecture Schedule

W #	Lecture Topics
1	Introduction to Course, Number Systems
2	Binary Arithmetic, binary codes, error detection and correction (Parity code, Hamming code)
3	Boolean Algebra: Laws, Theorems, and Simplification
4	Logic gates & circuit construction (AND, OR, NOT, NAND, NOR, XOR)
5	Simplification Methods (K-Map, Quinn Mc-Cluskey method)
6	Combinational Circuits and designs I: Adders, Subtractors, Comparators
7	Combinational Circuits and designs II: Decoders, Encoders, Mux, DeMux

W #	Lecture Topics
8	Sequential Circuits and designs: Flip Flops and Latches
9	Asynchronous and Synchronous circuits
10	Shift Registers
11	Counters
12	Triggered devices & its types
13	Arithmetic Circuits Arithmetic Circuits: Ripple-Carry, Carry Look-Ahead Adders, ALU Basics
14	Memory Elements
15	Introduction Programmable Logic Devices (CPLD, FPGA)
16	Introduction to HDL/VHDL

CC211- Data Structures**Credit Hours:** 4(3+1) **Prerequisites:** CC102**Course Category:** Core Computing Course**Description**

The course is designed to teach students structures and schemes, which allow them to write programs and algorithms to efficiently manipulate, store, and retrieve data. Students are exposed to the concepts of time and space complexity of computer programs.

Textbook(s)

- Algorithms, 4th Edition, Robert Sedgewick & Kevin Wayne, 2022, ISBN 978-0134218200
- Mark Weiss. Data Structures and Algorithm Analysis in Java 3rd Edition, Pearson, (2011), ISBN: 978-0132576277
- Adam Drozdek. Data Structures and Algorithms in C++ 4th Edition, Cengage Learning, (2012), ISBN: 978-1133608424

Other supplemental materials

- Grokking Data Structures, Marcello La Rocca, 2024, ISBN 978-1633436992
- <https://stepik.org/course/579/promo>
- <https://dev.to/ayabouchiha/free-resources-to-master-algorithms-data-structure-2nfj>

Course Learning Objectives

#	Course Learning Outcomes (CLOs)	BT/ PLO #
CLO 1	Implement various data structures and their algorithms and apply them in implementing simple applications	C3 1, 10
CLO 2	Analyze simple algorithms and determine their complexities	C4 2, 6
CLO 3	Apply the knowledge of data structure to other application domains	C3 3,4, 8, 10
CLO 4	Design new data structures and algorithms to solve problems	C5 4,5, 8

Week wise Lecture Schedule

W #	Lecture Topics
1	Introduction to Data structures
2	Abstract data types
3	Complexity analysis, Big Oh notation,
4	Arrays, traversing in a Linear Array, Searching in a Linear Array, Insertion in a Linear Array. Deletion in a Linear Array
5	linked list & its various types, sorted linked list,
6	searching an unsorted array, binary search for sorted arrays, hashing and indexing, open addressing and chaining,

W #	Lecture Topics
7	Memory management and garbage collection.
8	Stacks (linked lists and array implementations), Recursion and analyzing recursive algorithms, divide and conquer algorithms,
9	Sorting algorithms (selection, insertion, merge, quick, bubble, heap, shell, radix, bucket),
10	queue, dequeuer, priority queues (linked and array implementations of queues),
11	trees and tree traversals, binary search trees, ,
12	Representation of Binary Tree, Using Linked List, Sequential Representation
13	Algorithms for Binary Tree Traversing(Preorder, Inorder, Post Order)
14	Insertion, Deletion and Searching operations with algorithms
15	Graphs, breadth-first and depth-first traversal, topological order, shortest path,
16	adjacency matrix and adjacency list implementations

CC221 - Database Systems**Credit Hours: 4(3+1) Prerequisites: None****Course Category:** Core Computing Course**Description**

The course aims to introduce basic database concepts, different data models, data storage and retrieval techniques and database design techniques. The course primarily focuses on relational data model and DBMS concepts. A brief introduction to NoSQL systems is also given.

Textbook(s)

- Database System Concepts, Abraham Silberschatz, Henry F. Korth, & S. Sudarshan – 7th Edition (2020), ISBN: 978-0078022159

- Thomas Connolly. Database Systems: A Practical Approach to Design, Implementation, and Management 6th Edition, Pearson, (2015), ISBN: 978-0132943260.

Other supplemental materials

- Jeffrey D. Ullman Hector Garcia-Molina, Jennifer Widom, Database Systems, Pearson, (2013), ISBN: 978-9332518674.

Course Learning Objectives

#	Course Learning Outcomes (CLOs)	BT/ PLO #
CLO 1	Explain fundamental database concepts.	C2 1,2, 10
CLO 2	Design conceptual, logical and physical database schemas using different data models.	C5 1,2, 4
CLO 3	Identify functional dependencies and resolve database anomalies by normalizing database tables.	C2 3,4
CLO 4	Use Structured Query Language (SQL) for different operations in any DBMS	C3 4,5
CLO 5	Build different SQL databases	C5 4,5, 10

Week wise Lecture Schedule

W #	Lecture Topics
1	Basic database concepts, Database approach vs. file-based system
2	Database architecture, three level schema architecture, data independence
3	Relational data model, attributes, schemas, tuples, domains, relation instances,
4	Keys of relations, integrity constraints
5	Relational algebra, selection, projection, Cartesian product, types of joins

W #	Lecture Topics
6	Entity relationship model, entity sets, attributes, relationship, entity-relationship diagrams
7	Normalization, functional dependencies, normal forms
8	Structured Query Language (SQL)
9	Data Query Language, Data Management Language
10	Data Definition Language, Data Control Language
11	Joins and sub-queries in SQL
12	Indexing, Triggers, Store Procedures
13	Transaction Management, ACID Properties
14	Concurrency, Database backup and recovery
15	NoSQL systems
16	NoSQL systems - MongoDB

CC231 - Computer Organization and Assembly Language

Credit Hours: 3(2+1) **Prerequisites:** CC131

Course Category: Core Computing Course

Description

This course focuses on machine-independent assembly language programming and basic architecture of computer systems including fundamental concepts such as register structure, memory organization, addressing modes, organization of peripherals, and machine-level operations. These concepts are integrated through the use of assemblers, linkers and loaders. Students will learn how to write and debug programs using assembly language programming

Textbook(s)

- Introduction to Computer Organization, Robert G. Plantz, 2022, ISBN 978-1718500099
- Kip Irvine. Assembly Language for Intel Based Computers, 5th edition, Prentice Hall, (2007), ISBN: 978-0132383103
- J. Duntemann. Assembly Language Step by Step, 2nd edition, Wiley publishers, (2011), ISBN: 978-0470497029

Other supplemental materials

- Assembly Lang Tutorial:
<http://www.osdata.com/topic/language/asm/asmintro.htm>
- ASCII Table: <http://www.asciitable.com>
- <http://en.tldp.org/HOWTO/Unix-and-Internet-Fundamentals-HOWTO/core-formats.html>

Course Learning Objectives

#	Course Learning Outcomes (CLOs)	BT/ PLO #
1	CLO Describe the basic functionalities and limitations of computer system	C2 1,10
2	CLO Define the concepts related to computer architecture	C2 3,10
3	CLO Explain how high-level languages implement arithmetic expressions, loops, and logical structures at the machine level	C3 4,6

Week wise Lecture Schedule

W #	Lecture Topics
1	Introduction to the course. The need to study assembly language. Resources (books, assemblers, emulators etc, General Knowledge about microcomputers and microprocessors
2	Registers, User visible registers, their categories, 16, 32 and 64-bit registers, Introduction to the Flag register also called as FLAGS or program status word
3	Description of various flags

W #	Lecture Topics
4	Bus, bus categories, (Address, Data and Control Lines of a Bus), Bus Types
5	Memory organisation and Structure, Linear and segmented memory models
6	Our first English language program and its equivalent in the assembly language, Register programming: data movement, performing simple arithmetic
7	Introduction to the assembler and debugger: Advanced full screen debugger, NASM
8	Addressing modes: register direct addressing with various label types
9	Addressing modes: Base register indirect addressing
10	Jumps: jnz condition and associated flags in the flag register affected by jnz
11	Looping and jumps
12	Addressing modes: Indexed register indirect addressing
13	Addressing modes: Base register indirect + offset addressing
14	Addressing modes: Indexed register indirect + offset addressing
15	Comparison and conditional jumps, Unconditional jumps (types of jumps => near, short, far jumps
16	Sorting – bubble sort algorithm implementation

CC232 - Operating Systems

Credit Hours: 3(3+0) **Prerequisites:** None

Course Category: Core Computing Course

Description

To help students gain a general understanding of the principles and concepts governing the functions of operating systems and acquaint students with the layered approach that makes design, implementation and operation of the complex OS possible.

Textbook(s)

- Abraham Silberschatz, Greg Gagne, Peter B. Galvin. Operating System Concepts 10th Edition, Wiley, (2021), ISBN: 978-1119800361

- Modern Operating Systems (5th Edition), Andrew S. Tanenbaum & Herbert Bos, 2022, ISBN 978-0137618828

Other supplemental materials

- Unix online Tutorial -

<https://www.tutorialspoint.com/unix/index.htm>

- Operating Systems -

https://www.tutorialspoint.com/operating_system/index.htm

Course Learning Objectives

#	Course Learning Outcomes (CLOs)	BT/ PLO #
1	CLO Understand the characteristics of different structures of the Operating Systems and identify the core functions of the Operating Systems	C2 1, 10
2	CLO Analyze and evaluate the algorithms of the core functions of the Operating Systems and explain the major performance issues with regard to the core functions	C6 2, 8
3	CLO Demonstrate the knowledge in applying system software and tools available in modern operating systems.	C3 3, 5

Week wise Lecture Schedule

W #	Lecture Topics
1	Operating systems basics
2	System calls, process concept and scheduling, inter-process communication
3	Multithreaded programming, multithreading models, threading issues
4	Process scheduling algorithms, thread scheduling, multiple-processor scheduling
5	Synchronization, critical section, synchronization hardware, synchronization problems
6	Deadlocks, detecting and recovering from deadlocks

W #	Lecture Topics
7	Memory management, swapping, contiguous memory allocation
8	segmentation & paging, segmentation strategies and page replacement strategies
9	Virtual memory management, demand paging, thrashing, memory-mapped files
10	File systems, file concept, directory and disk structure, directory implementation
11	Free space management, disk structure and scheduling, swap space management,
12	System protection
13	Operating system security
14	Introduction to Unix/Linux
15	Working in Bash Shell (Basic bash commands)
16	Technical Review of Famous Operating systems like MS Windows, MacOS, Linux, Android etc.

CC251 - Software Engineering**Credit Hours:** 3(3+0) **Prerequisites:** None**Course Category:** Core Computing Course**Description**

This course provides a comprehensive introduction to software engineering, covering key topics such as professional software development, software process models, agile development techniques, requirements engineering, design and implementation, software testing and quality assurance, software evolution, project management, and configuration management. Students will gain a solid understanding of software engineering principles and practices, preparing them for successful software development projects.

Textbook(s)

- Sommerville I., Software Engineering, 10th Edition, Pearson Inc., (2017), ISBN: 978-9332582699

Other supplemental materials

- Roger Pressman, Bruce Maxim. ISE SOFTWARE ENGINEERING: A PRACTITIONERS APPROACH, McGraw-Hill Education, (2019), ISBN: 978-1260548006

Course Learning Objectives

#	Course Learning Outcomes (CLOs)	BT/ PLO #
CLO 1	Describe various software engineering processes and activates	C1 1, 10
CLO 2	Apply the system modeling techniques to model medium size software systems	C3 2, 4, 5
CLO 3	Apply software quality assurance and testing principles to medium size software systems	C3 2,3
CLO 4	Discuss key principles and common methods for software project management such as scheduling, size estimation, cost estimation and risk analysis	C2 3,10

Week wise Lecture Schedule

W #	Lecture Topics
1	Nature of Software, The Unique nature of WebApp, Software Engineering, The software process, Software Engineering practice, software myth
2	Software Process Models: Generic and Prescriptive Models
3	Specialized Process Models, CMM, Process Assessment and Improvement
4	Agile Methods: Extreme Programming, Scrum, and Agile Principles
5	Unified Process Model, Personal and Team Process Models

W #	Lecture Topics
6	Software Engineering Core principles, Understanding Requirements
7	Requirements modeling scenarios, information and analysis classes
8	UML diagrams
9	Data and Process Modeling Diagram, DFDs, Logical Vs Physical Models
10	Design Concepts: Principles, Component-level Design
11	Architectural Design: Styles, Patterns, and Case Studies
12	Quality concepts, Review Techniques
13	Software Testing Strategy and testing conventional Application
14	Software Configuration management
15	Introduction to project management, Project scheduling, Gantt Chart, Risk Management: Proactive VS Reactive Risk Management strategies
16	Maintenance, Reengineering, and Risk Management

CC261 - Computer Networks**Credit Hours:** 3(2+1) **Prerequisites:** None**Course Category:** Core Computing Course**Description**

This course introduces the basic concept of computer network to the students. The network functionalities, procedures, protocols and various terminologies are discussed by taking the layered network models (OSI, TCP/IP). A simulation tool is also discussed. In the practical work the packet tracer is used for building different types of networks.

Textbook(s)

- James F. Kurose, Keith Ross. Computer Networking 8th Edition, Pearson, (2020), ISBN: 978-0136681557
- William Stallings. Data and Computer Communications, 10th Edition, Pearson, (2013), ISBN: 978-0133506488
- Behrouz A. Forouzan. Data Communication and Computer Networks, 5th Edition, McGraw Hill, (2012), ISBN: 978-0073376226

Other supplemental materials

- Packet Tracer Tutorial -
<https://www.packettracernetwork.com/tutorials/>

- Online Networking Tools -
<https://manytools.org/network/>

Course Learning Objectives

#	Course Learning Outcomes (CLOs)	BT/ PLO #
1	CLO Describe the key terminologies and technologies of computer networks	C2 1,2,10
2	CLO Explain the services and functions provided by each layer in the Internet protocol stack.	C2 1,2,10
3	CLO Analyze various internetworking devices and protocols and their functions.	C4 4,5
4	CLO Build Computer Network on various Topologies	P3 4,5,6 ,8

Week wise Lecture Schedule

W #	Lecture Topics
1	Basic concepts of networking, Peer to peer and client server architectures,
2	Network topologies, LAN, MAN and WAN
3	Layered architecture, TCP/IP, OSI
4	Physical layer functionality, Broadcast vs. Point-to-Point Communication, Directional and Omni Directional Communication, Modes of Communication (Simplex, Half Duplex and Full Duplex).

W #	Lecture Topics
5	Media: Bounded Media, Coaxial Cable, Twisted Pair (Shielded and Unshielded), Optical Fiber, Unbounded Media, Radio Waves, Microwave, Infrared, Satellites
6	Data link layer functionality, MAC addressing (unicast, multicast, broadcast) , CSMA, Types of Transmission (Asynchronous and Synchronous Transmission)
7	Multiplexing: Frequency Division Multiplexing, Time Division Multiplexing, Synchronous TDM & Statistical TDM
8	IEEE 802.x standards (Wired Networks)
9	Networking devices, Network layer protocols (IPv4 and IPv6)
10	Transport layer protocols, ports and sockets, TCP and UDP
11	Application layer protocols
12	Wireless Networks standards and technologies
13	Decentralized wireless networks
14	Cellular Networks, Generations and advancements
15	Latest trends in computer networks
16	Case studies

CC252 - Artificial Intelligence

Credit Hours: 3(2+1) **Prerequisites:** None

Course Category: Core Computing Course

Description

Artificial Intelligence has emerged as one of the most significant and promising areas of computing. This course focuses on the foundations of AI and its basic techniques like Symbolic manipulations, Pattern Matching, Knowledge Representation, Decision Making and Appreciating the differences between Knowledge, Data and Code. AI programming language Python has been proposed for the practical work of this course.

Textbook(s)

- Russell, S. and Norvig, P. Artificial Intelligence. A Modern Approach, 3rd ed, Prentice Hall, (2015), ISBN: 978-9332543515

- Philip Robbins. Python Programming for Beginners: The Complete Guide to Mastering Python, (2023), ISBN: 979-8376161821

- Kaplan, J. Artificial Intelligence: What Everyone Needs to Know 1st Edition, Oxford University Press, (2016), ISBN: 978-1493682225

Other supplemental materials

- Python Tutorial - <https://www.learnpython.org>

- Online Python Compiler -

<https://www.programiz.com/python-programming/online-compiler/>

Course Learning Objectives

#	Course Learning Outcomes (CLOs)	BT/ PLO #
CLO 1	Understand the fundamental constructs of Python programming language.	C2 5,10
CLO 2	Understand key concepts in the field of artificial intelligence	C2 1, 2,10
CLO 3	Implement artificial intelligence techniques and case studies	C3 3, 5, 6

Week wise Lecture Schedule

W #	Lecture Topics
1	An Introduction to Artificial Intelligence and its applications
2	AI history, branches of AI
3	Intelligent agents, PEAS
4	types of agents and architectures
5	Knowledge Based Systems, Introduction to Reasoning and Knowledge Representation

W #	Lecture Topics
6	Problem Solving by Searching (Informed searching, Uninformed searching, Heuristics, Local searching)
7	Problem Solving by Searching (Min- max algorithm, Alpha beta pruning, Game-playing)
8	Case Studies: General Problem Solver, Eliza, Student, Macsyma
9	Rule based system, Channing method, Fuzzy Logic System
10	Machine Learning, Types (supervised and Unsupervised) Reinforcement Learning.
11	Learning from examples; ANN and Computer Vision
12	More topics on Searching: Hill climbing, Simulated Annealing, Tabu Search, Min-Max Algorithm
13	Recent trends in AI and applications of AI algorithms
14	Introduction to Python. Setting up Python environment (installation). Python interpreter and basic commands. Variables, data types, and basic operations. Print statements and user input
15	Python: Control Flow and Loops. If-else statements. While loops and For loops. Break and continue statements. Nested loops. Defining and calling functions. Arguments and return values. Scope and global variables. Lambda functions
16	Case study – 1 (e.g., AGI: J. Reed, et. Al, “A Generalist Agent”, May 19, 2022)

CC262 - Information Security**Credit Hours:** 3(3+0) **Prerequisites:** None**Course Category:** Core Computing Course**Description**

This course provides a broad overview of the threats to the security of information systems, the responsibilities and basic tools for information security, and the levels of training and expertise needed in organizations to reach and maintain a state of acceptable security. It covers concepts and applications of system and data security. Areas of particular focus include secure network design, implementation and transition issues, and techniques for responding to security breaches.

Textbook(s)

- Computer and Information Security Handbook (4th Ed.), edited by John R. Vacca, Morgan Kaufmann (2024), ISBN: 978-0443132230.

- William Stallings, Lawrie Brown,. Computer Security: Principles and Practice 4th Edition, Pearson, (2017), ISBN: 978-0134794105

- Michael E. Whitman, Herbert J. Mattord.Principles of Information Security 6th Edition, Cengage Learning, (2017), ISBN: 978-1337102063

Course Learning Objectives

#	Course Learning Outcomes (CLOs)	BT/ PLO #
CLO 1	Explain key concepts of information security such as design principles, cryptography, risk management, and ethics	C2 1,10
CLO 2	Discuss legal, ethical, and professional issues in information security	A2 2,3,7 ,8,9
CLO 3	Apply various security and risk management tools for achieving information security and privacy	C3 4,5
CLO 4	Identify appropriate techniques to tackle and solve problems in the discipline of information security	C4 2,3,4

Week wise Lecture Schedule

W #	Lecture Topics
1	Information security Fundamentals: Define Security, Role & Significance, Applications, Beneficiaries, Characteristics of Information i.e. Confidentiality, Integrity, Availability, Authenticity, Utility etc, Terminologies of Information Security
2	Memory exploits, i.e. Buffer overflow (Stack Overflow, Heap Overflow, Integer Overflow) , Format String

W #	Lecture Topics
3	Threats, its Types & Defenses. Virus, Malware, Spyware, Trojan, Phishing, Ransomware, Breaches etc.
4	Attacks, its Types & Defenses. Masquerade, Modification, Repudiation, Eavesdropping, Spoofing, DoS etc.
5	Crimes, its Types & Defenses. Crime, Computer Crime, Cyber Crime. Types of Cyber Crimes: Financial Fraud, Counterfeiting, ID Theft, Cyber Terrorism, Cyber Extortion, Cyber Warfare, Crypto Jacking, Cyber Bullying, Cyber Talking, Cyber Espionage, Cyber Squatting, Digital Piracy, IP Infringement, Privacy Invasion, Doxing, BEC & Exit Scam, Spam & Salami, Hacking, Botnet & Predator etc. Cyber Crimes Prevention and Combating Approaches
6	System Security: Defining System Security and its Objectives. Violations via Vulnerabilities, Threats, and Attacks. System Breaches via Confidentiality, Integrity, Authentication, ToS, and DoS
7	Information security Policies, Standard and practices . Types of security policies (Access control lists, Access control Metrix, Configuration Rules)
8	Access Control, Its types and Elements of Access Control . Access control Models (Mandatory access control, Nondiscretionary access control, Discretionary access control)
9	Network Security
10	Cryptography, Definitions and Concepts,
11	Cipher Methods (Bit Stream, Block Stream). Substitution Cipher (Monoalphabetic substitution, Polyalphabetic substitution, Vigenère cipher)
12	Transposition Cipher (Caesar Matrix, Exclusive OR (XOR), Varnam Cipher, Book or Running Key Cipher)
13	Symmetric Key Cryptography. Private Key Distribution Approaches. Stream Ciphers vs. Block Ciphers. DES Algorithm vs. AES Algorithm
14	Asymmetric Key Cryptography. Public Key Distribution Approaches. RSA Algorithm vs. PGP Application. Comparison of Symmetric and Asymmetric Key Cryptography
15	Intrusion Detection and Prevention Systems
16	Signature based IDPS, Behavior Based IDPS, Network-based IDPS, Host-based IDPS

CC411 - Analysis of Algorithms**Credit Hours:** 3(3+0) **Prerequisites:** CC211**Course Category:** Core Computing Course**Description**

Detailed study of the basic notions of the design of algorithms and the underlying data structures. Several measures of complexity are introduced. Emphasis on the structure, complexity, and efficiency of algorithms.

Textbook(s)

- Thomas H. Corman, Charles E. Leiserson, Ronald L. Rivest and Clifford Stein. Introduction to Algorithms, 3rd edition, MIT Press, (2009), ISBN: 978-0262033848
- Robert Sedgewick, Kevin Wayne. Analysis of Algorithms, Fourth Edition, Deluxe Edition, (2020), ISBN: 978-0134384689

Other supplemental materials

- CS502- Fundamentals of Algorithms, Open Courseware, Virtual University of Pakistan
- <https://www.coursera.org/learn/analysis-of-algorithms>

Course Learning Objectives

#	Course Learning Outcomes (CLOs)	BT/ PLO #
CLO 1	Understanding Algorithms and Asymptotic Notations	C2 1,10
CLO 2	Apply algorithm Design Techniques	C3 2,3
CLO 3	Exploring Graph Algorithms and Complexity Classes	C4 4
CLO 4	List and contrast standard complexity classes	C4 4,6,10

Week wise Lecture Schedule

W #	Lecture Topics
1	Introduction; role of algorithms in computing
2	Analysis on nature of input and size of input Asymptotic notations; Big-O, Big Ω , Big Θ , little-o, little- ω
3	Recursion and recurrence relations Divide-and conquer technique: Merge Sort
4	Divide-and conquer technique: Quick Sort
5	Heaps, Heap Sort, Hashing
6	Algorithm Design Techniques, Brute Force Approach
7	Understand the concepts of Dynamic programming
8	Dynamic programming: Edit Distance
9	Dynamic programming: Chain Matrix Multiplication

W #	Lecture Topics
10	Dynamic programming: Knapsack Problem
11	Greedy approach: Coin Change
12	Greedy approach: Huffman Encoding, Activity Selection
13	Graph algorithms, Graph traversing
14	Understand the concepts of MST and their algorithms: Kruskal's Algorithm, Prim's Algorithm
15	Understand the algorithms for computing shortest path: Dijkstra's Algorithm, Bellman-Ford Algorithm, Floyd-Warshall Algorithm
16	Introduction to complexity classes

CS311 - Theory of Automata

Credit Hours: 3(3+0) **Prerequisites:** None

Course Category: Computer Science Domain

Description

This is an introductory course on Theory of Automata. Students are introduced to the concept of Formal Languages using Automata and Compiler. In Automata they learn about finite automata (deterministic; non-deterministic), transition graphs and pushdown automata (deterministic; non-deterministic). Introduces the foundations of formal language theory, computability, and complexity. This Course shows the relationship between automata and various classes of languages. Addresses the issue of which problems can be solved by computational means, and studies complexity of solutions

Textbook(s)

- Peter Linz Jones & Bartlett. Introduction to Formal Languages and Automata 6th Edition, Learning publisher (Latest Edition), ISBN: 978-1284077247
- Daniel I.A Cohen John. Introduction to Computer Theory, Willy & Sons Inc (1986), ISBN: 0-471-80271-9
- John E. Hopcroft, Rajeev Motwani, Jeffery D. Ullman. Introduction to Automata Theory, language and Computation, 3rd Edition, Pearson Publisher (Latest Edition), ISBN: 9788131720479

Other supplemental materials

- https://www.tutorialspoint.com/automata_theory/index.htm
- <https://www.javatpoint.com/automata-tutorial>

Course Learning Objectives

#	Course Learning Outcomes (CLOs)	BT/ PLO #
1	CLO Understand the fundamental concepts of formal languages and automata theory, including terminologies, descriptive and recursive definitions, and regular expressions, to effectively analyze and manipulate languages.	C2 1,2
2	CLO Demonstrate proficiency in designing and implementing Finite Automata (FA) and Deterministic Finite Automata (DFA) using transition tables, enabling accurate language recognition and processing.	C3 3,4
3	CLO Develop a comprehensive understanding of advanced topics such as Non-Deterministic Finite Automata (NFA), conversion of NFA to DFA, and context-free grammars, enabling the analysis and construction of complex language structures.	C4 1,3

Week wise Lecture Schedule

W #	Lecture Topics
1	Introduction, Terminologies of Languages, Operations on languages
2	Descriptive definition
3	Recursive Definition
4	Regular Expressions (RE)
5	Transition Table, Finite Automaton (FA)
6	Deterministic Finite Automata- DFA
7	Martin Technique
8	Non-Deterministic Finite Automata- NFA
9	Conversion of NFA to DFA
10	Union of Two DFAs,
11	Concatenation of Two DFAs
12	Transition Graph-TG
13	Generalize Transition Graph (GTG)
14	Context Free Grammar-CFG
15	Total Language Tree
16	Turing machine

CS315 - Compiler Construction

Credit Hours: 3(3+0) **Prerequisites:** None

Course Category: Computer Science Domain

Description

This course covers programming language differentiation, front-end and back-end roles in compilers, grammar recognition, lexical analysis, parsing strategies (top-down and bottom-up), LL (k) and LR (k) grammars, semantic analysis (type and scope checking), runtime environments, code generation, and optimization techniques. Practical implementation and hands-on experience are emphasized.

Textbook(s)

- Compiler Construction: Fundamentals and Applications
By Hanspeter Mössenböck, 2025 ISBN: 978-3031848124
- Kenneth C. Louden and Glagotia. Compiler Construction,
Cengage Learning (1997), ISBN: 978-0534939724

Other supplemental materials

- Andrew W. Appel, Maia Ginsburg, Contributor Maia
Ginsburg. Modern Compiler Implementation in C, -
Cambridge University Press, (2004), ISBN: 978-
0521820608
- John I Moore. Introduction to Compiler Design: An
Object-Oriented Approach Using Java(R), Softmoore
Consulting (2022), ISBN: 978-1734139129

Tutorial: <https://t4tutorials.com/passes-and-phases-of-compiler-design/>

Course Learning Objectives

#	Course Learning Outcomes (CLOs)	BT/ PLO #
CLO 1	Describe the architecture of a compiler and function of its components. (C1-Knowledge)	C1 1,10
CLO 2	Choose and apply appropriate formal notations to define a programming language.	C3 2,3
CLO 3	Design and implement lexical and syntax analyzers by using various algorithms.	C3 2,3
CLO 4	Select and use appropriate code generation and optimization techniques.	C4 2,3, 6, 10

Week wise Lecture Schedule

W #	Lecture Topics
1	Introduction to Compiler Construction: Overview of high-level languages and their need for translators. Phases of the compilation process
2	Language Specification: Syntax and semantic specification of a language. Chomsky's hierarchy of grammars and their classification

W #	Lecture Topics
3	Lexical Analysis: Design and implementation of a lexical analyzer. Tokenization and regular expressions
4	Parsing Techniques: Top-down parsing strategy. Bottom-up parsing strategy
5	Handling Ambiguous Grammars: Identification and resolution of ambiguities in grammars
6	LL(k) and LR(k) Grammars: Introduction to LL(k) and LR(k) grammars. Constructing LL(k) and LR(k) parsing tables
7	Recursive-Descent Parsing: Implementation of recursive-descent parsing, LL(1) parsing and its properties
8	Eliminating Left Recursion and Left Factoring: Techniques for removing left recursion, strategies for left factoring
9	Shift-Reduce Parsing: Introduction to shift-reduce parsing techniques, SLR (Simple LR) parsing
10	Look-Ahead LR Parsing: LALR (Look-Ahead LR) parsing, Building LALR(1) parsing tables
11	Semantic Analysis: Introduction to semantic analysis phase, Role of attributed grammars in semantic analysis
12	Syntax-Directed Translation: Semantic actions and syntax-directed translation rules, Constructing annotated parse trees
13	Intermediate Code Generation: Generation of intermediate code from annotated parse tree, Representation of intermediate code
14	Code Optimization Techniques: Overview of code optimization, Common code optimization techniques
15	Code Generation: Introduction to code generation phase, mapping intermediate code to target machine code
16	Review and Conclusion: Recapitulation of key concepts, Future directions in compiler construction and language processing

CS321 - Advanced Database Management Systems

Credit Hours: 3(2+1) **Prerequisites:** CC221

Course Category: Computer Science Domain

Description

This course provides a comprehensive overview of the fundamental concepts, principles, and practical techniques involved in managing and building modern databases. From traditional relational databases to the emerging NoSQL technologies, students will gain a solid understanding of data modeling, transaction management, concurrency control, query optimization, and security in both SQL and NoSQL database environments. Through a combination of theoretical knowledge and hands-on exercises, students will develop the skills necessary to design, implement, and optimize efficient and scalable databases, meeting the demands of today's data-intensive applications.

Textbook(s)

- A. Silberschatz, H. F. Korth, S. Sudarshan, Database System Concepts, Seventh Edition McGraw-Hill, (2019) ISBN 9780078022159

- Database Systems: The Complete Book 2nd Edition by Hector Garcia-Molina, Jeffrey Ullman and Jennifer Widom, (2013) ISBN-13: 978-0131873254

Other supplemental materials

- MongoDB tutorial -

<https://www.w3schools.com/mongodb/>

- Apache Cassandra's documentation -

<https://cassandra.apache.org/doc/latest/>

Course Learning Objectives

#	Course Learning Outcomes (CLOs)	BT/ PLO #
1	CLO Apply advanced concepts and techniques of Relational Database Management Systems (RDBMS).	C3 1,2,4 5,10
2	CLO Analyze different file organizations and indexing techniques for optimizing data retrieval and storage.	C4 1, 3
3	CLO Apply and utilize advanced database management concepts and technologies, including NoSQL databases, big data management systems, and database security.	C3 4,5,6 7,8
4	CLO Design and implement efficient and scalable NoSQL databases using appropriate data models and technologies.	C5 1,4,5, 6,10

Week wise Lecture Schedule

W #	Lecture Topics
1	Course overview, Transaction Management, ACID properties, TCL, Transaction log

W #	Lecture Topics
2	Concurrency Control
3	DBMS File Organizations and Indexes: Heap Files, Ordered Files, Hash Files, Single-Level Ordered Indexes, Multi-Level Indexes
4	Views, triggers and indexes: Usage and advantages
5	Query Optimization: Basic concepts and techniques, examples
6	Parallel algorithms on relations, Map reduce
7	Distributed Databases, Characteristics and types, Distribution of data, Practical example like MySQL cluster
8	NoSQL Databases: NoSQL Overview and Data Models Document Model, Key-Value Model, Column Family, Aggregates, Graph Model, Triple Stores), CAP Theorem, BASE vs ACID, NoSQL Data Modelling Concepts
9	Introduction to NoSQL DBMS like MongoDB or Cassandra
10	Big Data Management Systems: Introduction. 3Vs and 5Vs. Use cases. Types of organization of data
11	Big data technologies used by various companies/organization like Facebook Amazon etc. Data warehousing. Tools used for Big data analytics
12	Business Intelligence Technologies: Introduction. Framework for Business Intelligence, BI tools and applications
13	Datamining, Tools and techniques, Web scraping. Examples for datamining and web scraping
14	Introduction to Database Security, Threats and Countermeasures. Controls and policies.
15	Case studies/Practical implementation for security in SQL and NoSQL databases (e.g. MySQL & MongoDB)
16	Case studies: Data management in a NoSQL database (e.g. MongoDB)

CS331 - Computer Architecture

Credit Hours: 3(3+0) **Prerequisites:** CC131

Course Category: Computer Science Domain

Description

The course in Computer Architecture explores the fundamental principles that govern the design and functioning of modern computer systems. Students will explore the evolution of key architectures, performance considerations, interconnection structures, memory systems, and processor organization. This course provides a comprehensive understanding of the underlying mechanisms that contribute to the efficiency and functionality of computer systems.

Textbook(s)

- Computer Organization and Architecture 10th Edition by William Stallings, (2016), ISBN: 978-0134101613.
- Introduction to Computer Organization: An Under the Hood Look at Hardware and x86-64 Assembly by Robert Plantz, (2022), ISBN: 978-1718500099.

Other supplemental materials

- Kip Irvine. Assembly Language for Intel Based Computers, 5th edition, Prentice Hall, (2007), ISBN: 978-0132383103.

Course Learning Objectives

#	Course Learning Outcomes (CLOs)	BT/ PLO #
CLO 1	Develop a foundational understanding of computer architecture basics.	C4 1, 2, 10
CLO 2	Examine principles for designing high-performance computer systems, including considerations for multicore processors, MICs, and GPGPUs.	C5 1, 6
CLO 3	Acquire the skills to measure and evaluate computer performance, utilizing fundamental metrics and performance indicators to gain insights into the efficiency and functionality of computer systems.	C2 1, 2, 4,10
CLO 4	Understand the top-level view of computer function and interconnection.	C1 1, 2,4
CLO 5	Understand the memory hierarchy, covering cache memory principles, semiconductor main memory, and external memory options. Gain insights into processor structure, organization, registers, and the instruction cycle, forming a holistic understanding of computer architecture and organization.	C1 1, 2,3, 10

Week wise Lecture Schedule

W #	Lecture Topics
1	Basics: Evolution of Intel x86 Architecture, Embedded Systems, ARM Architecture, Introduction to Cloud Computing
2	Performance Issues: Designing for Performance, Multicore Architectures, MICs (Many Integrated Cores), and GPGPUs
3	Performance Metrics: Amdahl's Law and Little's Law, Basic Measures of Computer Performance
4	Introduction to Computer Architecture: Top-Level View of Computer Function, Main Components, and Functional Overview
5	Interconnection Structures: Bus Interconnection, System Bus Structure, Control and Timing, PCI Express
6	Point-to-Point Interconnects: Comparison with Bus Systems, High-Speed Interconnect Technologies
7	Computer Memory System Overview: Memory Hierarchy, Characteristics of Memory Systems
8	Cache Memory Principles: Cache Mapping Techniques, Replacement Policies, Write Policies
9	Elements of Cache Design: Associativity, Block Size, Hit Rate, and Latency
10	Pentium 4 Cache Organization: L1/L2 Cache, Cache Coherence, Performance Considerations
11	Internal Memory: Types of Semiconductor Memory, SRAM vs DRAM, DDR DRAM, ECC (Error Correction Code)
12	External Memory: Overview of Magnetic Disks, Disk Performance Parameters
13	RAID Concepts: RAID Levels, Performance and Reliability Tradeoffs
14	Solid State Drives (SSD): Architecture, Performance Benefits, Limitations; Optical Memory Basics
15	Processor Structure: Basic Processor Components, Control Unit, ALU, Datapath
16	Instruction Cycle and Execution: Register Organization, Instruction Fetch and Decode, Micro-operations

CS342 – HCI and Computer Graphics

Credit Hours: 3(2+1) **Prerequisites:** None

Course Category: Computer Science Domain

Description

This course provides an introductory yet comprehensive overview of Human-Computer Interaction (HCI) and Computer Graphics. Students will explore fundamental principles for designing, implementing, and evaluating user interfaces that are efficient, intuitive, and visually engaging. In the HCI component, the course emphasizes usability, user-centered design, prototyping, and accessibility. In the computer graphics component, students will be introduced to key techniques in 2D and 3D graphics, rendering, lighting, shading, animation, and immersive technologies such as Virtual and Augmented Reality. Through hands-on practice and critical evaluation, students will acquire the skills to build and assess user-centric and visually rich applications.

Textbook(s)

- Designing Interfaces: Patterns for Effective Interaction Design by Jenifer Tidwell (2nd Edition, 2011).
- Interaction Design: Beyond Human-Computer Interaction by Helen Sharp, Yvonne Rogers, and Jenny Preece (5th Edition, 2019).
- Computer Graphics: Principles and Practice by James D. Foley, Andries van Dam, Steven K. Feiner, and John F. Hughes (3rd Edition, 2013).

Course Learning Objectives

#	Course Learning Outcomes (CLOs)	BT/ PLO #
CLO 1	Understand the principles and techniques of human-computer interaction and computer graphics.	C2
CLO 2	Understand the process of rendering, animation, and 3D graphics.	C2
CLO 3	Apply principles of visual design to user interfaces.	C3
CLO 4	Evaluate and analyze user interfaces and identify areas for improvement.	C4

Week wise Lecture Schedule

W #	Lecture Topics
1	Introduction to Human-Computer Interaction (HCI): Definitions, Importance, and Applications
2	Basic Principles and Guidelines of HCI
3	User-Centered Design and Usability Testing: Processes and Case Studies
4	Designing Effective User Interfaces: Interaction Styles and Design Patterns
5	User Interface Design Principles and Guidelines (Consistency, Feedback, Error Handling)

W #	Lecture Topics
6	User Interface Prototyping: Tools and Techniques
7	Prototyping through Wireframes and Low/High-Fidelity Models
8	Designing for Accessibility and Mobile/Responsive Interfaces
9	Visual Design Principles: Layout, Typography, Color Theory
10	Introduction to Computer Graphics: History, Applications, and Graphics Pipeline
11	2D and 3D Graphics and Rendering Concepts
12	Animation Techniques: Keyframing, Morphing, and Intro to VR/AR
13	Lighting and Shading Techniques in 3D Graphics
14	Rendering Algorithms and Techniques: Ray Tracing, Rasterization, etc.
15	Usability Testing and Evaluation Methods
16	User Feedback, Analytics, and User Experience (UX) Metrics

CS461 - Parallel & Distributed Computing

Credit Hours: 3(3+0) **Prerequisites:** None

Course Category: Computer Science Domain

Description

"Parallel and Distributed Computing" is an intensive course covering essential topics in high-performance computing. Explore concepts such as concurrency control, fault tolerance, GPU programming, interconnection topologies, and load balancing. Gain practical experience with tools like Cuda, OpenMP, Hadoop, and more. Develop expertise in parallel algorithms, performance analysis, and scalability. By the end, you'll be equipped to design and optimize efficient computing systems for tackling complex computational challenges.

Textbook(s)

- Maarten van Steen, Andrew S. Tanenbaum . Distributed Systems, (2023) ISBN: 978-9081540636

- Ajit Singh. Parallel And Distributed Computing Kindle Edition, (2021), B09DMP9JT4

Other supplemental materials

Course Learning Objectives

#	Course Learning Outcomes (CLOs)	BT/ PLO #
CLO 1	Learn about parallel and distributed computers.	C2 1,2, 10
CLO 2	Write portable programs for parallel or distributed architectures using Message-Passing Interface (MPI) library.	C3 2,3, 4
CLO 3	Analyze complex problems with shared memory programming with openMP.	C4 4,5, 6,10

Week wise Lecture Schedule

W #	Lecture Topics
1	Asynchronous/synchronous computation/communication
2	Concurrency control, fault tolerance,
3	GPU architecture and programming
4	Heterogeneity, interconnection topologies,
5	Load balancing,
6	Memory consistency model, memory hierarchies,
7	Message passing interface (MPI), MIMD/SIMD,
8	Multithreaded programming,
9	Parallel algorithms & architectures,
10	Parallel I/O, performance analysis and tuning,

W #	Lecture Topics
11	Programming models (data parallel, task parallel, process-centric, shared/distributed memory),
12	Scalability and performance studies,
13	Scheduling, storage systems, Synchronization
14	Quantum Computing Basics
15	Introduction to tools (Cuda, Swift, Globus, Condor, Amazon AWS, OpenStack, Cilk, gdb, threads, MPICH, OpenMP, Hadoop, FUSE)
16	Introduction to tools (Cuda, Swift, Globus, Condor, Amazon AWS, OpenStack, Cilk, gdb, threads, MPICH, OpenMP, Hadoop, FUSE)

CS371 - Web Development

Credit Hours: 3(2+1) **Prerequisites:** None

Course Category: Computer Science Elective

Description

This course provides an introduction to the fundamentals of web development. Students will learn about front-end and back-end development basics, including HTML, CSS, JavaScript, and server-side programming. They will acquire the skills to create interactive and user-friendly web interfaces. The course also covers database integration, full-stack development, content management systems, deployment and hosting considerations, as well as version control and collaboration practices. By the end of the course, students will have a solid understanding of web development principles and the ability to create functional web applications.

Textbook(s)

- Riaz Ahmed. Full Stack Web Development for Beginners: Learn Ecommerce Web Development Using HTML5, CSS3, Bootstrap, JavaScript, MySQL, and PHP, (2021), ISBN: 979-8738951268

- Frank Zammetti. Modern Full-Stack Development: Using TypeScript, React, Node.js, Webpack, Python, Django, and Docker 2nd ed. Edition (2020), ISBN: 978-1484257371

Other supplemental materials

Course Learning Objectives

#	Course Learning Outcomes (CLOs)	BT/ PLO #
CLO 1	Demonstrate an understanding of the fundamentals of web development, including front-end and back-end development basics	C2 1,2, 10
CLO 2	Apply front-end development skills to create interactive and user-friendly web interfaces	C3 2, 4, 5,6
CLO 3	Understand the process of deploying and hosting web applications, including version control and collaboration	C2 3, 5, 6, 10

Week wise Lecture Schedule

W #	Lecture Topics
1	Introduction to Web Development: Overview of web technologies, Introduction to HTML and CSS, Setting up a development environment
2	Front-End Development Basics: HTML markup and structure, HTML simple tags
3	Front-End Development Basics: HTML advance tags
4	Front-End Development Basics: CSS styling and layout
5	Front-End Development Basics: Responsive design, CSS frameworks (e.g., Bootstrap)

W #	Lecture Topics
6	Front-End Development Basics: JavaScript syntax and data types, Working with variables, arrays, and objects
7	Front-End Development Basics: Applying JQuery, handling actions, DOM manipulation and event handling
8	Back-End Development Basics: Introduction to server-side programming languages (e.g., Node.js, PHP), Installing a webserver and understanding how a page request is processed, Understanding the language syntax – variables, arrays, functions, Handling HTTP requests and responses
9	Database Integration: Introduction to databases (e.g., MySQL), Working with databases and SQL, querying databases and retrieving data
10	Full-Stack Development: Combining front-end, back-end components, and database components, submitting form data to a database, retrieving data from database, and displaying on a webpage
11	Full-Stack Development: Performing CRUD operations, Deployment and hosting considerations
12	Full-Stack Development: Building a complete web application, A complete study to understand form submission, retrieval data, display data, edit form, giving a single page/form multi-view
13	A complete case study from front-end design to dynamic server-side functionality
14	An introduction to Content Management Systems: E.g., WordPress, Joomla
15	Deployment and hosting considerations (deploy static sites via Git Hub and run dynamic pages locally.
16	Version Control and Collaboration: Introduction to version control systems (e.g., Git) Managing code repositories and branching, Collaborative development workflows Portfolio Development

CS372 - Advanced Programming**Credit Hours:** 3(2+1) **Prerequisites:** CC111**Course Category:** Computer Science Elective**Description**

This advanced programming course typically focuses on advanced programming concepts, techniques, and best practices in high level language C#.NET which will help students enhance their programming skills and build complex software applications for desktop, web and games. They will learn how to write maintainable, scalable, and readable code.

Textbook(s)

- Joyce Farrell. Microsoft Visual C# 2015: An Introduction to Object-Oriented Programming 6th Edition, Cengage Learning Publisher (2015), ISBN: 978-1285860237

- Gabriel Baptista, Francesco Abbruzzese. Software Architecture with C# 10 and .NET 6 Develop software solutions using microservices, DevOps, EF Core, and design patterns for Azure, Packt Publishing 3rd Edition,(2022), ISBN: 978-1803235257

- Jon Skeet. C# in Depth, Manning Publisher, Fourth Edition, (2019), ISBN: 978-1617294532

Other supplemental materials

Online Tutorial - <https://www.w3schools.com/cs/index.php>

Online Compiler - <https://www.programiz.com/csharp-programming/online-compiler/>

Course Learning Objectives

#	Course Learning Outcomes (CLOs)	BT/ PLO #
CLO 1	Describe and explain the functional mechanisms of Windows platform	C2 1, 10
CLO 2	Identify the objects & their relationships to build object-oriented solution	C3 2, 6
CLO 3	Model a solution for a given problem using object-oriented principles	C3 3,4, 8,
CLO 4	Examine an object-oriented solution	C4 5,6, 8

Week wise Lecture Schedule

W #	Lecture Topics
1	Introduction: Introduction to Visual Studio(VS) .NET IDE, VS installation and setting environment for C# development, Menu Bar and Toolbar
2	Solution Explorer: Toolbox, Properties Window, Using Help, Simple Program: Displaying Text and an Image Introduction to C# programming: Simple Program: Printing a line of text, Another Simple Program: Adding Integers

W #	Lecture Topics
3	Memory Concepts: Arithmetic, Decision Making: Equality and Relational Operators, Control Structures: Looping, Making Decision
4	Control Structures: Arrays in C#, Array Initialization in different ways, Different Types of Arrays in C#, Jagged Arrays, Foreach Loop
5	Object Oriented Programming with C#, Structure of .Net classes, Defining Namespaces, Classes, Abstract Classes
6	Methods: Properties, Defining Access Level, Assemblies, Class View and Object Browser, Static Classes, Static Constructors, Static Members, Static vs Instance Members
7	Exception Handling, Try, Catch, and Finally blocks, C# Exception Classes, Throwing Exception
8	Graphical User Interface, Windows Forms, Event Handling Model, Delegates, Basic Controls of Windows Forms, Advance Controls of Windows Forms, Listbox, CheckedList, RadioList
9	Multiple Document Interfaces (MDI), MenuStrip, ToolStrip, StatusStrip. TreeView, Data Access with ADO.Net, Namespaces and Classes for Database access, Connecting to database
10	SqlConnection, SqlCommand, Insert, Update, Delete commands. ExecuteNonQuery. Fetching Data from database, DataReader, Going through fetched data. Displaying data
11	Binding controls to database, GridView control, DropDownList, RadioList, CheckBoxList, Editing data in GridView control. Layering and its important, Data Link Layer (DLL), Business Login Layer (BLL), Presentation Layer
12	Introduction to LINQ and Object Relational Mapping (ORM). Files and Streams, Static Classes, FileInfo (Create, Delete) File, Append Text to File
13	DirectoryInfo(Create, Delete) Directory, Look for files, File Explorer. Working with Streams: Threads, Thread States: Life Cycle of a Thread, Thread Priorities
14	MultiThreading, Delegates, MultiThreaded Applications: Thread Synchronization, ThreadPool
15	Graphical Device Interface (GDI) and GDI+Drawing basic graphic elements, Working with Convas, Complex graphics objects manipulation
16	Windows Based Services, Files and Registry Operations. Windows Services, Windows Service Architecture. Creating Windows Services

CS373 - Mobile Application Development

Credit Hours: 3(2+1) **Prerequisites:** None

Course Category: Computer Science Elective

Description

Mobile app development creates software for smartphones and tablets, mainly on Android and iOS. It's rapidly growing as industries—from retail to healthcare—need real-time, convenient access for users. Mobile apps are now the primary way people and businesses connect online. To stay competitive, organizations must meet these demands despite device limitations and distribution challenges. Following best practices can simplify this development process.

Textbook(s)

- Bill Phillips and Brian Hardy. Android Programming: The Big Nerd Ranch Guide, Addison-Wesley Professional, (2013), ISBN: 978-0321804334

- Eric Windmill, Flutter in Action, Manning, ISBN: 978-1617296147

- Dawn Griffiths, David Griffiths. Head First Android Development: A Learner's Guide to Building Android Apps with Kotlin 3rd Edition, O'Reilly Media, (2021), ISBN: 978-1492076520

Other supplemental materials

Android Developer Documentation -

<https://developer.android.com>

Flutter Documentation - <https://docs.flutter.dev>

Course Learning Objectives

#	Course Learning Outcomes (CLOs)	BT/ PLO #
CLO 1	Describe Mobile Application Development fundamentals and flow on multiple devices and publishing it online	C2 1,3,4, 5,10
CLO 2	Develop Mobile Application using provided assets with basic functionality	C5 2,3, 6, 10
CLO 3	Generate Mobile application that uses hardware and software resources like sensors and configuration etc. and evaluate functionality	C5 1,3, 5, 7,10
CLO 4	Deploy mobile applications to relevant app stores (e.g., Google Play Store, Apple App Store)	C3 2,4,5 ,10

Week wise Lecture Schedule

W #	Lecture Topics
1	Overview of Android platform and its architecture.

W #	Lecture Topics
	Setting up the development environment (Android Studio, SDK installation).
	Introduction to Java and Kotlin programming languages for Android development.
2	Understanding Android layouts and views Creating interactive UI elements (buttons, text fields, etc.)
3	Handling user input and events Designing responsive and adaptive layouts
4	Working with SQLite database for local data storage. Using Shared Preferences for simple data persistence.
5	Fetching and storing data from/to web services (RESTful APIs) Managing data using Content Providers
6	Making HTTP requests using Retrofit or Volley libraries Parsing and processing JSON/XML responses
7	Handling network connectivity and error scenarios Authentication and security considerations
8	Integrating multimedia elements (images, audio, video) Accessing device features like camera, sensors, and location
9	Utilizing permissions and requesting user consent Working with notifications and background services
10	Preparing an app for deployment (debugging, testing) Publishing an app on the Google Play Store
11	App store guidelines and submission process Monetization strategies (in-app purchases, ads, subscriptions)
12	Advanced UI techniques (custom views, animations) Performance optimization and memory management
13	Testing and debugging Android applications Exploring emerging trends in Android development (e.g., Jetpack Compose)
14	Applying the concepts learned throughout the course to develop a complete Android application
15	Iterative development, testing, and refinement of the project
16	Demonstrate critical thinking and problem-solving skills when facing challenges and complexities in mobile app development projects.

CS374 - Game Design and Development

Credit Hours: 3(2+1) **Prerequisites:** CC102

Course Category: Computer Science Elective

Description

A potential economic contributor to our country is the computer games industry. This interactive course is a great setting for students to design and development their gaming graphics and implementation. Students will learn how to develop games and program them using software tools in addition to the theoretical components. In the process of creating video games, practical experience is prioritized.

Textbook(s)

- Jeremy Gibson. Introduction to Game Design, Prototyping, and Development from Concept to Playable Game—with Unity and C#, Addison-Wesley Professional, (2017), ISBN: 978-0134659862

- Eric Smith. Game Development with Rust and WebAssembly: Learn how to run Rust on the web while building a game, Packt Publishing, (2022), ISBN: 978-1801070973

- Joe Hocking Unity in Action Multiplatform game development in C#, Third Edition, Manning, (2022), ISBN: 978-1617299339

Other supplemental materials

Online Tutorial -

https://www.w3schools.com/cpp/cpp_oop.asp

Online Compiler - <https://ideone.com/xA0en6>

Course Learning Objectives

#	Course Learning Outcomes (CLOs)	BT/ PLO #
CLO 1	Defining the principles of game design and development, applying the mathematics and physics to game design.	C2 1, 10
CLO 2	Learn to structure and define the duties of the game development team, practice art creation, music and animations and various tools needed to create assets. gain programming capability to develop games.	C3 2, 6
CLO 3	To apply basic artificial intelligence to developing computer games. Model a solution for a given problem using object-oriented principles	C3 3,4, 8, 10
CLO 4	To manage porting issues on various platforms, Understand principles of object-oriented paradigm	C4 5, 6, 8

Week wise Lecture Schedule

W #	Lecture Topics
1	Definition of games, game development life cycle, game loop, history of video games, game types taxonomy, basic components.

W #	Lecture Topics
2	Converting ideas to game concepts, player and design team roles, game design documentation types.
3	Game genre taxonomy, genre-specific elements, classification, interface requirements, audience targeting.
4	Game world concepts and dimensions, UI design principles, level design, development tools, technology trends.
5	2D games: sprites, animations, coordinate systems, input handling, Unity intro, basic game physics.
6	Object collision, environments, particle systems, first full game development, PC/mobile deployment.
7	Lighting, particle effects (fire, smoke, water), terrain creation, object placement, texturing custom worlds.
8	Acquiring/importing assets from online sources and Unity Store, building sample games.
9	Advanced game physics: velocity, acceleration, collision, momentum, Unity physics engine, materials, constraints.
10	Game optimization, efficiency improvements, platform-specific considerations.
11	3D humanoid creation: design, texturing, optimization, rigging.
12	Animations and use of humanoids in 2D/3D games.
13	Puzzle design, Scott Kim's concepts, puzzle examples in action games.
14	Game balance, probability math, randomizers, weighted distributions, permutations, feedback loops.
15	Game testing: playtesting importance, tester roles, methods.
16	Publishing on app stores, monetization models.

CS375 - Software Testing and Quality Assurance

Credit Hours: 3(3+0) **Prerequisites:** CC251

Course Category: Computer Science Elective

Description

The course aims to focus on exploring the essential concepts, methodologies, and practices related to ensuring the quality of software through effective testing.

Textbook(s)

- Paul C. Jorgensen, Byron DeVries. Software Testing A Craftsman's Approach, Fifth Edition, Auerbach Publications (2022), ISBN: 978-0367767624

- Andrew Pollner, Mark Fewster, Ina Schieferdecker, Test Automation Engineer Guide to the ISTQB Advanced Level Certification Rocky Nook, (2022), ISBN: 978-1937538088

Course Learning Objectives

#	Course Learning Outcomes (CLOs)	BT/ PLO #
CLO 1	To Understand how Software testing as a primary means to ensure software quality.	C2 1, 10
CLO 2	To discover other alternatives for quality assurance (QA), including defect prevention, process improvement, inspection, formal verification, fault tolerance, safety assurance, and damage control	C3 2, 6
CLO 3	Getting an understanding of the principles, procedure and protocol pertaining to the implementation of Agile Software Development methodology including its potential and impact	C3 3, 4, 8, 10
CLO 4	Understanding the measurement and analysis to close the feedback loop for quality assessment and quantifiable improvement.	C2 5, 6, 8

Week wise Lecture Schedule

W #	Lecture Topics
1	Basics of Software Testing – Important Concepts, Software Testing - Coverage, Limitations, Software Quality Engineering - Defect Prevention, Reduction, Containment.
2	Testing Object-oriented Software – What's Special? Testing Object-oriented Software, Models of the system, Test Model, Finite State Machines, Testing using FSMs.
3	Techniques for Test case generations using FSMs, The FREE State Model, Test generation using State Transition Trees
4	Testing using Class information, Testing using Combinational Models, Integration Testing for Model

W #	Lecture Topics
	Based Software Testing , Regression Testing using Models
5	Model-based Test Data Generation, Test Oracles for MBT, Test Automation & Designing Test Harness for MBT, Test Adequacy Assessment
6	Mission of test groups, Generic Test Process, Test Concepts, What to Test: Black Box vs. White Box, Stopping Criteria: Coverage Based vs. Usage Based
7	An Initial Definition of (Software) Quality, Standard Definitions of (Software) Quality, Quality Problems in Large Software, Turning Failure to Success: SQA for Meeting Quality Expectations, The SQA Process
8	Different views of quality, Quality Frameworks, Defect Management,
9	Detailed Look at Major Testing Activities, Test Management: Responsibilities and Roles, Test Documentation
10	Fault Tolerance via recovery block (RB) and N version programming (NVP) , Safety Assurance Techniques/Strategies
11	FSM-Based Testing, Markov Chains as Enhanced FSMs, Unified Markov Models (UMM) for Testing
12	Quality Models and Measurements, Quality Models in QA Activities, Direct & Indirect Quality Measurements, Data Requirement by Models
13	Checklist-Based Testing, Partitions and Partition Testing, Usage-Based Testing OP Development
14	Ad hoc (non-systematic) testing, Checklist-based testing, Partition-based coverage testing, Boundary testing (BT), Control flow testing (CFT)
15	Quality Standards and Frameworks, Defect Handling in Quality Engineering. Software quality standards, ISO/IEC 25010, ISO/IEC 9126, Capability Maturity Model Integration (CMMI), IEEE 730, IEEE 1061, Common Criteria for Information Technology Security Evaluation (ISO/IEC 15408)
16	Case studies

CS376 - Advanced Web Development

Credit Hours: 3(2+1) **Prerequisites:** CS371

Course Category: Computer Science Elective

Description

Expand your web development skills with this advanced course. Discover advanced techniques in HTML, CSS, and JavaScript, enabling you to create visually stunning and interactive web pages. Explore popular frameworks like Laravel, Node.js, and React to build robust full-stack applications. By the end of this course, you'll have the expertise to develop professional-grade web applications and take your web development skills to new heights.

Textbook(s)

- Christopher John Pecoraro. Mastering Laravel, Packt Publishing, (2015), ISSN: 978-1785285028

- Matt Stauffer. Laravel: Up & Running: A Framework for Building Modern PHP Apps, 2nd Edition, O'Reilly Media, (2019), ISSN: 978-1492041214

- Sebastian Springer. Node.js: The Comprehensive Guide to Server-Side JavaScript Programming, Rheinwerk Computing, (2022), ISSN: 978-1493222926

Other supplemental materials

<https://laracasts.com/series/laravel-8-from-scratch>

<https://laracasts.com/topics/react>

Course Learning Objectives

#	Course Learning Outcomes (CLOs)	BT/ PLO #
CLO 1	Demonstrate proficiency in advanced HTML and CSS techniques	C3 1, 2, 10
CLO 2	Apply advanced JavaScript concepts to develop interactive and dynamic web applications	C4 3, 4
CLO 3	Build full-stack applications using modern frameworks such as Laravel, Node.js, and React	C5 2, 5, 6, 10

Week wise Lecture Schedule

W #	Lecture Topics
1	Advanced HTML and CSS Techniques: HTML5 advanced features and APIs. CSS preprocessors (e.g., SASS, LESS), Responsive design and media queries
2	Advanced JavaScript Concepts: Advanced JavaScript syntax and features, Promises and async/await, Manipulating the DOM with advanced techniques
3	Introduction to Laravel Framework: Overview of Laravel and its ecosystem, Setting up a Laravel project, Initial environment setup and composer

W #	Lecture Topics
4	Laravel Basics: Understanding MVC architecture, understanding various file/folders in Laravel, routing requests, and linking
5	Laravel Basics: Including CSS and Javascript in Laravel, Blade Layout
6	Laravel Database Integration: Working with databases using Laravel's ORM (Eloquent), Database migrations and seeding
7	Advanced Laravel Features: Authentication and authorization in Laravel, Caching and performance optimization techniques, Building RESTful APIs with Laravel
8	Introduction to Node.js: Overview of Node.js and its architecture, Asynchronous programming with Node.js, Building a basic Node.js server
9	Express.js Framework: Introduction to Express.js framework, Routing and middleware in Express.js, Handle different types of requests (GET, POST, PUT, DELETE, etc.)
10	Express.js Framework: Integrating databases with Node.js using ORMs (e.g., Sequelize)
11	Advanced Node.js Development: Implementing authentication and authorization in Node.js
12	Introduction to React: Understanding the React component model, JSX syntax and rendering elements
13	Introduction to React: State management and props in React
14	React Components and Hooks: Building reusable components in React, React hooks for state and side-effects
15	React Components and Hooks: React router for client-side routing
16	Building Full-Stack Applications: Integrating React and Node.js (MERN stack). Creating RESTful APIs with Node.js and Express, building a complete web application using the MERN stack

CS377 - Machine Learning**Credit Hours:** 3(3+0) **Prerequisites:** None**Course Category:** Computer Science Elective**Description**

Machine learning is one of the fastest growing areas of computer science, with far-reaching applications. The aim of this course is to: a) Present the basic machine learning concepts; b) Present a range of machine learning algorithms along with their strengths and weaknesses; c) Apply machine learning algorithms to solve problems of moderate complexity.

Textbook(s)

- Alpaydin, E. Introduction to Machine Learning. Cambridge, MA: The MIT Press. (2020), ISBN: 978-0262043793

- Bishop, C. M. Pattern Recognition and Machine Learning. New York, NY: Springer. (2016), ISBN: 978-1493938438

Other supplemental materials

- Kaggle - <https://www.kaggle.com>

- Google colab

https://colab.research.google.com/?utm_source=scs-index

Course Learning Objectives

#	Course Learning Outcomes (CLOs)	BT/ PLO #
CLO 1	Describe basic machine learning concepts, theories and applications.	C1 1,10
CLO 2	Apply supervised learning techniques to solve classification problems of moderate complexity.	C3 6,7,8, 9,10
CLO 3	Apply unsupervised learning techniques to solve clustering problems of moderate complexity	C3 2,3, 4, 5
CLO 4	Apply reinforcement learning algorithms to environments with complex dynamics.	C3 1,2, 3
CLO 5	Develop a reasonable size project using suitable machine learning technique	C6 1,2, 3,4

Week wise Lecture Schedule

W #	Lecture Topics
1	Introduction to Machine Learning, Types of learning (supervised, unsupervised, semi-supervised, reinforcement), Real-world applications
2	Concept learning, General-to-specific ordering of hypotheses, Hypothesis spaces
3	Version Spaces Algorithm, Candidate Elimination Algorithm, Inductive bias

W #	Lecture Topics
4	Introduction to Supervised Learning, Decision trees, ID3 algorithm, Splitting criteria
5	Naive Bayes classifier, Assumptions, Training and prediction, Use cases
6	Artificial Neural Networks, Perceptron learning, Activation functions
7	Support Vector Machines (SVM), Linear and non-linear classification, Kernels
8	Overfitting and underfitting, Dealing with noisy data, Pruning techniques
9	Measuring classifier performance, Confusion matrix, Precision, Recall, F1-score, ROC curve
10	Linear Regression, Least squares method, Cost functions
11	Logistic Regression, Sigmoid function, Gradient descent for optimization
12	Random Forest,
13	Ada-boost, XG Boost,
14	Unsupervised Learning, Hierarchical Agglomerative Clustering, Dendrograms
15	Case Study/Project/Demo
16	Presentation/Peer Review/Q&A Sessions/Demo

CS378 - Wireless Networks**Credit Hours:** 3(3+0) **Prerequisites:** CC261**Course Category:** Computer Science Elective**Description**

The course introduces the basics of wireless communication, the evolution of modern wireless cellular networks and the design principles of cellular network. It will also demonstrate the architecture and protocols of GSM, GPRS and WCDMA cellular systems. The course will develop concepts of the emerging wireless networks: Wireless LAN, Wireless Mesh Networks, Wireless Personal Area Networks, Wireless Sensor Networks, WiMAX. The emphasis will also be on exploring research issues in emerging wireless networks.

Textbook(s)

- Enabling Technologies for Next Generation Wireless Communications, Mohammed Usman, Mohd Wajid, Mohd Dilshad Ansari, 2024, ISBN: 978-0367689643

- Daniel Collins, Clint Smith. Wireless Networks, , 3rd Edition, McGraw-Hill Education, (2014), ISBN: 0071819835, 9780071819831

Other supplemental materials

- Shafullah Khan, Al-Sakib Khan Pathan. Wireless Networks and Security, Springer Berlin Heidelberg, (2013), ISBN: 978-364236168-5.

- William Stallings, Cory Beard. Wireless Communication Networks and Systems, Pearson, (2016), ISBN: 0133594173, 9780133594171

Course Learning Objectives

#	Course Learning Outcomes (CLOs)	BT/ PLO #
CLO 1	Explain the principles and components of wireless communication systems	C2 1,2, 10
CLO 2	Analyze the operation and characteristics of analog mobile phone systems	C4 2,3
CLO 3	Evaluate the features and benefits of EDGE (Enhanced Data Rate for GSM Evolution) technology	C6 4,5, 6

Week wise Lecture Schedule

W #	Lecture Topics
1	Introduction to Wireless Communication, Wired vs. Wireless Communication
2	Design Challenges, Wireless Transmission
3	Evolution of Wireless Networks, Fundamentals of Cellular Concepts. Analog Mobile Phone System
4	Identifiers in GSM Network, Call Routing in GSM, GPRS

W #	Lecture Topics
5	EDGE: Enhanced Data Rate for GSM Evolution, WCDMA
6	UMTS, CDMA 2000
7	Wireless LAN/IEEE 802.11
8	IEEE 802.16: Worldwide Interoperability for Microwave Access
9	Short range wireless personal area networks
10	Wireless body area networks and applications
11	Mobile ad hoc networks and application
12	Wireless sensor networks and applications
13	Internet of Things and applications
14	Wireless networks security
15	Latest trends in wireless communication
16	Case studies, latest trends

CS379 - Search Engine Optimization

Credit Hours: 3(2+1) **Prerequisites:** None

Course Category: Computer Science Elective

Description

This course introduces the basic concept of Search Engine Optimization to the students. On-Page SEO, Off-Page SEO and Technical SEO strategies are part of the course.

Textbook(s)

- Eric Enge, Stephan Spencer & Jessie Stricchiola, The Art of SEO: Mastering Search Engine Optimization (4th Edition), 2023, ISBN: 978-1098102616.

- Peter Kent. SEO For Dummies, 6th Edition, For Dummies, (2015) ISBN-13: 9781119129554

- Matthew Capala. SEO Like I'm 5: The Ultimate Beginner's Guide to Search Engine Optimization, Maciej M. Capala Publishers, (2015) ISBN-978-1500865207

Other supplemental materials

MOZ Beginners guide to SEO - <https://moz.com/beginners-guide-to-seo>

Learn SEO practically - <https://www.seoblueprint.in/>

Course Learning Objectives

#	Course Learning Outcomes (CLOs)	BT/ PLO #
1	CLO Describe the key terminologies and technologies of SEO	C2 1,2, 10
2	CLO Explain the importance and challenges in White hat SEO	C2 1,2, 10
3	CLO Analyze various types of SEO, which one to adopt, which to avoid	C4 2,3, 4,9
4	CLO Analyze working and performance of key SEO technologies and algorithms	C4 3,4, 5,8
5	CLO Build a website, optimized for modern search engines	C5 4,5, 6,8

Week wise Lecture Schedule

W #	Lecture Topics
1	Introduction to SEO, Benefits of SEO, Challenges in SEO, Types Of SEO technique , Black hat SEO, White hat SEO, History Of SEO
2	On-Page SEO, On-page SEO checklist. Title Optimization, Content optimization, Header Tags Optimization, Meta Description optimization, outbound and Internal Links, URL optimization

W #	Lecture Topics
3	On page Vs. Off-Page Optimization, Introduction to Back Links: Why is it important for SEO? Anchor text and why it is important in link building, creating backlinks. Some proven ways to get backlinks organically, social media submission
4	What Is Technical SEO, Technical SEO Ranking factors, Type of meta tags and their effect on SEO, Site architecture Optimization, Breadcrumbs
5	How to perform the Page Speed Test, XML and HTML sitemap creation and optimization,
6	How to Add and verify a website in Google Console, Google Analytics setup and monitoring
7	Keyword Research Process, the importance of Keyword Research? Different types of keywords. How to do Analysis of keywords using Free & Paid Tools
8	How to find easy to rank keywords related to your website and business, How to do Analysis of Keyword Using Chrome Extension, What Is LSI Keywords : Easy Strategies To find LSI Keywords
9	Avoid Negative SEO, What falls under Negative SEO, Avoid duplicate content on your website, what is Black hat SEO? Common Mistake in SEO. Common Negative SEO practiced by your competitor
10	Local SEO, What is local SEO and why is it important, Important Local SEO factors, Setup of Google My Business (GMB). How to optimize GMB Listing
11	Pay-per-Click and SEO, Pay-per-Click and SEO, Pay-per-Click Categories, Understanding How PPC Affects SEO Keyword,
12	Competitive Research, Choosing Effective Keywords
13	Increasing Keyword Success, Writing Keyword, Advertisement Text
14	Understanding and Using Behavioural Targeting What Is Behavioural Targeting? ,Taking Advantage of Behavioural Targeting,
15	Keyword Tools and Services Campaign management Reports Analytics Print ads
16	The Content Piece of the SEO How Does Web-Site Content Affect SEO?, Duplicate Content, Doorway pages, Cloaking, Hidden links, Considerations for Multi-Lingual Sites

CS381 - Human Computer Interaction

Credit Hours: 3(3+0) **Prerequisites:** None

Course Category: Computer Science Elective

Description

Human-Computer Interaction (HCI) is the study of the principles and methods with which one builds effective interfaces for users. A basic precept of HCI is that users should be able to focus on solving problems, rather than dealing with the intricacies of complex software. Interfaces must be accessible, meaningful, visually consistent, comprehensive, accurate, and oriented around the tasks that users tend to perform. The course will provide a balance of practical and theoretical knowledge, giving students experience ordinarily not provided by other courses in computer science.

Textbook(s)

- Gerard Jounghyun Kim. Human-Computer Interaction: Fundamentals and Practice, , Auerbach Publications, (2015), ISBN: 9781482233896

- Helen Sharp, Jennifer Preece, Yvonne Rogers. Interaction Design: Beyond Human-Computer Interaction, Wiley; 5th edition (2019), ISBN-13 : 978-1119547259

- Alan Dix, Janet Finlay. Human Computer Interaction, 3rd Ed, (2016) ISBN: 9780130461094.

Other supplemental materials

Course Learning Objectives

#	Course Learning Outcomes (CLOs)	BT/ PLO #
CLO 1	Acquire fundamental concepts of computer components functions regarding interaction with human and vice versa.	C2 1,10
CLO 2	Analyze interface problems to recognize what design approach and interaction styles is required in the light of usability standards.	C4 2,3,10
CLO 3	Apply basic concepts to construct a user-interaction strategy for a given problem its usability evaluation and	C3 2,3,4, 10
CLO 4	Ability to design and develop an interface by using appropriate HCI techniques that are preferred by the user.	C5 5, 6,10

Week wise Lecture Schedule

W #	Lecture Topics
1	Introduction to HCI, The Goals of HCI Usability, HCI & its Evolution. Input/Output Channel
2	Human Senses and Characteristics. Eyesight, Hearing, Touch, Human Memory, Emotions etc.
3	The Computer, Text Entry Devices
4	Positioning, Pointing & Drawing Devices, Display, Devices

W #	Lecture Topics
5	Physical Controls, sensors & special devices
6	The Interaction, What are Goals, Domain, Intention. Task Analysis, User & Task Language, and Intention
7	Models of Interaction, The Execution-evaluation cycle, The interaction
8	Frame Work, Ergonomics, Interaction Styles, Command line, Menus, WIMP, Natural Language, Point & Click, and Forms Fill-Ins etc
9	Elements of WIMP, Paradigm & Interaction Design Basics, Paradigm Introduction, Paradigm for Interaction, Time Sharing, VDUs
10	Programming Toolkit, Personal Language Vs Actions, Hypertexts, Multimodality, Computing, Direct Manipulations
11	Computer supported Cooperative Works, The WWW, Agent Based Interfaces, Ubiquitous Computing Sensor Based and Context aware Interaction, Interaction Design Basics, The Process of design, Design Goals Constraints and Trade-Offs, Golden Rule of Design, User Focus, Scenario
12	Navigation Design, Screen design and layout Iteration & Prototype, Design Rules, Principles to support usability, Standards
13	Guidelines, Golden Rules & Heuristics, HCI Patterns
14	Evaluation Techniques, What is Evaluation? Goals of Evaluation, Evaluation through expert analysis
15	Evaluation through user participation, Observational Techniques, Choosing an Evaluation method
16	Universal Design, Universal Design Principles, Multi-Modal Interaction, Design for diversity

CS382 - Computer Graphics

Credit Hours: 3(2+1) **Prerequisites:** None

Course Category: Computer Science Domain

Description

Computer Graphics is a foundational course that explores how images and visual content are created, manipulated, and displayed using computational methods. The course begins with an overview of the field, its evolution, and its integration with art, design, and human perception. It covers essential hardware and software components, mathematical foundations like vectors and matrices, and widely used graphics libraries such as OpenGL. Students will learn techniques for rendering geometric shapes, performing 2D transformations, and implementing clipping and raster algorithms. The course also delves into color models, lighting, shading, and advanced topics like image segmentation, binary image analysis, triangulation, animation, and interactive environments using Unity 3D—equipping students with both theoretical insight and practical skills for modern graphics applications.

Textbook(s)

- Donald Hearn, M. Pauline Baker, and Warren R. Carithers. Computer Graphics with Open GL, 4th Edition, Pearson, (2011), ISBN: 978-0136053583

- Steve Marschner, Peter Shirley. Fundamentals of Computer Graphics, 5th Edition, CRC Press, (2021), ISBN: 978-0367505035

Course Learning Objectives

#	Course Learning Outcomes (CLOs)	BT/ PLO #
CLO 1	To introduce students with fundamental concepts and theory of computer graphics.	C2 1, 10
CLO 2	To Identify the important drawing algorithm, polygon fitting, clipping and 2D transformation curves and an introduction to 3D transformation.	C3 2, 6
CLO 3	provides the basics of OpenGL application programming interface which allows students to develop programming skills in CG.	C3 3,4, 8, 10
CLO 4	Examine a computer graphics system.	C4 5,6,8

Week wise Lecture Schedule

W #	Lecture Topics
1	A Brief History, Relationship of Graphics to Art, Design, and Perception.
2	Computer Graphics Hardware: Video Display Devices, Raster-Scan Systems, Graphics Workstations and Viewing Systems, Input Devices, Hard-Copy Devices, Graphics Networks, Graphics on the Internet.
3	Maths Preliminaries: Coordinate Reference Frames, Points and Lines, Vector, Vector Operations, Matrices, Matrix Operations.

W #	Lecture Topics
4	Computer Graphics Software: Coordinate Representations, Graphics Functions, Software Standards, Other Graphics Packages, Introduction to OpenGL.
5	OpenGL point, line, curve, and fill-area functions; polygon and vertex arrays; pixel and character primitives; display lists and picture partitioning.
6	Basic 2D transformations, matrix and homogeneous representations, inverse and composite transformations, coordinate system mappings, and OpenGL-based transformation functions with programming examples.
7	Two-Dimensional Viewing: The Two-Dimensional Viewing Pipeline, The Clipping Window, Normalization and Viewport Transformations, OpenGL Two-Dimensional Viewing Functions, Clipping Algorithms,
8	Line Drawing Algorithms: Incremental line algorithm, DDA line algorithm, Bresenham line algorithm, Examples.
9	Circle Drawing Techniques: Circle , A Simple Circle Drawing Algorithm , Eight-Way Symmetry , Mid-Point circle drawing algorithm , Examples
10	Color Models and Color Applications: Properties of Light, Color Models, Standard Primaries and the Chromaticity Diagram, The RGB Color Model, The YIQ and Related Color Models, The CMY and CMYK Color Models, The HSV Color Model, The HLS Color Model, Color Selection and Applications.
11	Lighting: Blinn-Phong Lighting Model, Ambient Lighting Model, Diffuse Lighting Model, Specular Lighting Model, Light source
12	Shading: Shading model, some real word examples, Flat shading, Smooth Shading, OpenGL Examples of Flat and Smooth Shading. Triangulation: Problem of Triangulation, Tribulation, Calculation, Trifocal Tensor (Tritensor), Photogrammetry.
13	Binary Image Analysis: Binary Image, Applications of Binary Image Analysis, Key Techniques in Binary Image Analysis(Thresholding, Morphological Operations, Connected Component Labeling, Contour Detection and Analysis, Region Properties), Example Use Case
14	Image Segmentation: Segmentation, Segmentation Applications, Human Vision: Grouping and Gestalt, Gestalt Factors, Parallel curves, symmetric Course Descriptions Bachelor of Studies in Computer Science 47 Week Number Topic groups, continuous curves, Closure Curves, Visual illusions (examples), Simple Segmentation Techniques: Background Subtraction, Shot Boundary Detection, Segmentation by Clustering, displaying object
15	Computer Animation: Computer animation, History of Computer Animation, Applications of Computer Animation, Methods/Techniques(Frame by Frame (Traditional Method), Procedural, Behavioral, Key Framing, Motion Capture, Dynamics)
16	Unity 3D: Introduction to Unity 3D, Scripting in Unity 3D via Visual C#.NET

CS471 - Distributed Wireless Networks

Credit Hours: 3(3+0) **Prerequisites:** None

Course Category: Computer Science Elective

Description

By the end of this course, students will have a comprehensive grasp of Wireless Distributed Systems, Ad Hoc and Sensor Networks, and the various applications and implications of IoT in different domains. They will also be equipped to evaluate security challenges and propose effective countermeasures to ensure the integrity and confidentiality of data in distributed wireless networks.

Textbook(s)

- Ashish Bagwari, Geetam Singh Tomar, Jyotshana Bagwari (Editor), Jorge Luis Victória Barbosa and Musti K.S. Sastry. Advanced Wireless Communication and Sensor Networks: Applications and Simulations 1st Edition, Kindle Edition (2023), ISBN: 978-1032347189

- Savo G. Glisic. Advanced Wireless Networks: Technology and Business Models 3rd Edition, Kindle Edition Wiley. (2016), ISBN:978-1119096856

Other supplemental materials

- Ling Lyu, Xinping Guan, Nan Cheng, Xuemin Sherman Shen. Advanced Wireless Technologies for Industrial Network Systems (Wireless Networks), Springer (2023). ISBN:978-3031269622

Course Learning Objectives

#	Course Learning Outcomes (CLOs)	BT/ PLO #
CLO 1	Understand the fundamental concepts of Wireless Distributed Systems (WDS), including centralized and decentralized/distributed network architectures, and analyze their respective advantages and limitations.	C1 1,2 10
CLO 2	Investigate the Internet of Things (IoT) and its diverse applications	C4 1,2,3 ,10
CLO 3	Assess security threats, challenges and countermeasures in wireless distributed networks.	C6 3,4
CLO 4	Analyze the principles and characteristics of Ad hoc and similar networks	C4 2,3

Week wise Lecture Schedule

W #	Lecture Topics
1	Wireless distributed systems (WDS), basic concepts
2	Centralized versus decentralized/distributed networks
3	Architecture of distributed networks
4	Ad hoc networks

W #	Lecture Topics
5	MANET and VANET
6	Sensor networks
7	Internet of Things (IoT)
8	Internet of Vehicles (IoV)
9	Industrial IoT
10	Internet of Medical Things (IoMT)
11	Social IoT
12	Smart homes and smart cities
13	Security threats and challenges
14	Security countermeasures
15	Case study 1
16	Case study 2

CS472 - Internet of Things and Applications

Credit Hours: 3(3+0) **Prerequisites:** CC261

Course Category: Computer Science Elective

Description

This course provides a comprehensive introduction to the basic concepts of the Internet of Things (IoT) and explores its wide range of applications. Students will learn the basic concepts, principles, and technologies behind IoT systems, as well as their impact on various industries. The course will cover essential topics such as sensors and actuators, connectivity protocols, data analytics, security, and privacy considerations. Students will also receive practical experience through practical exercises and projects involving IoT devices and platforms.

Textbook(s)

- Rajkumar Buyya, Amir Vahid Dastjerdi, and Anton Y. Dong . Internet of Things: Principles and Paradigms, Morgan Kaufmann Publishers, (2016), ISBN: 978-0128053959

- Internet of Things Applications for Sustainable Development, edited by Niranjana Lal, Shamimul Qamar, Sanyam Agarwal, et al., 2023, ISBN: 978-1032128986.

- Pethuru Raj and Anupama C. Raman. The Internet of Things: Enabling Technologies, Platforms, and Use Cases, CRC Press, (2017), ISBN: 978-1032339719

Other supplemental materials

- Research papers on IoT

- <https://theinternetofthings.eu/>

- <https://www.techtarget.com/iotagenda/>

Course Learning Objectives

#	Course Learning Outcomes (CLOs)	BT/ PLO #
1	CLO Understand the fundamental concepts, principles, and architectures of the Internet of Things.	C2 1, 10
2	CLO Identify and evaluate different types of sensors, actuators, and connectivity protocols used in IoT systems.	C2 2,3
3	CLO Examine real-world applications and case studies of IoT across different industries.	C3 3,4,10

Week wise Lecture Schedule

W #	Lecture Topics
1	Internet of Things Definition and characteristics, Key concepts of IoT systems
2	IoT architecture
3	IoT application in different domains

W #	Lecture Topics
4	Enabling Technologies in IoT, History of IoT, About Things in IoT
5	IoT frameworks, IoT and M2M.
6	Sensors: Types of sensors and their characteristics
7	Actuators and their functionalities
8	Interfacing sensors and actuators with IoT platforms
9	Wireless communication protocols (e.g., Wi-Fi, Bluetooth, Zigbee)
10	IoT device connectivity options (e.g., RFID, NFC)
11	Edge computing and fog computing in IoT
12	IoT security challenges and threats Authentication and access control in IoT
13	Smart homes and buildings Industrial IoT and smart manufacturing (research papers)
14	Healthcare and wearable devices Smart cities and transportation (research papers)
15	Agriculture and environmental monitoring
16	Analysis of real-world IoT case studies Presentations and discussions on IoT projects

CS473 - Sensor Networks and Applications

Credit Hours: 3(3+0) **Prerequisites:** CC261

Course Category: Computer Science Elective

Description

In this course, students will explore the unique constraints, advantages, and driving applications of wireless sensor networks. Topics covered include MAC protocols, routing protocols, QoS, energy management, IoT, layer-wise security, TinyOS, and real-world case studies. By the end of this course, students will be able to develop a solid understanding of ad hoc and sensor networks, enabling them to design and analyze efficient wireless communication systems.

Textbook(s)

- Seong-Eun Yoo, Taehong Kim. Industrial Wireless Sensor Networks: Protocols and Applications, Mdp AG, (2020), ISBN: 978-3039436057

- Jing He, Mr. Shouling Ji, Yingshu Li, Yi Pan. Wireless Ad Hoc and Sensor Networks: Management, Performance, and Applications 1st Edition, (2019), ISBN: 978-0367379681

- Vibha Soni. IoT for Beginners: Explore IoT Architecture, Working Principles, IoT Devices, and Various Real IoT Projects, BPB Publications, (2021), ISBN: 978-9355510068

Other supplemental materials

<https://www.springeropen.com/collections/WSN>

https://www.mdpi.com/topics/Wireless_Sensor_Networks

Course Learning Objectives

#	Course Learning Outcomes (CLOs)	BT/ PLO #
1	CLO Understand the unique constraints, challenges, and advantages of sensor networks and their applications.	C2 1, 8, 10
2	CLO Analyze the design issues and architectural considerations in wireless sensor networks, including MAC protocols and their classification.	C4 2, 7
3	CLO Evaluate different routing protocols used in wireless sensor networks and their classifications.	C6 4, 7
4	CLO Explore the concepts of Quality of Service (QoS) and energy management in sensor networks and the Internet of Things (IoT)	C5 3, 10

Week wise Lecture Schedule

W #	Lecture Topics
1	Introduction to adhoc/sensor networks: unique constraints and challenges, advantages of sensor network, driving applications,

W #	Lecture Topics
2	issues in wireless sensor networks, issues in design of sensor network, sensor network architecture, data dissemination and gathering
3	MAC protocols for wireless sensor networks, design goals,
4	classification of MAC protocols, location discovery, quality, other issues,
5	S-MAC, IEEE 802.15, WBAN
6	Routing protocol: classification of routing protocols, table-driven
7	Routing protocols: on-demand, hybrid, flooding, hierarchical, and power aware routing protocols.
8	QoS and Energy Management: Issues and Challenges in providing QoS, classifications, Energy harvesting
9	QoS frameworks, need for energy management, classification, battery, transmission power, and system power management schemes.
10	Internet of Things, Sensors based IoT and different variants
11	Routing protocols used for IoT
12	Industrial IoT, Internet of Vehicles, Social IoT
13	Layer wise security in wireless sensor networks
14	Introduction to TinyOS
15	Case Studies – Applications of WSNs
16	Case studies – Applications of IoT Applications: Healthcare, Industrial Monitoring, Forests Monitoring, Warfare, smart homes, surveillance and security, ocean monitoring, Clustering

CS474 - Artificial Neural Networks

Credit Hours: 3(3+0) **Prerequisites:** CC351

Course Category: Computer Science Elective

Description

Artificial Neural Networks (ANNs) are a fundamental concept in the field of artificial intelligence and machine learning. This 16-week course is designed to provide a comprehensive understanding of ANNs, their architecture, training algorithms, and applications. Students will gain hands-on experience in designing, training and evaluating neural networks through a combination of theoretical lectures, practical exercises, and projects. By the end of the course, students will be equipped with the knowledge and skills necessary to apply ANNs to solve real-world problems.

Textbook(s)

- Martin T Hagan, Howard B Demuth, Mark H Beale, Orlando De Jesús. Neural network design (2nd ed.). Martin Hagan. (2014), ISBN: 978-0971732117
- Artificial Neural Networks and Machine Learning – ICANN 2024, Edited by Michael Wand, Kristina, Malinová, Jürgen Schmidhuber & Igor V. Tetko, 2024 (33rd ICANN conference proceedings)
- Charu C. Aggarwal. Neural Networks and Deep Learning: A Textbook, Springer, (2018), ISBN: 978-3319944623
- Haykin, S. Neural networks and learning machines, 3rd edition. Pearson. (2016), ISBN: 978-9332570313

Other supplemental materials

- TensorFlow Documentation and Tutorials - <https://www.tensorflow.org/>
- PyTorch Tutorials - <https://pytorch.org/tutorials/>

Course Learning Objectives

#	Course Learning Outcomes (CLOs)	BT/ PLO #
1	CLO Understand the fundamental concepts of artificial neural networks, including their architecture, activation functions, and backpropagation algorithm.	C2 1, 9,10
2	CLO Apply appropriate data preprocessing techniques, regularization methods, and optimization algorithms to train neural networks effectively.	C2 2,3,4, 7,8,10
3	CLO Design and implement various types of neural networks, including feedforward, and backpropagation, for solving different types of problems.	C3 3,4,5, 6,10
4	CLO Evaluate and analyze the performance of trained neural networks, interpret their results, and make informed decisions for model improvement and deployment.	C6 3,6, 8,10

Week wise Lecture Schedule

W #	Lecture Topics
1	Introduction to Neural Networks
2	Structure and components of a Neural Network
3	Feedforward Neural Networks
4	Activation functions and their properties
5	Forward propagation and matrix operations
6	Backpropagation Algorithm
7	Error functions and loss calculation
8	Gradient descent and chain rule
9	Backpropagation algorithm for weight updates
10	Training Neural Networks
11	Training set, validation set, and test set
12	Overfitting, regularization, and early stopping
13	Optimization techniques
14	Hyperparameter tuning and grid search
15	Practical Project
16	Practical Project

CS475 - Recommender Systems

Credit Hours: 3(3+0) **Prerequisites:** CC351

Course Category: Computer Science Elective

Description

The course provides an in-depth exploration of recommender systems' theory, algorithms, and applications. Recommender systems play a crucial role in today's digital landscape, helping users discover personalized recommendations for products, services, and content. This course covers various techniques, including content-based filtering, collaborative filtering, and hybrid approaches. Throughout the course, students will gain a comprehensive understanding of the fundamental concepts and methodologies used in building recommender systems. They will learn how to collect and preprocess data, design and implement various recommendation algorithms, and evaluate the effectiveness and performance of recommender systems. By the end of the course, students will have the practical skills to develop personalized, diverse, and fair recommender systems while addressing privacy and ethical concerns. This course is suitable for undergraduate students interested in AI, data science, and user-centric applications.

Textbook(s)

- Ricci, F., Rokach, L., Shapira, B., & Kantor, P. B. Recommender systems handbook (3rd ed.). Springer. (2022), ISBN: 978-1617292705
- Hastie, T., & Tibshirani, R. The Elements of Statistical Learning: Data Mining, Inference, and Prediction (2nd ed.). Springer. (2017), ISBN: 978-0387848570
- Aggarwal, C. C. Recommender Systems: The Textbook. Springer. (2016), ISBN: 978-3319296579

Other supplemental materials

- KDnuggets - <https://www.kdnuggets.com/>
- Towards Data Science - <https://towardsdatascience.com/>

Course Learning Objectives

#	Course Learning Outcomes (CLOs)	BT/ PLO #
CLO 1	Understand the principles and concepts of recommender systems	C2 1, 10
CLO 2	Apply recommender system techniques and algorithms	C3 2, 6, 7, 8, 10
CLO 3	Design and develop recommender systems	C5 2,3,4, 5,6
CLO 4	Evaluate and analyze recommender systems	C6 2,3, 4,6,10

Week wise Lecture Schedule

W #	Lecture Topics
1	Introduction to Recommender Systems
2	Types of recommender systems: content-based, collaborative filtering, and hybrid approaches
3	Data Collection and Pre-processing
4	Dealing with missing data and data sparsity
5	Content-Based Recommender Systems
6	TF-IDF and cosine similarity for content similarity calculation
7	Building and evaluating a content-based recommender system
8	Collaborative Filtering
9	Matrix factorization methods: singular value decomposition (SVD) and alternating least squares (ALS)
10	Handling scalability and sparsity issues in collaborative filtering
11	Hybrid Recommender Systems
12	Weighted hybrid approaches and ensemble methods
13	Context-Aware Recommender Systems
14	Evaluation of Recommender Systems
15	Practical Project
16	Practical Project

CS476 - Blockchain Technologies

Credit Hours: 3(3+0) **Prerequisites:** None

Course Category: Computer Science Elective

Description

In this course the students will learn the basics concepts and importance s of Blockchain. Moreover, architecture, design, mining and consensus algorithm of blockchain especially in bitcoin, Ethereum and hyperledger fabric are also included in this course. The students will be able Implement different applications using blockchain technology at the end of this course

Textbook(s)

- Lukas Engelbrecht. Introduction to Blockchain Technology (2023), ASIN: B0BSXSSTWP
 - Joseph J. Bambara, Paul R. Allen, Kedar Iyer, Rene Madsen, Solomon Lederer, Michael Wuehler Blockchain: A Practical Guide to Developing Business, Law, and Technology Solutions, 1st Edition, McGraw Hill (2018), ISBN: 9781260115871

Other supplemental materials

- Jeffery Macs. Blockchain Basics: A Non-Technical Introduction for Beginners, (2023), ASIN: B0C8RWZKJ

Course Learning Objectives

#	Course Learning Outcomes (CLOs)	BT/ PLO #
CLO 1	Acquire the basic concepts and uses of blockchain with different applications/Systems	C1 1, 10
CLO 2	Describe and apply different stages of blockchain development using different algorithms	C3 2,3, 10
CLO 3	Identify the problems and apply blockchain solutions	C2 2,3, 10
CLO 4	Build blockchain environment using tools	P3 4,5,6, 10

Week wise Lecture Schedule

W #	Lecture Topics
1	Basic Introduction of the course and the agenda Introduction to useful terminologies
2	Hash and its usage, Digital Signature, Authenticity and managing integrity. Need to manage and check integrity
3	Blockchain, how cryptocurrency is related to Blockchain? History of cryptocurrency
4	How cryptocurrency works, Transactions, Blocks, Mining, and the Blockchain, Transactions, Transactions Inputs and Outputs

W #	Lecture Topics
5	Constructing a Transactions, cryptocurrency Mining
6	Centralization vs Decentralization
7	Distributed Consensus, Consensus without identity using a blockchain
8	Security and attacks. The Root of Trust, User Security Best Practices, Physical Storage ▪ Hardware wallets
9	Incentives and proof of work, Mining and proof of work
10	Mechanics of bitcoin, bitcoin scripts and its applications
11	Structure of a Block, Block Header, Block Identifiers: Block Header Hash and Block Height, The Genesis Block, Linking Blocks in the Blockchain, Merkle Trees and Simplified Payment Verification (SPV)
12	Payment services with bitcoin
13	Blockchain Applications, Building Blocks (Primitives)
14	Applications from Building Blocks, Colored Coins, Colored Coins Transactions Payment Channels and State Channels
15	Making Trustless Channels, Routed Payment Channels (lightning Network)
16	Recent research articles about the applications of Blockchains

CS477 - Object Oriented Analysis and Design

Credit Hours: 3(3+0) **Prerequisites:** None
Course Category: Computer Science Elective

Description

The course aims to focus on object-oriented concepts, analysis and software development. The basic concept of OOAD is covered in this course.

Textbook(s)

- John W. Satzinger, Robert B. Jackson, Stephen D. Burd. Systems Analysis and Design in a Changing World, 7th Edition, Cengage Learning, (2016) ISBN: 978-1305117204
 - Object-Oriented Analysis and Design for Information Systems (2nd Ed.), Raul Sidnei Wazlawick, 2024, ISBN: 978-0443137402
 - Russ Miles & Kim Hamilton. Learning UML 2.0, O'Reilly (2006), ISBN: 978-0596009823.

Other supplemental materials

Online Tutorial -
<https://learn.oracle.com/ols/course/object-oriented-analysis-and-design-using-uml/42305/54507>
 Online UML Tool - <https://creately.com/lp/uml-diagram-tool/>
 Online UML Diagram Tool - <https://online.visual-paradigm.com/diagrams/features/uml-tool/>
 Online UML Diagram Tool -
<https://www.smartdraw.com/uml-diagram/uml-diagram-tool.htm>
 Online UML Diagram Maker -
https://www.lucidchart.com/pages/examples/uml_diagram_tool

Course Learning Objectives

#	Course Learning Outcomes (CLOs)	BT/ PLO #
CLO 1	Understand principles of object oriented paradigm	C2 1, 10
CLO 2	Identify the objects & their relationships to build object oriented solution	C3 2, 6
CLO 3	Model a solution for a given problem using object oriented principles	C3 3,4, 8,10
CLO 4	Examine an object oriented solution	C4 5,6, 8

Week wise Lecture Schedule

W #	Lecture Topics
1	Introduction to Object-Oriented Design, History and Advantages of Object-Oriented Design, Comparison of Traditional and Object-Oriented Approaches
2	Understanding the Object-Oriented Concepts and Terminology
3	Introducing the Modeling and Software Development Process, Unified Modeling Language Fundamentals and the Modeling Process
4	Essentials of Rational Unified Process, RUP Disciplines, Phases & Iterations
5	Requirement Elicitation and Analysis, Understanding Requirements, Requirement Types, Use Case Modeling
6	Creating Use Case Diagrams, Creating Use Case Scenarios and Forms
7	Creating Activity Diagrams, Determining the key Abstractions
8	Constructing the Problem Domain Model, Identifying and Modeling Classes, Objects, and their Relationships.
9	Visualizing the Domain Models (UML Class Diagrams), Object-Oriented Analysis and Design Software Tools, CASE Tools overview
10	SOLID Principles
11	Transitioning from Analysis to Design using Interaction Diagrams
12	Implementation of Sequence, Collaboration, Analysis Use Case Diagrams, Object-Oriented Design Patterns
13	State Chart Diagrams and Implementation, Designing for Reusability and Extensibility (DRY, KISS, YAGNI)
14	Object-Oriented Analysis and Design and Agile Software Development
15	Applying Object-Oriented Analysis and Design to Real-World Projects
16	Software Testing and OOAD

SE321- Software Requirement Engineering

Credit Hours: 3(3+0) **Prerequisites:** None
Course Category: Software Engineering Domain

Description

"Software Requirements Engineering" Is the first course in our "Software Engineering" series. If you are involved in software projects, especially large ones, you know that dealing with software requirements is the biggest problem the software practitioner faces almost at every project. Not working effectively with software requirements could be the number one reason for many software projects' failures. The course will discuss concepts for systematically establishing, defining, and managing the requirements for large, complex, changing, and software-intensive systems from technical, organizational, and management perspectives. The course will consider the past, present, and future software requirements effectively and efficiently. In this course, you will learn various techniques to capture requirements, analyse and validate requirements, control, and manage requirements change requests, and deliver a solid requirements document. You will understand the requirements analysts' role in a software project, which will help you know what you need from them and what they need from you.

Textbook(s)

- Modern Software Engineering by Dave Farley (2022), ISBN: 978-0137314911.
- Karl Wieggers. Software Requirements , Microsoft Press ,(2003) ISBN: 978-0735618794
- Roger Pressman "Software Engineering: A Practitioner's Approach" McGraw Hill publisher, (2019), ISBN: 978-1260548006

Other supplemental materials

- Elizabeth Hull, Ken Jackson and Jeremy Dick. "Requirements Engineering" 4th edition, Springer-Verlag London Limited, (2017), ISBN: 978-3319869971

Course Learning Objectives

#	Course Learning Outcomes (CLOs)	BT/ PLO #
CLO 1	Understand of the importance of following a systematic requirement engineering process	C2 1, 4, 10
CLO 2	Effectively gather and analyse software requirements for the development of cost-effective and efficient technical solutions	C3 6, 7, 8, 9, 10
CLO 3	Use system modelling techniques for requirements analysis and requirements presentation	C4, 3 5, 10

Week wise Lecture Schedule

W #	Lecture Topics
1	Introduction to software Requirement Engineering, Software Requirements, Types of Software Requirements
2	Another view of requirements, Processes and Process Models, Requirements Engineering Process
3	Actors and stakeholders in the requirements engineering process, Social and Cultural Issues in Requirements Engineering
4	Requirements Elicitation, Requirements Elicitation Techniques
5	Requirements Analysis, Requirements Analysis and Negotiation
6	Requirements Errors, types of Requirements Errors,
7	Requirements Validation, Requirement Validation Techniques
8	Requirements Management, Requirements Traceability, , Introduction to Jira for Requirements Tracking
9	Prototyping, Writing Requirements, Requirements Document
10	Quality Attributes of Requirements Documents, Use Case Modeling, Case Study and Use Cases for Case Study
11	Software Modeling, Object-Oriented Modeling with UML
12	Object-Oriented Static Modeling of the Banking System
13	Functional Modelling, Real-Time Structured Analysis and Design Technique (RSTAD) and Application of Banking Case
14	Structured Analysis and Design Technique, Dynamic Modelling, Dynamic Modelling of Banking System Case Study - I
15	Dynamic Modelling of Banking System Case Study – II, Requirements Document for the Banking System, Requirements Engineering in Agile Using BDD (Behavior-Driven Development)
16	Requirements Engineering for Web-Based Information Systems, Requirements Engineering for Product Lines, Requirements Prioritization

SE331 - Software Construction and Development

Credit Hours: 3(3+0) **Prerequisites:** CC251
Course Category: Software Engineering Domain

Description

In the course the student will master software development processes, engineering infrastructure, and system life cycle models. They will be able to implement improvements, practice code salvaging, and adopt Martin Fowler's refactoring concepts. Learn about legacy code handling, fault tolerance, and configuration management. Embrace peer reviews and personal feedback for enhanced expertise.

Textbook(s)

- Adair Dingle. Software Essentials: Design and Construction (Chapman & Hall/CRC Innovations in Software Engineering and Software Development Series) 1st Edition (2014), ISBN:978-1439841204
- Software Engineering A Complete Guide - 2023 Edition.(2023) by The Art of Service - Software Engineering Publishing ,(2023) ISBN: 978-1038807991

Other supplemental materials

- Steve McConnell. Code Complete: A Practical Handbook of Software Construction, Microsoft Press; (2004) 2nd Edition. ISBN: 978-0735619678

Course Learning Objectives

#	Course Learning Outcomes (CLOs)	BT/ PLO #
CLO 1	Understand the role of design and its major activities within the OO software development process, with focus on the Unified process.	C2 1,2, 10
CLO 2	Develop Object-oriented design models and refine them to reflect implementation details	C5 2,3
CLO 3	Evaluate different architectures for a medium size software.	C6 3,4, 5
CLO 4	Implement design model using an object-oriented programming language.	C3 3, 5,6, 7, 10

Week wise Lecture Schedule

W #	Lecture Topics
1	Software development process, Software engineering process infrastructure
2	Software engineering process improvement, Systems engineering life cycle model
3	Process implementation, Levels of process definition
4	Life cycle model characteristics, Individual and team software process

W #	Lecture Topics
5	Lehman's Laws, code salvaging, and configuration management
6	Martin Fowler's refactoring concepts and their application to small projects
7	Apply Michael Feathers' "legacy code" concepts
8	Exception handling, making methods robust by having them check their inputs sent from calling objects
9	Software configuration management, Release management,
10	Software configuration management processes
11	Software deployment processes, Distribution and backup, Evolution processes and activities
12	Basic concepts of evolution and maintenance, Working with legacy systems
13	Refactoring, Error handling
14	exception handling, and fault tolerance
15	Personal reviews (design, code, etc.)
16	Peer reviews (inspections, walkthroughs, etc.)

SE332 - Software Design and Architecture

Credit Hours: 3(3+0) **Prerequisites:** CC251
Course Category: Software Engineering Domain

Description

Introduction to OO Design, System design and software architecture, Object design, Mapping design to code, User interface design, Persistent layer design, Web applications design, State machine diagrams and modeling, Agile software engineering, Exploring inheritance, Interactive systems with MVC architecture, Software reuse.

Textbook(s)

- Len Paul Bass, Clements, Rick Kazman, Software Architecture in Practice, Addison-Wesley Professional, (2021), ISBN: 978-0136886099
 - Moises Gamio. Software Design Principles: A Practical Guide, (2022), ISBN: 979-8848053401

Other supplemental materials

- Ludwin Barbin. SOFTWARE DESIGN PRINCIPLES Kindle Edition by, (2013), B00EHWS180
<https://www.gofpatterns.com/index.php>

Course Learning Objectives

#	Course Learning Outcomes (CLOs)	BT/ PLO #
1	CLO Understand the role of design and its major activities within the OO software development process, with focus on the Unified process	C2 1, 4, 10
2	CLO Design OOD models and refine them to reflect implementation details	C5 3, 4, 5
3	CLO Implement the design model using an object-oriented programming language	C5 2, 3, 5

Week wise Lecture Schedule

W #	Lecture Topics
1	Introduction to Software Design: Introduction to Software Architecture, Introduction to Software Design, Design Vs. Architecture
2	Software Architecture Core Concepts: Define Software Architecture, Architecture Address NFR, Architectures and Technologies
3	Software Architecture Business Cycle: Where do architecture comes from, Software process and architecture business cycle, what makes a good architecture
4	Software Architecture Business Cycle: Activities in software process and architecture business cycle, Architectural Structures and views
5	Quality Attributes in Software Architecture: Motivation, what are the different Quantity attributes,

W #	Lecture Topics
	Performance, Scalability, Modifiability, Security, Availability, Robustness, How to Achieve quality attributes at architectural level
6	Software Architecture Process: Process Outline, Architecture Design, Validation
7	Case Study (solution): Architecture Patterns, Structural View, Behavioral View, Implementation Issues
8	Documenting Software Architecture: Architecture Requirements, Solution, Architecture Analysis
9	Introduction to design: Introduction to Unified modelling language (UML), Structural and behavioral diagrams
10	Introduction to design: Introduction to Unified modelling language (UML), Structural and behavioral diagrams
11	Activity: Modeling structural (class diagram) and behavioral diagram (use case, sequence diagram and state machine diagram) of online shopping application
12	Introduction to Software Product Line engineering: Overview, what makes SPL works, Architectures for Product Lines
13	Aspect Oriented Architecture: Introduction to Aspect-Oriented Programming (basic concepts), Aspect oriented architecture
14	Aspect Oriented Architecture: Aspect oriented Modelling using UML, Aspect Oriented Modelling tools
15	Model-Driven Architecture: What is MDA? Why MDA?
16	Model-Driven Architecture: State of the Art Practices and Tools

SE351 - Software Quality Engineering

Credit Hours: 3(3+0) **Prerequisites:** CC251
Course Category: Software Engineering Domain

Description

This course gives you the overview of the concepts, philosophies, tools, techniques, guidelines, and framework for software quality assurance, verification and validation of software, measurement of quality, quality factors, quality costs at different stages of software development lifecycle, Software testing and usability evaluation methods Software quality standards.

Textbook(s)

- A Guide to Software Quality Engineering by Shravan Pargaonkar, 2024, ISBN 978-1032694412
- Ivan Mistri Richard M Soley John Grundy Bedir Tekinerdogan and Nour Ali. Software Quality Assurance, Elsevier Inc, (2016), ISBN: 978-0-12-8023013.

Other supplemental materials

- Nina S. Godbole, Software Quality Assurance: Principles and Practices 2nd Revised Edition, Alpha Science International Ltd, (2016) ISBN-10: 184265702X, ISBN-13: 978-1842657027

Course Learning Objectives

#	Course Learning Outcomes (CLOs)	BT/ PLO #
CLO 1	Demonstrate an understanding of software quality assurance principles and practices	C2 1, 2, 10
CLO 2	Apply software quality assurance techniques for verification and validation	C3 1, 3, 4, 10
CLO 3	Evaluate the role of automated tools and coding standards in ensuring software quality	C6 4, 5

Week wise Lecture Schedule

W #	Lecture Topics
1	Introduction to Software Quality Assurance: Course Introduction Course Objectives, Course Contents, Quality expectations, An Initial Definition of (Software) Quality, Standard Definitions of (Software) Quality, Quality Problems in Large Software
2	SQA Expectation : Turning Failure to Success: SQA, SQE for Meeting Quality Expectations, The SQE Process, Scope of Major SQE Activities
3	Software Quality Frameworks : Quality: Perspectives and Expectations, Quality Frameworks and ISO-9126,

W #	Lecture Topics
	Correctness and Defects: Definitions, Properties, and Measurements , A Historical Perspective of Quality
4	Software Quality Assurance Activities: Quality Assurance (QA), Classification Scheme for QA as Dealing with Defects, Defect Prevention, Defect Reduction (Defect Detection and Removal), Defect Containment
5	Software Quality through Verification and Validation: QA in context, QA as Defect Handling, QA in Software Process Context, QA activities in V&V Context, V&V in Software Process, QA: Defect View vs. V&V View
6	SQE embedded in Software Process: Quality Engineering, Key SQE Activities, SQE in Software Process
7	Testing Overview: Mission of test groups, Generic Test Process, Test Concepts, What to Test: Black Box vs. White Box, Stopping Criteria: Coverage Based vs. Usage Based, Lifecycle Testing
8	Testing Activities: Detailed Look at Major Testing Activities, Test Management: Responsibilities and Roles, Testing Automation, Test Documentation
9	Boundary Testing : Checklist-Based Testing, Partitions and Partition Testing, Usage-Based Testing with Musa's Ops, OP Development, Procedure/Examples
10	Coverage and Usage Testing Based on FSMs and Markov Chains: Finite-State Machines (FSMs), FSM-Based Testing, Markov Chains as Enhanced FSMs, Unified Markov Models (UMM) for Testing
11	Software Inspection: Basic Concept and Generic Process, Fagan Inspection, Other Inspection and Related Activities, Other Issues.
12	Defect Prevention and Process Improvement: Basic Concepts and Generic Approaches, Root Cause Analysis for Defect Prevention, Education and Training for Defect Prevention, Other Techniques for Defect Prevention, Focusing on Software Processes
	Fault Tolerance and Safety Assurance: Basic Concepts, Fault Tolerance via RB and NVP, Safety Assurance Techniques/Strategies, Summary and Perspectives
	Quality Through Automated Tools: UML generation , UML conversion into Coding
	Coding Standards: Coding Standards Rules, Best Practices
	Comparing QA Alternatives: General Areas/Questions for Comparison, Applicability, Effectiveness, and Cost, Summary and Recommendations
	Quality in GUI: Quality in user behavior Pattern , Guidelines for Quality interface designing.

SE461 - Software Project Management

Credit Hours: 3(3+0) **Prerequisites:** None
Course Category: Software Engineering Domain

Description

This course equips learners with essential project management principles and techniques for IT projects. Participants will critically evaluate real-world project management issues and collaborate in teams to develop comprehensive project plans. Through practical examples, learners will gain confidence in managing projects effectively.

Textbook(s)

- Bob Hughes and Mike Cotterell. Software Project Management, McGraw-Hill Education; 5th Edition (2009). ISBN-10: 0077122798
 - Project management Institute. A Guide to the Project Management Body of Knowledge, 3rd Edition (PMBOK Guides), (2021) ISBN-13: 978-1930699458

Course Learning Objectives

#	Course Learning Outcomes (CLOs)	BT/ PLO #
1	CLO Demonstrate an understanding of project lifecycle principles and the ability to identify suitable project scenarios to share this knowledge with learners effectively.	C2 1, 10
2	CLO Engage in critical discussions with course participants and learners to assess the challenges and practical applications of project management in real-world settings	C6 2, 3, 10
3	CLO Apply project management techniques for IT projects, including initiation, planning, execution, and evaluation, while collaborating in teams to develop a comprehensive project plan. This plan should encompass key tasks, critical path analysis, task dependencies, and a realistic timeline.	C3 3, 4, 5
4	CLO Present practical strategies for building confidence in project management by illustrating simple project planning examples.	C5 4,5,6

Week wise Lecture Schedule

W #	Lecture Topics
1	Introduction to Software Project Management: Importance in IT, fundamentals of People, Process, Product, and Technology.
2	Project Planning & Structures: Planning, tracking, measurement; organizational structures and their influence.

W #	Lecture Topics
3	Initiating a Project: Project Charter, Statement of Work (SOW), Request for Proposal (RFP).
4	Planning Phase: Development lifecycle models, creating project plans.
5	Project Selection & Financial Analysis: NPV, ROI, Payback, Weighted Scoring Model.
6	Estimation & Budgeting: Effort and cost estimation, developing project budgets.
7	Scheduling: Network diagrams, PERT, Gantt charts, Critical Chain Scheduling.
8	Risk & Change Management: Risk analysis/mitigation, change control processes.
9	Development Management: Team models, requirements process, configuration management.
10	Metrics, Tools, & Team Management: Software metrics, programming tools, conflict management, motivation.
11	Project Control: Status reporting, project metrics, Earned Value Analysis (EVA).
12	Communication & Process Improvement: Communication techniques, identifying process improvements.
13	MS Project: Resource leveling, various MS Project views.
14	System Testing: Test specifications, black-box and white-box testing.
15	Final Phases: Project recovery, documentation, cutover/migration, post-project reviews.
16	Project Success Factors: Management support, expectation handling, success metrics.

SE371 - Software Testing and Quality Assurance

Credit Hours: 3(3+0) **Prerequisites:** CC251
Course Category: Software Engineering Elective

Description

The course aims to focus on exploring the essential concepts, methodologies, and practices related to ensuring the quality of software through effective testing.

Textbook(s)

- Paul C. Jorgensen, Byron DeVries. Software Testing A Craftsman's Approach, Fifth Edition, Auerbach Publications (2022), ISBN: 978-0367767624
 - Andrew Pollner, Mark Fewster, Ina Schieferdecker, Test Automation Engineer Guide to the ISTQB Advanced Level Certification Rocky Nook, (2022), ISBN: 978-1937538088

Course Learning Objectives

#	Course Learning Outcomes (CLOs)	BT/ PLO #
CLO 1	To Understand how Software testing as a primary means to ensure software quality.	C2 1, 10
CLO 2	To discover other alternatives for quality assurance (QA), including defect prevention, process improvement, inspection, formal verification, fault tolerance, safety assurance, and damage control	C3 2, 6
CLO 3	Getting an understanding of the principles, procedure and protocol pertaining to the implementation of Agile Software Development methodology including its potential and impact	C3 3, 4, 8, 10
CLO 4	Understanding the measurement and analysis to close the feedback loop for quality assessment and quantifiable improvement.	C2 5, 6, 8

Week wise Lecture Schedule

W #	Lecture Topics
1	Basics of Software Testing – Important Concepts, Software Testing - Coverage, Limitations, Software Quality Engineering - Defect Prevention, Reduction, Containment.
2	Testing Object-oriented Software – What's Special? Testing Object-oriented Software, Models of the system, Test Model, Finite State Machines, Testing using FSMs.
3	Techniques for Test case generations using FSMs, The FREE State Model, Test generation using State Transition Trees
4	Testing using Class information, Testing using Combinational Models, Integration Testing for Model

W #	Lecture Topics
	Based Software Testing , Regression Testing using Models
5	Model-based Test Data Generation, Test Oracles for MBT, Test Automation & Designing Test Harness for MBT, Test Adequacy Assessment
6	Mission of test groups, Generic Test Process, Test Concepts, What to Test: Black Box vs. White Box, Stopping Criteria: Coverage Based vs. Usage Based
7	An Initial Definition of (Software) Quality, Standard Definitions of (Software) Quality, Quality Problems in Large Software, Turning Failure to Success: SQA for Meeting Quality Expectations, The SQA Process
8	Different views of quality, Quality Frameworks, Defect Management,
9	Detailed Look at Major Testing Activities, Test Management: Responsibilities and Roles, Test Documentation
10	Fault Tolerance via recovery block (RB) and N version programming (NVP) , Safety Assurance Techniques/Strategies
11	FSM-Based Testing, Markov Chains as Enhanced FSMs, Unified Markov Models (UMM) for Testing
12	Quality Models and Measurements, Quality Models in QA Activities, Direct & Indirect Quality Measurements, Data Requirement by Models
13	Checklist-Based Testing, Partitions and Partition Testing, Usage-Based Testing OP Development
14	Ad hoc (non-systematic) testing, Checklist-based testing, Partition-based coverage testing, Boundary testing (BT), Control flow testing (CFT)
15	Quality Standards and Frameworks, Defect Handling in Quality Engineering
16	Software quality standards, ISO/IEC 25010, ISO/IEC 9126, Capability Maturity Model Integration (CMMI), IEEE 730, IEEE 1061, Common Criteria for Information Technology Security Evaluation (ISO/IEC 15408)

SE372 - Agile Software Development

Credit Hours: 3(3+0) **Prerequisites:** None
Course Category: Software Engineering Elective

Description

In this course, the students will explore the principles, practices, and techniques of Agile Software Development in the context of software development. Agile methodologies have gained significant popularity in recent years due to their ability to deliver high-quality software solutions in a flexible and collaborative manner.

Textbook(s)

- Agile Software Development by Susheela Hooda, Vandana Mohindru Sood, Yashwant Singh, Sandeep Dalal, Manu Sood (2023), ISBN: 978-1119896395
- Jim Highsmith. Agile Project Management: Creating Innovative Products, 2nd edition, Addison-Wesley Professional(2009) ISBN-10: 0321658396
- Scott M. Graffius. Agile Transformation: A Brief Story of How an Entertainment Company Developed New Capabilities and Unlocked Business Agility to Thrive in an Era of Rapid Change (2019), ISBN-10: 1072447967

Other supplemental materials

Online Tutorial -
<https://www.altexsoft.com/whitepapers/agile-project-management-best-practices-and-methodologies/>
 Online Tutorial - <https://www.atlassian.com/agile/project-management>

Course Learning Objectives

#	Course Learning Outcomes (CLOs)	BT/ PLO #
1	CLO Understand how to encapsulate Agile Software Development Life Cycle and principles in their respective organizational environments	C2 1, 10
2	CLO Discover how to assimilate prerequisites pertaining to the implementation of Agile Software Development principles into a contextual framework of localized objectives	C3 2, 6
3	CLO Applying the principles, procedure and protocol pertaining to the implementation of Agile Software Development methodology including its potential and impact	C3 3, 4, 8
4	CLO Examining the roles, functions and responsibilities of various actors, organizations, and institutions in the Agile Software Development genre	C4 5, 6, 8

Week wise Lecture Schedule

W #	Lecture Topics
1	Introduction to Agile: History of agility and software development lifecycle.

W #	Lecture Topics
2	Agile KPIs & Philosophy: Key performance metrics, Agile philosophy and principles.
3	Agile Methodology: Phases, benefits, frameworks, and practices.
4	Agile Frameworks Overview: Scrum, Kanban, Hybrid, Lean, Bimodal, XP, Crystal.
5	Scrum Deep Dive: Roles, artifacts, sprints, meetings, and when to use.
6	Kanban & Hybrid: Principles, workflows, usage scenarios.
7	Bimodal & Lean: Principles, comparison with traditional approaches.
8	Extreme Programming (XP): TDD, refactoring, CI, pair programming, coding standards.
9	Agile Implementation: Iteration workflow, Wrike, sprint planning, Gantt charts.
10	Agile Teams: Structure, roles, responsibilities, traditional vs agile.
11	Scaled Agile Framework (SAFe): Core values, principles, benefits.
12	Lean-Agile Portfolio Mgmt: Spotify model, squads, tribes, and guilds.
13	Agile Transformation: Adoption vs transformation, ROI, challenges, Agile work structure.
14	Agile Estimation: Story points, planning poker, best practices.
15	Case Studies: Real-world Agile projects, lessons learned, best practices.
16	Project Success: Agile success factors, common pitfalls.

SE373 - Usability Engineering

Credit Hours: 3(3+0) **Prerequisites:** None
Course Category: Software Engineering Elective

Description

This course gives you the overview about, What Usability Engineering actually is? What is its importance in software development? What are Rules/principles to make a usable interface? It also teaches how to conduct usability testing of software.

Textbook(s)

- NurayAykin, Usability and Internationalization of Information Technology, LAWRENCE ERLBAUM Associates CRC Press, (2015), ISBN: 978-14106-1109-3
 - Sreedhar G. Design Solutions for Improving Website Quality and Effectiveness, IGI Global, (2016), ISBN: 9781466669765
 - Carol M. Barnum, Morgan Kaufmann. Usability Testing Essentials: Ready, Set ...Test!: Ready, Set...Test!, 2nd edition (July 31, 2020), ISBN: 9780128169421

Other supplemental materials

- Jeffrey Rubin, Dana Chisnell and Jared Spool. Handbook of Usability Testing: Howto Plan, Design, and Conduct Effective Tests, Wiley, (2008), ISBN-13: 978-0470185483.

Course Learning Objectives

#	Course Learning Outcomes (CLOs)	BT/ PLO #
CLO 1	Understand and identify usability issues in User interfaces.	C2 1, 10
CLO 2	Design user interfaces according to the standards.	C5 2,3,4
CLO 3	Evaluate user interfaces using Heuristic Evaluation and Thinking aloud Test.	C6 3,4,5, 10
CLO 4	Demonstrate skills to collaborate in a team for justifying identified problems and to write interface related reports as per the standards.	C3 3,4, 10

Week wise Lecture Schedule

W #	Lecture Topics
1	Introduction, Background, what is usability? What are its considerations, Testing Usability slogans(Know your user, less is more, etc.)
2	Discount Usability Engineering techniques, Usability attributes: Learnability, Memorability, Few Errors
3	Usability attributes: Satisfaction, Efficiency, Cost saving gained from Usability Testing , Usability trade-offs

W #	Lecture Topics
4	Categories of users and individual user differences , Generation of User Interfaces, Batch system, Line oriented Interfaces, Full Screen Interfaces, GUI
5	Long term trends in Usability, Usability Engineering Lifecycle, Know the user, Competitive analysis, Goal settings, Parallel Design.
6	Usability Engineering Lifecycle: Participatory design, Interface evaluation, Prototyping, Heuristic Usability Guidelines: Simple dialogue, Speak the user language.
7	Heuristic Usability Guidelines, minimize user memory load, Consistency, Clearly marked exits
8	Heuristic Usability Guidelines, Short cuts, Prevent errors , Heuristic Usability Guidelines, Good error messages, Help and documentations
9	Usability Testing, Test Goals, Test Plans, Test budgets
10	Getting test users , Choosing experimenters
11	Ethical aspects of tests with human , Test Stages, Preparation, Introduction
12	The test itself , Debriefing
13	Interface standards, National and International Standards
14	Producing usable In-house standards, International Usability Engineering
15	Guidelines for Internationalization , Usability Assessment Methods beyond, Observation
16	Questionnaires, Interviews

SE374 - Software Metrics

Credit Hours: 3(3+0) **Prerequisites:** None
Course Category: Software Engineering Elective

Description

This course is a step-by-step description of the software metrics. It includes introduction to foundations of measurement theory, models of software engineering measurement, software products metrics, software process metrics and measuring.

Textbook(s)

- Software Metrics: A Rigorous and Practical Approach (3rd Edition), Norman Fenton & James Bieman, 2020, ISBN: 978-0367659028
- Alain Abran, Software Metrics and Software Metrology, Wiley, (2015), ISBN: 978-0470597200
- Norman Fenton, James Bieman. Software Metrics: A Rigorous and Practical Approach, Third Edition, CRC Press (2014), ISBN: 978-1439838228

Other supplemental materials

- Norman Fenton and James Bieman, Software Metrics: A Rigorous and Practical Approach, Third Edition (Chapman & Hall/CRC Innovations in Software Engineering and Software Development Series) (2014), ISBN-10: 1439838224

Course Learning Objectives

#	Course Learning Outcomes (CLOs)	BT/ PLO #
CLO 1	Demonstrate an understanding of fundamental concepts and principles of measurement and experimentation in the context of software engineering	C2 1,2, 10
CLO 2	Apply measurement techniques and models to gather, analyse, and interpret software metrics data	C3 3, 4
CLO 3	Evaluate the effectiveness of software measurement for decision support and quality assessment	C6 4,5

Week wise Lecture Schedule

W #	Lecture Topics
1	Fundamentals of Measurement and Experimentation, Measurement: What Is It and Why Do It? Measurement in Everyday Life, Measurement in Software Engineering Scope of Software Metrics
2	The Representational Theory of Measurement, Measurement and Models. Measurement Scales and Scale Types, Meaningfulness in Measurement
3	A Goal-Based Framework for Software Measurement
4	Classifying Software Measures, Determining What to Measure Applying the Framework, Software

W #	Lecture Topics
	Measurement Validation, Performing Software Measurement Validation
5	Empirical Investigation: Principles of Empirical Studies, Planning Experiments, Planning Case Studies as Quasi-Experiments, Relevant and Meaningful Studies
6	Software Metrics Data Collection: Defining Good Data, Data Collection for Incident Reports, How to Collect Data, Reliability of Data Collection Procedures
7	Analyzing Software Measurement Data: Statistical Distributions and Hypothesis Testing, Classical Data Analysis Techniques, Examples of Simple Analysis Techniques, More Advanced Methods, Multicriteria Decision Aids, Overview of Statistical Tests
9	Metrics for Decision Support: The Need for Causal Models from Correlation and Regression to Causal Models. Bayes Theorem and Bayesian Networks, Applying Bayesian Networks to the Problem of Software Defects Prediction, Bayesian Networks for Software Project Risk Assessment and Prediction
10	
11	Metrics for Decision Support: The Need for Causal Models. From Correlation and Regression to Causal Models. Bayes Theorem and Bayesian Network Applying Bayesian Networks to the Problem of Software Defects Prediction. Bayesian Networks for Software Project Risk Assessment and Prediction
12	
13	Measuring Internal Product Attributes: Structure Aspects of Structural Measures, Control Flow Structure of Program Units Design-Level Attributes, Object-Oriented Structural Attributes and Measures, No Single Overall "Software Complexity" Measure
14	
15	Measuring External Product Attributes: Modeling Software Quality Measuring Aspects of Quality, Usability Measures Maintainability Measures, Security Measures
16	Software Reliability: Measurement and Prediction, Basics of Reliability Theory, The Software Reliability Problem, Parametric Reliability Growth Models, Predictive Accuracy: Recalibration of Software Reliability Growth

SE375 - Fundamentals of UX

Credit Hours: 3(3+0) **Prerequisites:** None
Course Category: Software Engineering Elective

Description

Fundamentals of User Experience (UX) introduces students to the principles, processes, and practices of designing intuitive, user-centered digital products. This course emphasizes the importance of usability, accessibility, and human-centered design in software development. Students will learn how to evaluate and improve user interfaces through research, prototyping, and usability testing.

Textbook(s)

- Don't Make Me Think: A Common Sense Approach to Web Usability, Steve Krug, 3rd Edition, 2014, ISBN: 978-0321965516, New Riders
- About Face: The Essentials of Interaction Design, Alan Cooper, Robert Reimann, David Cronin, Christopher Noessel, 4th Edition, 2014, ISBN: 978-1118766576, Wiley
- The Design of Everyday Things, Don Norman, Revised and Expanded Edition, 2013, ISBN: 978-0465050659, Basic Books
- Anthony E Sanchez, 2024 - 2025 Newbies Guide to UI/UX Design Using Figma, 2024, ISBN: 978-1763504516.
- Designing the User Interface: Strategies for Effective Human-Computer Interaction, Ben Shneiderman, Catherine Plaisant, Maxine Cohen, Steven Jacobs, 6th Edition, 2016, ISBN: 978-0134380384, Pearson

Other supplemental materials

- Industry standard content on: UX, UsabilityHeuristics, User testing, Accessibility: www.nngroup.com
- Visually explains psychological principles in UX : www.lawsofux.com

Course Learning Objectives

#	Course Learning Outcomes (CLOs)	BT/ PLO #
CLO 1	Describe the fundamental principles of usability and user-centered design	C1 1,2, 10
CLO 2	Apply UX design guidelines to create intuitive and user-friendly layouts and navigation structures.	C3 1,2, 10
CLO 3	Analyze and evaluate web interfaces for usability based on user behavior and design principles	C4 2,3,5, 9
CLO 4	Conduct and interpret simple usability tests using observation and think-aloud protocols.	C3 3,8
CLO 5	Demonstrate the ability to revise and improve UX designs based on user feedback and test results.	C5 4,6, 8

Week wise Lecture Schedule

W #	Lecture Topics
1	Introduction to UX What is UX?, Usability vs. User Experience, The "Don't Make Me Think" philosophy
2	Principles of Usability, What makes something usable?, The importance of intuitive design, Users don't read—they scan
3	Conventions and Visual Design, Web conventions and why they matter, Visual hierarchy: Making things obvious, First impressions of design
4	Clickability and Clarity, Designing things that look clickable, Making choices obvious, Reducing cognitive load
5	Navigation Design – Part 1, Importance of good navigation, Types of navigation (global, local, utility), How users look for information
6	Navigation Design – Part 2, Creating clear navigation menus, Page names and "you are here" cues, Breadcrumbs and orientation,
7	Site Structure & Information Architecture, Organizing content logically, Sitemaps and hierarchy, Designing for scan-ability
8	Home Page Design, Goals of the homepage, Communicating purpose and identity, Case studies of good homepages
9	Writing for the Web, Microcopy and meaningful text, Writing clear instructions and labels, Eliminating unnecessary words
10	Accessibility and Inclusive UX, Why accessibility matters, Inclusive design principles, Designing for diverse users
11	Intro to Usability Testing, Why test? What to test? When to test?, Types of usability tests (formative, summative), Recruiting test users
12	Conducting a Usability Test, Thinking aloud protocol, Observing and recording behaviors, Common testing mistakes
13	Interpreting Test Results, Making sense of findings, Fixing the obvious first, Prioritizing usability issues
14	Responsive and Mobile Design, Mobile-first thinking, Consistency across devices, Adapting UX to different screen sizes
15	Collaborative Design and Feedback, Handling design criticism, UX team roles and communication, Presenting design work
16	UX Mindset, Applying what you've learned, UX is never finished: The mindset of iteration

SE376 - Introduction to User-Centered Design

Credit Hours: 3(3+0) **Prerequisites:** None
Course Category: Software Engineering Elective

Description

This course introduces students to the principles and practices of User-Centered Design (UCD), emphasizing the importance of designing interactive systems that prioritize the needs, behaviors, and limitations of real users.

Textbook(s)

- The Design of Everyday Things, Revised and Expanded Edition, 2013, ISBN-13: 978-0465050659
- Fundamentals of User-Centered Design: A Practical Approach, Brian Still & Kate Crane, 2024, ISBN: 978-1315200927
- Don't Make Me Think, Revisited: A Common Sense Approach to Web Usability, 3rd Edition, 2014, ISBN-13: 978-0321965516
- About Face: The Essentials of Interaction Design, 4th Edition, 2014, ISBN-13: 978-1118766576
- Universal Principles of Design, 2nd Edition, 2010, ISBN-13: 978-1592535873

Other supplemental materials

- In-depth, industry-recognized, academic approach to UCD: www.interaction-design.org
- Gold standard in UX research and UCD principles: www.nngroup.com

Course Learning Objectives

#	Course Learning Outcomes (CLOs)	BT/ PLO #
CLO 1	Define key principles of User-Centered Design (UCD) and explain their importance in product design.	C1 1,2, 10
CLO 2	Explain the importance and challenges in UCD	C2 1,2 10
CLO 3	Analyze and evaluate digital products based on UCD principles, identifying usability issues.	C4 2,3, 5,9
CLO 4	Demonstrate the ability to conduct basic user research and usability testing.	C3 3, 8
CLO 5	Propose design solutions that integrate UCD principles and iterate based on user feedback	C5 4, 6, 8

Week wise Lecture Schedule

W #	Lecture Topics
1	Introduction to design and usability, the importance of human-centered design, understanding good vs. bad design

W #	Lecture Topics
2	Principles of discoverability, the role of feedback in interaction design, making systems understandable
3	Affordances vs. perceived affordances, the role of signifiers, examples from everyday objects
4	Conceptual models, mental models, natural mapping, visibility of system state
5	The Seven Stages of Action, goal formation to execution and evaluation, memory and attention in user behavior
6	Gulf of Execution and Evaluation, Bridging the gaps between what users want to do and what the system allows, simplifying interactions
7	Knowledge in the Head vs. Knowledge in the World, Design implications of internal vs. external knowledge, minimizing memory burden through better design
8	Constraints in Design, Types of constraints—physical, cultural, semantic, and logical—and how they guide users intuitively
9	Errors in Human Interaction, Types of user errors—slips vs. mistakes, understanding the causes of error
10	Designing for Error, Design strategies to prevent, detect, and recover from errors, the use of forcing functions
11	Simplifying Complexity, Making complex systems simple through design, layering, progressive disclosure
12	Design Thinking and Iteration, Introduction to design thinking, importance of iterative design, user-centered iteration cycles
13	Feedback Loops and User Testing, Designing feedback mechanisms, role of observation and testing in improving usability
14	System Image and User Expectations, Aligning user expectations with system behavior, importance of consistency in interface design
15	Real-World Case Studies, Analyzing everyday objects from a UCD perspective, breakdown of what works and what doesn't
16	Ethics and the Evolution of Design, Ethical implications of design, the evolution of products and systems, designing responsibly for the future

SE471 - Software Verification and Validation

Credit Hours: 3(3+0) **Prerequisites:** CC251
Course Category: Software Engineering Elective

Description

An in-depth study of verification and validation strategies and techniques as they apply to the development of quality software. Topics include test planning and management, testing tools, technical reviews, formal methods and the economics of software testing.

Textbook(s)

- Fisher, Marcus S .Software Verification and Validation, An Engineering and Scientific Approach, (Latest Edition). ISBN 978-0-387-47939-2

Other supplemental materials

- Gerard Blokdyk. Software verification and validation: A Project-Based Tutorial Paperback – (2017), ISBN-10: 1979885079, ISBN-13: 978-1979885072

Course Learning Objectives

#	Course Learning Outcomes (CLOs)	BT/ PLO #
CLO 1	Demonstrate an understanding of the principles and practices of verification and validation in systems engineering and software engineering	C2 1, 10
CLO 2	Apply modeling languages, such as UML, for systems engineering and software engineering design	C3 2,3
CLO 3	Implement verification and validation techniques for different testing levels and domains, including system testing, real-time testing, and user-interface testing	C3 4,5, 6

Week wise Lecture Schedule

W #	Lecture Topics
1	Introduction to Verification and Validation, Systems Engineering, Systems Engineering Standards
2	Systems Engineering Modeling Languages, UML History
3	UML Diagrams, class diagram, component diagram, Composite structure diagram
4	Deployment diagram, object diagram, package diagram, Use case diagram, state machine diagram, sequence diagram,
5	Verification, Validation, and Accreditation, V&V Techniques Overview, Inspection, Testing
6	Reference Model Equivalence Checking, Verification Techniques for Object-Oriented Design

W #	Lecture Topics
7	Design Perspectives, Software Engineering Techniques, Formal Verification Techniques, Program Analysis Techniques
8	V & V of system engineering design models, Verification and Validation of UML Behavioral Diagram, Configuration Transition System
9	Model Checking of Configuration Transition Systems, V&V of UML State Machine Diagram
10	Introduction to Verification and Validation, Systems Engineering, Systems Engineering Standards
11	V&V of UML Sequence Diagram, V&V of UML Activity Diagram
12	Semantic Model Derivation, Model-Based Testing, Model-Based Test Development, Black-Box Testing Approaches
13	White-Box Testing Approaches, Automatic Test Generation, Component and Integration Level Testing, Integration Strategies and Integration Level Testing
14	Test Configuration, Test Components, and Emulators, System and Acceptance Level Testing, Unit Test Plan (UTP) and System Level Testing, Data-Driven Testing, UTP and Data-Driven Testing
15	Value Specification, Usability testing, Path Testing, Parameterization of Tests and Data Pools, Encoding and Decoding of Data, Real-Time and Performance Testing, Real-Time Testing Concerns.
16	UTP and Real-Time Testing, Hard Real-Time Concerns Soft Real-Time Concerns, Performance Testing Concerns UTP and Performance Testing, User-Interface Testing, Issues in User-Interface Testing Planning UI Test Activities, User Interface Context, Logical Aspects, Cyclomatic Complexity and Path Testing

SE472 - Software Engineering Economics

Credit Hours: 3(3+0) **Prerequisites:** None
Course Category: Software Engineering Elective

Description

Explore the economics of software engineering in this comprehensive course. Learn to estimate software development effort, analyze cost drivers, optimize performance models, and cope with uncertainties in decision-making. Delve into Detailed COCOMO for accurate estimation and understand how to handle unquantifiable goals effectively. Gain valuable insights into managing software projects economically and making informed decisions based on subjective probabilities and utility functions.

Textbook(s)

- Capers Jones. Estimating Software Costs: Bringing Realism to Estimation, McGrawHill Osborne Media; 2nd Edition (2019). ISBN-10: 0071483004.

Other supplemental materials

- Shari Lawrence Pfleeger. Software Cost Estimation and Sizing Methods, Issues, and Guidelines, Rand Publishing (2005). ISBN-10: 0833037137.

Course Learning Objectives

#	Course Learning Outcomes (CLOs)	BT/ PLO #
1	CLO Analyze software development effort estimation techniques and apply them effectively.	C4 1,2,10
2	CLO Evaluate cost drivers and performance models to optimize software projects and make informed decisions.	C6 2,3,4
3	CLO Manage uncertainties and unquantifiable goals in software engineering using appropriate strategies.	C3 2,4,5, 10
4	CLO Understand the implications of software engineering decisions, considering subjective probabilities and utility functions.	C2 2,10

Week wise Lecture Schedule

W #	Lecture Topics
1	Introduction to Software Engineering Economics: Importance of cost estimation and effort/maintenance estimation.
2	Model-Product Level Estimates: Interpolation, extrapolation, and adapting existing software.
3	Intermediate COCOMO Effort Equations: Overview and application in effort estimation.
4	Component Level Estimation: CLEF methodology and using it with adapted components.

W #	Lecture Topics
5	Performance & Cost Effectiveness Models: Optimizing performance and analyzing cost-benefit.
6	Economics of Scale: Discrete production functions and (dis)economies of scale.
7	Unquantified Goals: In-house vs vendor development, handling non-measurable criteria.
8	Uncertainties in Decision-Making: Subjective probabilities, utility functions, decision rules.
9	Detailed COCOMO: Overview, software hierarchy estimating forms and procedures.
10	Detailed COCOMO Examples: Step-by-step estimation and schedule adjustment.
11	Cost Drivers – Product Attributes: Reliability, DB size, complexity.
12	Cost Drivers – Computer Attributes: Execution time, storage, VM volatility, turnaround time.
13	Cost Drivers – Personnel Attributes: ACAP, AEXP, PCAP, VEXP, LEXP.
14	Agile Cost Estimation: Effort, duration, and cost in Agile software projects
15	SE Implications I: Value of information and subjective probability in SE decisions.
16	SE Implications II: Utility functions, applying decision rules under uncertainty.

SE473 - Software Re-Engineering

Credit Hours: 3(3+0) **Prerequisites:** None
Course Category: Software Engineering Elective

Description

This course explains the 'state-of-the-art' reengineering existing software systems. This includes an introduction to the recent research, as well as an overview of the principles techniques and skills applied in practice today. Students will acquire a range of principles, techniques and skills that are currently being used for reengineering existing software systems. Consequently, the course has a practical ring to it with a minimal theoretical content (taught as reengineering patterns), several lab-sessions (experimenting with a suite of reengineering tools) and one project (restructuring an existing large software system).

Textbook(s)

- Serge Demeyer, Stéphane Ducasse, Oscar Marius Nierstrasz. Object Oriented Reengineering Patterns (2016), ISBN: 9781558606395,
- Chris Birchall. Re-Engineering Legacy Software, Manning (2016), ISBN: 978-1617292507

Other supplemental materials

- Kshirasagar Naik and Priyadarshi Tripathy. Software Evolution and Maintenance, Wiley, (2016) ISBN: 978-0470603413

Course Learning Objectives

#	Course Learning Outcomes (CLOs)	BT/ PLO #
CLO 1	Demonstrate an understanding of software reengineering concepts and techniques.	C2
CLO 2	Utilize reengineering techniques to manage and adapt software systems effectively.	C3 2,3,4
CLO 3	Evaluate and comprehend maintenance-related challenges in object-oriented software systems.	C6 3,4, 10
CLO 4	Proficiently tackle complex design reengineering and reverse engineering tasks.	C3 2,5,6 ,10

Week wise Lecture Schedule

W #	Lecture Topics
1	Introduction to Software Re-Engineering
2	Software Evolution
3	Legacy Systems
4	RE-Engineering Techniques overview (reverse engineering, restructuring & forward propagation)
5	Reverse Engineering & its techniques
6	Refactoring code to analysis artifacts

W #	Lecture Topics
7	Refactoring code to architecture
8	Object Oriented Re-engineering Patterns
9	Program Comprehension
10	Code restructuring
11	Quality issues in re-engineering processes
12	Tool support for Re-engineering, Challenges & Stakeholder aspiration
13	Software maintenance and re-engineering economics
14	Software Documentation and Knowledge Management in Re-engineering
15	Model-Driven Re-engineering
16	Case studies

AI311 - Programming for AI

Credit Hours: 3(2+1) **Prerequisites:** CC102

Course Category: Artificial Intelligence Domain

Description

This course aims to introduce standard programming practices and to help develop programming skills necessary for designing and implementing Artificial Intelligence systems. The course introduces classical as well as modern state of the art programming language for Artificial Intelligence (Lisp, Prolog, Python, and R), and builds up the necessary programming background for the main courses like Machine Learning, Artificial Neural Networks & Deep Learning, Natural Language Processing, and Speech Processing. This course will help the students of Artificial Intelligence develop the programming acumen and style. The ultimate aim of this course is to help students in using the AI programming languages to solve problems of interest to them.

Textbook(s)

- Miller, B.N., Ranum, D.L. and Anderson, J., 2019.
"Python programming in context." Jones & Bartlett Pub.

Other supplemental materials

- Joshi, P., 2017. "Artificial intelligence with python."
Packt Publishing Ltd.
- Russell, S. and Norvig, P. "Artificial Intelligence. A Modern Approach", 3rd ed, Prentice Hall, Inc., 2015.

Course Learning Objectives

#	Course Learning Outcomes (CLOs)	BT/ PLO #
CLO 1	Understand the fundamental constructs of Lisp, Prolog, and Python programming languages.	C2
CLO 2	Comprehend the fundamental constructs of programming languages for data analysis and representation.	C2
CLO 3	Understand and apply the Object-oriented concepts in the programming languages.	C2
CLO 4	Apply various libraries for plotting, interpreting and analyzing data in Python.	C3

Week wise Lecture Schedule

W #	Lecture Topics
1	Introduction to Programming for AI, AI applications, Python setup with Anaconda, Jupyter Notebooks, Git basics.
2	Python essentials recap, Control structures, Functions and modules, Clean code practices, AI-focused examples

W #	Lecture Topics
3	Data structures for AI, Lists and dictionaries, List comprehensions, Nested structures, Dataset parsing
4	NumPy for array manipulation, Indexing and slicing, Broadcasting, Performance tuning, Image-as-array examples
5	Data visualization with Matplotlib, Seaborn basics, Histograms and scatter plots, Heatmaps, Feature visualization
6	Data handling with Pandas, Reading/writing files, DataFrame operations, Cleaning and transforming data
7	Object-Oriented Programming for AI, Classes and objects, Inheritance and polymorphism, Modeling ML components
8	Working with files and JSON, Reading/writing formats, Accessing APIs, Parsing AI-related data
9	Linear algebra for AI, Vectors and matrices, Dot product and norms, NumPy-based matrix operations
10	Intro to Scikit-learn, ML workflows, Data preprocessing, Pipelines, Hands-on with simple ML tasks
11	PyTorch/TensorFlow basics, Tensors and gradients, Building basic networks, Linear regression implementation
12	Image processing with OpenCV and PIL, Reading and transforming images, Preprocessing for vision tasks
13	Data pipelines for AI, Custom dataset creation, Batch loading, Image augmentation techniques
14	Handling large datasets, Memory-efficient loading, Profiling code, Optimizing data pipelines
15	Case Study/Project/Demo
16	Presentation/Peer Review/Q&A Sessions/Demo

AI322 - Machine Learning

Credit Hours: 3(2+1) **Prerequisites:** None
Course Category: Artificial Intelligence Domain

Description

Machine learning is one of the fastest growing areas of computer science, with far-reaching applications. The aim of this course is to: a) Present the basic machine learning concepts; b) Present a range of machine learning algorithms along with their strengths and weaknesses; c) Apply machine learning algorithms to solve problems of moderate complexity.

Textbook(s)

- Alpaydin, E. Introduction to Machine Learning. Cambridge, MA: The MIT Press. (2020), ISBN: 978-0262043793
 - Bishop, C. M. Pattern Recognition and Machine Learning. New York, NY: Springer. (2016), ISBN: 978-1493938438

Other supplemental materials

- Kaggle - <https://www.kaggle.com>
 - Google colab
https://colab.research.google.com/?utm_source=scs-index

Course Learning Objectives

#	Course Learning Outcomes (CLOs)	BT/ PLO #
CLO 1	Describe basic machine learning concepts, theories and applications.	C1 1,10
CLO 2	Apply supervised learning techniques to solve classification problems of moderate complexity.	C3 6,7,8, 9,10
CLO 3	Apply unsupervised learning techniques to solve clustering problems of moderate complexity	C3 2,3, 4, 5
CLO 4	Apply reinforcement learning algorithms to environments with complex dynamics.	C3 1,2, 3
CLO 5	Develop a reasonable size project using suitable machine learning technique	C6 1,2, 3,4

Week wise Lecture Schedule

W #	Lecture Topics
1	Introduction to Machine Learning, Types of learning (supervised, unsupervised, semi-supervised, reinforcement), Real-world applications
2	Concept learning, General-to-specific ordering of hypotheses, Hypothesis spaces
3	Version Spaces Algorithm, Candidate Elimination Algorithm, Inductive bias

W #	Lecture Topics
4	Introduction to Supervised Learning, Decision trees, ID3 algorithm, Splitting criteria
5	Naive Bayes classifier, Assumptions, Training and prediction, Use cases
6	Artificial Neural Networks, Perceptron learning, Activation functions
7	Support Vector Machines (SVM), Linear and non-linear classification, Kernels
8	Overfitting and underfitting, Dealing with noisy data, Pruning techniques
9	Measuring classifier performance, Confusion matrix, Precision, Recall, F1-score, ROC curve
10	Linear Regression, Least squares method, Cost functions
11	Logistic Regression, Sigmoid function, Gradient descent for optimization
12	Random Forest,
13	Ada-boost, XG Boost,
14	Unsupervised Learning, Hierarchical Agglomerative Clustering, Dendrograms
15	Case Study/Project/Demo
16	Presentation/Peer Review/Q&A Sessions/Demo

AI331 - Artificial Neural Networks & Deep Learning

Credit Hours: 3(2+1) **Prerequisites:** None
Course Category: Artificial Intelligence Domain

Description

This course will introduce Artificial Neural Networks and Deep Learning. ANN's basic architecture and how they mimic the human brain using simple mathematical models. Many of the important concepts and techniques around brain computing and the major types of ANN will also be introduced. Emphasis is made on the mathematical models, understanding learning laws, selecting activation functions and how to train the networks to solve classification problems. Deep neural networks have achieved state of the art performance on several computer vision and speech recognition benchmarks. This course will further build on the fundamentals of Neural networks and artificial intelligence and will introduce advanced topics in neural networks, convolutional and recurrent network structures, deep unsupervised and reinforcement learning.

Textbook(s)

- Christopher M. Bishop, Hugh Bishop, Deep Learning: Foundations and Concepts, 2024, ISBN: 978-3031454677.
 - Neural Network Design, 2nd Edition, Martin T. Hagan, Howard, B. Demuth, Mark Hudson Beale and Orlando De Jesus, Publisher: Martin Hagan; 2 edition (September 1, 2014), ISBN-10: 0971732116

Other supplemental materials

An Introduction to Neural Networks, James A Anderson, Publisher: A Bradford Book (March 16, 1995), ISBN-10: 0262011441

Course Learning Objectives

#	Course Learning Outcomes (CLOs)	BT/ PLO #
CLO 1	Understand the fundamentals of neural networks in AI	C2
CLO 2	Explain how simple ANNs can be designed	C2
CLO 3	Apply ANN for classification Problems	C3
CLO 3	Apply deep learning algorithms to real-world problems	C3
CLO 4	Analyze results from deep learning to select appropriate solutions	C4

Week wise Lecture Schedule

W #	Lecture Topics
1	Introduction and history of neural networks, Applications in AI
2	Basic architecture of neural networks, Perceptron, Adaline, Minimum Error Learning
3	Machine learning theory, Training vs. test sets, Evaluation metrics
4	Gradient descent (Delta) rule, Hebbian learning, Neo-Hebbian learning
5	Differential Hebbian Learning, Drive Reinforcement Theory
6	Kohonen Self-Organizing Maps, Associative memory, Bi-directional associative memory (BAM)
7	Energy surfaces, Boltzmann machines, Theory of generalization
8	Backpropagation algorithm, Multi-layer perceptrons
9	Feedforward networks, Deep convolutional networks
10	Computational complexity of neural networks, Optimization techniques
11	Autoencoders, Sparse coding, Unsupervised deep learning
12	Deep belief networks, Restricted Boltzmann Machines
13	Deep recurrent neural networks, BPTT, LSTM, GRU
14	GPU programming for deep learning, CuDNN basics
15	Case Study/Project/Demo
16	Presentation/Peer Review/Q&A Sessions/Demo

AI351 - Knowledge Representation & Reasoning

Credit Hours: 3(2+1) **Prerequisites:** None
Course Category: Artificial Intelligence Domain

Description

Knowledge representation is one of the fundamental areas of Artificial Intelligence. It is the study of how knowledge about the world can be represented and manipulated in an automated way to enable agents to make intelligent decisions. This course will provide an overview of existing knowledge representation frameworks developed within AI including but not limited to propositional and first-order logic, ontologies, planning, reasoning and decision making under uncertainty. The assignments component of the course would provide hands-on experience of software like Prolog, Protégé, probabilistic reasoning APIs and tools to support complex decision making. It is expected that after completing this course, students will understand (a) the foundations of Knowledge Representation & Reasoning and (b) which tools and techniques are appropriate for which tasks.

Textbook(s)

- Knowledge Graph Reasoning: A Neuro-Symbolic Perspective, Kewei Cheng & Yizhou Sun (Dec 2024, Springer), ISBN: 978-3031720079.
- Friedrich Bobach, Thomas Meyer, and Thomas Schaub, Knowledge Representation and Reasoning, Springer, 2020.

Other supplemental materials

- Ronald Brachman and Hector Levesque, Knowledge Representation and Reasoning, Morgan Kaufmann, 2004.

Course Learning Objectives

#	Course Learning Outcomes (CLOs)	BT/ PLO #
CLO 1	Understand the fundamentals of knowledge representation and reasoning in deterministic situations	C2
CLO 2	Understand the challenges in representing knowledge and reasoning under uncertainty	C2
CLO 3	Analyze different situations and apply appropriate knowledge representation frameworks	C4
CLO 4	Development of hybrid approaches by synergizing the existing framework to solve complex decision-making problem	C4

Week wise Lecture Schedule

W #	Lecture Topics
1	Propositional Logic, Syntax and Semantics, Truth Tables
2	First-order Logic, Quantifiers, Logical Equivalences

W #	Lecture Topics
3	Horn Clauses, Resolution in Propositional and FOL
4	Description Logic, Concepts and Roles, Basic Syntax
5	Reasoning using Description Logic, Concept Satisfiability
6	Forward Chaining, Backward Chaining, Rule-based Inference
7	Inference Engines, RETE Algorithm, Conflict Resolution
8	Semantic Networks, Concept Hierarchies, Frame-Based Systems
9	Ontologies, OWL, Ontology Languages, RDF
10	Logical Agents, Knowledge-Based Agents, Perception and Action
11	Planning, STRIPS, Partial Order Planning
12	Rule-Based Knowledge Representation, Production Rules
13	Reasoning Under Uncertainty, Probabilistic Logic
14	Bayesian Networks, Representation, Conditional Independence
15	Case Study/Project/Demo
16	Presentation/Peer Review/Q&A Sessions/Demo

AI441 - Computer Vision

Credit Hours: 3(2+1) **Prerequisites:** None
Course Category: Artificial Intelligence Domain

Description

With a single glance a human interprets the entire scene. How many objects are present in the scene and where they are located. Which person is present in the scene. What will happen next. However, computers lack this capability. We have seen only face detectors so far working in our mobile phones. What is the challenge in understanding the 3D scene, i.e., the identity, the location and the size of the objects present in the scene. In this course we will introduce the basic concepts related to 3D scene modelling from single view and multiple views.

Textbook(s)

Computer Vision: Algorithms and Applications (Texts in Computer Science) Second Edition 2022 by Richard Szeliski, ISBN: 978-3030343712.

Other supplemental materials

- Multiple View Geometry in Computer Vision, by Richard Hartley and Andrew Zisserman.
- Computer Vision: A Modern Approach, by David Forsyth and Jean Ponce.
- Digital Image Processing, by Rafael Gonzalez and Richard Woods.

Course Learning Objectives

#	Course Learning Outcomes (CLOs)	BT/ PLO #
CLO 1	Understanding the single view geometry concepts	C2
CLO 2	Understanding the multiple view geometry concepts	C2
CLO 3	Apply concepts of CV for solving real world problems	C3

Week wise Lecture Schedule

W #	Lecture Topics
1	Introduction to Computer Vision, Problems in CV, History and Modern Advancements
2	Fundamentals of Image Processing, Pixel operations, Color models
3	Image Filtering, Convolution, Gaussian and Median Filters
4	Image Pyramids, Fourier Transform in Image Processing
5	Hough Transform, Line and Circle Detection
6	Camera Models, Intrinsic and Extrinsic Parameters

W #	Lecture Topics
7	Camera Looking at a Plane, Horizon Line, Rotation about Camera Center
8	Transformation Estimation, Point Correspondences, Homographies
9	2D and 3D Transformations, Rotation Matrices, Euler Angles
10	Edge Detection, Canny, Sobel Operators
11	Corner Detection, Harris Corner Detector, Shi-Tomasi Method
12	Feature Descriptors, HoG, SIFT, SURF
13	Deep Learning for CV: Object Detection (RCNN, YOLO, RetinaNet)
14	Deep Learning for CV: Segmentation (UNet, MaskRCNN), Image Generation (GANs)
15	Case Study/Project/Demo
16	Presentation/Peer Review/Q&A Sessions/Demo

AIE371- Mathematics for Data Science & AI

Credit Hours: 3(3+0) **Prerequisites:** QR2

Course Category: AI Domain Elective

Description

This course is designed to provide a comprehensive understanding of mathematics and statistics for data science and artificial intelligence applications. Students will acquire the foundational knowledge needed to make informed decisions based on data, create mathematical and statistical models, and apply concepts to machine learning algorithms.

Textbook(s)

- Essential Math for Data Science by Thomas Nield, (2022), ISBN: ISBN: 9781098102937.

Other supplemental materials

- Probability and Statistics for Data Science: Math + R + Data (Chapman & Hall/CRC Data Science Series) 1st Edition by Norman Matloff. (2020), ISBN: 978-1138393295

Course Learning Objectives

#	Course Learning Outcomes (CLOs)	BT/ PLO #
CLO 1	Apply fundamental mathematical concepts, including calculus, probability, and statistics, to analyze and interpret data effectively.	C1 1,2,3
CLO 2	Utilize linear algebra techniques to solve problems related to data manipulation and transformation.	C2 2,4,5
CLO 3	Implement linear regression models for predictive analysis and understand the principles of model evaluation and validation.	C3 3,4,5
CLO 4	Apply logistic regression and classification algorithms to solve classification problems and evaluate model performance.	C2 1,3
CLO 5	Demonstrate proficiency in mathematical techniques essential for data science and artificial intelligence applications.	C4 2,3, 10
CLO 6	Critically analyze and interpret statistical results, considering practical implications and limitations in data-driven decision-making processes.	C3 3,4,10

Week wise Lecture Schedule

W #	Lecture Topics
1	Basic Math and Calculus Review: Number Theory, Summations, Exponents, Logarithms, Euler's Number and Natural Logarithms,

W #	Lecture Topics
2	Limits, Derivatives, Partial Derivatives, The Chain Rule, Integrals
3	Probability: Recall Probability, Probability Versus Statistics, Probability Math, Joint Probabilities, Union Probabilities, ,
4	Conditional Probability and Bayes' Theorem, Binomial Distribution, Beta Distribution
5	Descriptive and Inferential Statistics: What Is Data? Descriptive Versus Inferential Statistics, Populations, Samples, and Bias, Descriptive Statistics,
6	Mean and Weighted Mean, Median, Mode, Variance and Standard Deviation, The Normal Distribution, Inferential Statistics,
7	The Central Limit Theorem, Confidence Intervals, Understanding P-Values, Hypothesis Testing, Big Data Considerations and the Texas Sharpshooter Fallacy
8	Linear Algebra: What Is a Vector? Adding and Combining Vectors, Scaling Vectors, Span and Linear Dependence, Linear Transformations
9	Basis Vectors, Matrix Vector Multiplication, Matrix Multiplication, Determinants, Special Types of Matrices
10	Systems of Equations and Inverse Matrices, Eigenvectors and Eigenvalues
11	Linear Regression: A Basic Linear Regression, Residuals and Squared Errors, Finding the Best Fit Line, Closed Form Equation, Gradient Descent, Overfitting and Variance,
12	Stochastic Gradient Descent, The Correlation Coefficient, Statistical Significance, Coefficient of Determination, Standard Error of the Estimate,
13	Prediction Intervals, Train/Test Splits, Multiple Linear Regression
14	Logistic Regression and Classification: Understanding Logistic Regression, Performing a Logistic Regression, Logistic Function, Fitting the Logistic Curve,
15	Multivariable Logistic Regression, Understanding the Log-Odds, R-Squared, P-Values, Train/Test Splits, Confusion Matrices,
16	Bayes' Theorem and Classification, Receiver Operator, Characteristics/Area Under Curve, Class Imbalance.

AI372 - Data Science

Credit Hours: 3(3+0) **Prerequisites:** None
Course Category: Artificial Intelligence Elective

Description

Data science is a rapidly growing field that is transforming the way we live and work. Data scientists use their skills to collect, analyze, and interpret data to solve real-world problems. Data science is an interdisciplinary field that draws on knowledge from a variety of areas, including statistics, computer science, mathematics, and business. Data scientists need to be able to collect, clean, and analyze data, as well as communicate their findings to stakeholders. The demand for data scientists is high, and the field is expected to grow in the coming years. If you are interested in a career in data science, there are many online and in-person courses available to help you get started. Here are some of the key concepts in data science: Data collection: The process of gathering data from a variety of sources, such as surveys, social media, and sensors. Data cleaning: The process of removing errors and inconsistencies from data. Data analysis: The process of using statistical and machine learning techniques to extract insights from data. Data visualization: The process of communicating the results of data analysis using charts, graphs, and other visual representations. Communication and presentation: The ability to communicate your findings to a variety of audiences.

Textbook(s)

- Joel Grus. Data Science from Scratch: First Principles with Python, 2nd Edition, O'Reilly Media, (2019), ISBN: 978-1492041139
 - Wes McKinney. Python for Data Analysis, 3rd Edition, O'Reilly Media, (2022) ISBN: 978-1098104030

Other supplemental materials

- Trevor Hastie, Robert Tibshirani, and Jerome Friedman. The Elements of Statistical Learning, 2nd Edition, Springer, (2016), ISBN: 978-0387848570

Course Learning Objectives

#	Course Learning Outcomes (CLOs)	BT/ PLO #
CLO 1	Understand the basic concepts of data science, statistics, and probability and their application in understanding behavior of data.	CS 1,2,3,4
CLO 2	Apply basic tools for performing exploratory data analysis and visualization.	C3 1,2,3,4
CLO 3	Understand basic predictive modeling and data analysis methods.	CS 1,3

Learn Python for performing different data science steps.

CLO 4	Learn Python for performing different data science steps.	C3 2,4,5,6
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Week wise Lecture Schedule

W #	Lecture Topics
1	Introduction to data science
2	Data preprocessing
3	Statistical inferences and probability
4	Machine learning
5	Big data
6	Data visualization
7	Ethics in data science
8	Data distribution analysis
9	Correlation and covariance analysis
10	Feature selection and dimensionality reduction
11	Supervised learning (classification, regression)
12	Unsupervised learning (clustering, dimensionality reduction)
13	Model evaluation and validation techniques
14	Time series analysis and forecasting techniques
15	Case Study/Project/Demo
16	Presentation/Peer Review/Q&A Sessions/Demo

AI373 - Data Mining

Credit Hours: 3(3+0) **Prerequisites:** CC221
Course Category: Artificial Intelligence Elective

Description

Data Mining combines AI, statistics, and databases to uncover hidden patterns in large datasets. This course aims to develop students' ability to analyze and extract knowledge from data. It covers data preprocessing, summary statistics, frequent itemset generation, association and correlation measures, and core algorithms in classification, prediction, and clustering. Students will also apply programming and problem-solving skills to develop data mining applications.

Textbook(s)

- Data Mining: Concepts and Techniques (The Morgan Kaufmann Series in Data Management Systems) 4th Edition by Jiawei Han, Jian Pei, Hanghang Tong, 2022, ISBN: 978-0128117606.
- Jiawei Han & Micheline Kamber, Jian Pei (2011). Data Mining: Concepts and Techniques, 3rd Edition.
- Pang-Ning Tan, Michael Steinbach, and Vipin Kumar (2005). Introduction to Data Mining.

Other supplemental materials

- Charu C. Aggarwal (2015). Data Mining: The Textbook
- D. Hand, H. Mannila, P. Smyth (2001). Principles of Data Mining. MIT Press.

Course Learning Objectives

#	Course Learning Outcomes (CLOs)	BT/ PLO #
CLO 1	Apply preprocessing techniques on any given raw data.	C3 1,3,10
CLO 2	Select and apply proper data mining algorithm to discover interesting patterns	C3 1,3,4
CLO 3	Analyze and extract patterns to solve problems and point out how to deploy solution	C4 1,3, 4,5
CLO 4	Evaluate systematically supervised, semi supervised and unsupervised models and algorithms with respect to their accuracy	C4 3,4,5, 10

Week wise Lecture Schedule

W #	Lecture Topics
1	Introduction to Data Mining, Basic Concepts, and Applications
2	Data Preprocessing Techniques and Summary Statistics
3	Association Rule Mining: Apriori Algorithm

W #	Lecture Topics
4	Association Rule Mining: Frequent Pattern Trees (FP-Growth)
5	Introduction to Classification and Types
6	Supervised Classification: Decision Trees
7	Supervised Classification: Naïve Bayes and K-Nearest Neighbors
8	Supervised Classification: Support Vector Machines (SVM)
9	Introduction to Unsupervised Classification: Concepts and Use Cases
10	Unsupervised Classification: K-Means, K-Median
11	Unsupervised Classification: Hierarchical & Divisive Clustering
12	Unsupervised Classification: Kohonen Self-Organizing Maps
13	Outlier and Anomaly Detection Techniques
14	Web and Social Network Mining
15	Case Study/Project/Demo
16	Presentation/Peer Review/Q&A Sessions/Demo

AI374 - Natural Language Processing

Credit Hours: 3(3+0) **Prerequisites:** None
Course Category: Artificial Intelligence Elective

Description

Natural Language Processing (NLP) is the application of computational techniques to the analysis and synthesis of natural language and speech. This course is an introduction to NLP with prior programming experience in Python.

Textbook(s)

- Natural Language Processing in Action, Second Edition 2nd ed. Edition, by Hobson Lane, Maria Dyshel, , 2025, ISBN: 978-1617299445.
- Daniel Jurafsky and James H. Martin. 2018. Speech and Language Processing: An Introduction to Natural Language Processing. Third Edition. Prentice Hall

Other supplemental materials

Foundations of Statistical Natural Language Processing, Manning and Schütze, MIT Press. Cambridge, MA: May 1999

Course Learning Objectives

#	Course Learning Outcomes (CLOs)	BT/ PLO #
CLO 1	Understand techniques for information retrieval, language translation, and text classification.	C2
CLO 2	Understand the advantages of using standard corpora. Identify examples of current corpora for a variety of NLP tasks.	C2
CLO 3	Understand and contrast deterministic and stochastic grammars, providing examples to show the adequacy of each	C2
CLO 4	Solve classic and stochastic algorithms for parsing natural language.	C3

Week wise Lecture Schedule

W #	Lecture Topics
1	Introduction & History of NLP, NLP applications, Challenges in NLP
2	Parsing algorithms, Constituency and dependency parsing
3	Basic Text Processing, Tokenization, Normalization, Lemmatization, Stopword removal
4	Minimum Edit Distance, Levenshtein Distance, Applications in Spell Checking
5	Language Modeling, N-grams, Smoothing, Backoff techniques
6	Spelling Correction, Noisy Channel Model, Edit probabilities

W #	Lecture Topics
7	Text Classification, Naive Bayes, Logistic Regression for text
8	Deterministic and Stochastic Grammars, Context-Free Grammars (CFGs)
9	Representing Meaning, Semantic roles, Word Sense Disambiguation
10	Semantics and Vector Models, Word2Vec, GloVe, Embeddings
11	Sentiment Analysis, Polarity detection, Lexicon-based and ML approaches
12	Temporal Representations, Tense, Aspect, Event ordering
13	Corpus-based Methods, POS Tagging, Morphology
14	Information Retrieval, Vector Space Model, Precision and Recall
15	Case Study/Project/Demo
16	Presentation/Peer Review/Q&A Sessions/Demo

AI375 - Business Intelligence and Decision Support Systems

Credit Hours: 3(3+0) **Prerequisites:** None
Course Category: Artificial Intelligence Elective

Description

In the fast-paced business environment, effective decision-making is crucial for organizational success. This course delves into the realm of Decision Support Systems (DSS) and Business Intelligence (BI), providing insights into the tools and methodologies that empower informed decision-making. Participants will explore the key components of DSS, understand the nature of decision-making, and learn how data, models, and intelligence contribute to the strategic decision support process.

Textbook(s)

Decision Support Systems for Business Intelligence, 2nd Edition by Vicki L. Sauter, (2019), ISBN: 978-0-470-43374-4.

Other supplemental materials

Business Intelligence and Analytics: Systems for Decision Support 10th Edition by Ramesh Sharda, Dursun Delen, Efraim Turban, (2013), ISBN: 978-0133050905.

Course Learning Objectives

#	Course Learning Outcomes (CLOs)	BT/ PLO #
CLO 1	Analyze the role and uses of Decision Support Systems (DSS) in organizations, emphasizing the impact of Business Intelligence on decision-making processes.	C4 1,2,3
CLO 2	Examine different decision-making approaches, including rational decision-making, bounded rationality, muddling through, and the influence of bias in decision-making.	C5 3,4, 5,10
CLO 3	Evaluate the data component of DSS, considering characteristics such as timeliness, sufficiency, and level of detail. Explore database management systems, data warehouses, and data scrubbing techniques to ensure data relevance and cost efficiency.	C5 3,4,5
CLO 4	Explore the model component of DSS, focusing on models, analytics, and representation options. Investigate data mining, intelligent agents, and sensitivity analysis to enhance decision models' accessibility and understandability.	C6 1,2, 5, 10
CLO 5	Understand the intelligence aspect of Decision Support Systems, covering programming reasoning, backward-chaining, forward-chaining reasoning, and handling uncertainty in decision-making. Utilize	C2 1,2,10

probability theory and certainty factors to represent uncertainty.

Week wise Lecture Schedule

W #	Lecture Topics
1	Introduction to Decision Support Systems: Understanding Decision Support Systems (DSS)
2	The Role and Uses of DSS in Organizations, Business Intelligence and its Impact on Decision Making
3	Decision Making: Rational Decision-Making, Bounded Rationality and Muddling Through
4	The Nature of Managers in Decision Processes,
5	Bias in Decision Making, Group Decision Making and its Challenges.
6	Data Component: Characteristics of Information, Timeliness, Sufficiency, and
7	Level of Detail in Data Database Management Systems
8	Data Warehouses and Data Scrubbing, Ensuring Data Relevance and Cost Efficiency.
9	Model Component: Models and Analytics in Decision Support, Options for Models and Representation,
10	Data Mining and Intelligent Agents
11	Easy Access and Understandability of Results
12	Sensitivity Analysis in Decision Models
13	Intelligence and Decision Support Systems: Programming Reasoning in Decision Systems
14	Backward-Chaining and Forward-Chaining Reasoning
15	Case Study/Project/Demo
16	Presentation/Peer Review/Q&A Sessions/Demo

AI376 - Knowledge Based Systems

Credit Hours: 3(3+0) **Prerequisites:** None
Course Category: Artificial Intelligence Elective

Description

This course provides a foundation in Knowledge-Based Systems (KBS), focusing on the principles and techniques of knowledge representation, acquisition, and inference mechanisms used in expert systems. It covers problem-solving strategies under uncertainty, system development methodologies, and automation of knowledge extraction. Students will explore the architecture of knowledge-based systems and study real-world applications from various domains, along with current approaches to building intelligent systems.

Textbook(s)

Giarratano, Joseph and Riley, Gary, Expert Systems: Principles and Programming, Cengage Learning, 5th Edition, 2018.

Other supplemental materials

Stuart Russell, Peter Norvig, Artificial Intelligence: A Modern Approach, Prentice Hall, last edition.

Course Learning Objectives

#	Course Learning Outcomes (CLOs)	BT/ PLO #
CLO 1	Describe the principles of knowledge representation and reasoning in expert systems.	C2 1,3,4
CLO 2	Explain and apply knowledge acquisition and construction techniques for building knowledge bases.	C4 1,2, 3,10
CLO 3	Analyze the architecture and behavior of knowledge-based and expert systems in various domains.	C5 2,3, 5,6
CLO 4	Analyze the architecture and behavior of knowledge-based and expert systems in various domains.	C6 4,5, 6,10

Week wise Lecture Schedule

W #	Lecture Topics
1	Introduction to Knowledge-Based Systems and Expert Systems; History and Evolution
2	Characteristics of Expert Systems; Components: Knowledge Base, Inference Engine, User Interface
3	Problem-Solving Methods in Expert Systems; Rule-Based Systems Overview
4	Knowledge Representation: Semantic Nets, Frames, Production Rules
5	Logic and Inference: Propositional Logic, Predicate Logic, Forward and Backward Chaining
6	Certainty Factors and Uncertainty Handling in Expert Systems

W #	Lecture Topics
7	Fuzzy Logic and Reasoning with Uncertainty
8	Knowledge Acquisition: Manual and Automated Methods, Knowledge Engineering
9	System Development Methodologies: Life Cycle of Expert Systems, Planning and Design
10	Introduction to Expert System Shells; Architecture and Tool Support
11	Case Study: CLIPS-Based System Design; Rule Writing and Control Strategy
12	Advanced Inference Techniques; Explanation and Justification Mechanisms
13	Integration of Knowledge-Based Systems with Databases and Other Applications
14	Expert Systems in Practice: Medical Diagnosis, Engineering, Finance, and Customer Support
15	Case Study/Project/Demo
16	Presentation/Peer Review/Q&A Sessions/Demo

AI377 - Fuzzy Systems

Credit Hours: 3(3+0) **Prerequisites:** QR-2
Course Category: Artificial Intelligence Elective

Description

This course introduces students to the theory and applications of fuzzy sets and systems, which provide a framework for modeling uncertainty, vagueness, and imprecision in data-driven and decision-making processes. It covers both foundational concepts such as type-1 and type-2 fuzzy sets, fuzzy operations, similarity and distance measures, and advanced topics including fuzzy inference systems, fuzzy control, intuitionistic fuzzy sets, and their applications in algebraic structures, graphs, and differential equations.

Textbook(s)

- Fuzzy Logic and Computational Intelligence, edited by Himansu Das & Mahendra Kumar Gourisaria, February 2025, ISBN: 978-3-7258-3319-1
- Chen, G. and Pham, T.: Introduction to fuzzy systems (Chapman and Hall/CRC, 2005).
- Zimmermann, H. -J.: Fuzzy set theory and its application, (4th ed. Springer, 2001).
- Mordeson, J. N. and Malik, D.S.: Fuzzy semigroups (Springer, 2010).

Other supplemental materials

- Zadeh A., Fu, K.-S., Tanaka, K. and Shimura, M.: Fuzzy sets and their applications to cognitive and decision process (Academic Press, New York, 1975).

Course Learning Objectives

#	Course Learning Outcomes (CLOs)	BT/ PLO #
CLO 1	Understand and apply the mathematical foundations of fuzzy set theory and operations.	C1 1,2,3, 10
CLO 2	Analyze and implement fuzzy inference systems and decision-making models.	C5 3,4, 10
CLO 3	Compare type-1 and intuitionistic fuzzy sets using similarity and distance measures.	C6 1,2
CLO 4	Apply fuzzy and intuitionistic fuzzy systems to algebraic structures and practical modeling problems.	C3 2,4,5, 10

Week wise Lecture Schedule

W #	Lecture Topics
1	Introduction to Fuzzy Sets; Crisp vs. Fuzzy Logic; Historical Context (Zadeh, 1975)
2	Type-1 Fuzzy Sets: Membership Functions, Cardinality, Normality

W #	Lecture Topics
3	Fuzzy Set Operations: Union, Intersection, Complement
4	Distance Measures between Fuzzy Sets: Hamming, Euclidean, Normalized Forms
5	Similarity Measures and Applications in Pattern Matching
6	Fuzzy Relations and Composition; Max–Min and Max–Product Compositions
7	Fuzzy Numbers and Fuzzy Functions; Triangular and Trapezoidal Representations
8	Probability vs. Possibility Theory
9	Fuzzy Logic and Linguistic Variables
10	Fuzzy Inference Systems: Mamdani and Sugeno Models
11	Defuzzification Techniques: Centroid, Bisector, Mean of Maximum
12	Fuzzy Control Systems and Fuzzy Expert Systems
13	Intuitionistic Fuzzy Sets (IFSs): Definitions, Membership/Non-membership, Hesitation
14	Distances and Similarity Measures between IFSs; Level Cut Sets; IF Composition
15	Case Study/Project/Demo
16	Presentation/Peer Review/Q&A Sessions/Demo

AI378 - Fundamentals of Large Language Models

Credit Hours: 3(3+0) **Prerequisites:** None
Course Category: Artificial Intelligence Elective

Description

This course provides a beginner-friendly introduction to Natural Language Processing (NLP) and the world of Large Language Models (LLMs). In the first half, students will explore the fundamentals of NLP, including text preprocessing, tokenization, and language understanding. The second half focuses on prompt engineering, language models, how LLMs like ChatGPT work, and how to use them to build real-world AI applications.

Textbook(s)

- Natural Language Processing with Transformers: Building Language Applications with Hugging Face, by Lewis Tunstall, Leandro von Werra, and Thomas Wolf, 2022, ISBN-10: 1098103246

Other supplemental materials

- Tong Xiao, Jingbo Zhu, Foundations of Large Language Models, Submitted to arXiv on January 16, 2025
 - Natural Language Processing with Python: Analyzing Text with the Natural Language Toolkit
 - <https://stanford-cs336.github.io/spring2024/>
 - <https://online.stanford.edu/courses/xcs224n-natural-language-processing-deep-learning>

Course Learning Objectives

#	Course Learning Outcomes (CLOs)	BT/ PLO #
1	CLO Describe the basic concepts of Natural Language Processing (NLP) and its role in AI applications.	C2 1,2,10
2	CLO Apply text preprocessing techniques such as tokenization, stemming, and lemmatization.	C3 1,2,10
3	CLO Analyze word and sentence embeddings to understand semantic meaning in text data.	C4 2,3, 5,9
4	CLO Explain the architecture and functionality of transformers and large language models.	C2 3, 8
5	CLO Demonstrate prompt engineering techniques to solve tasks using pre-trained LLMs.	C5 4,6,8

Week wise Lecture Schedule

W #	Lecture Topics
1	Introduction to NLP: What is Natural Language Processing?, Importance of NLP in AI, Real-life applications of NLP
2	Understanding Text Data: Text as unstructured data, Challenges in language processing,

W #	Lecture Topics
	Linguistic basics: syntax, semantics, and pragmatics
3	Text Preprocessing Fundamentals: Tokenization and normalization, Removing stop words and punctuation, Stemming vs. lemmatization
4	Text Representation Techniques: Bag of Words model, TF-IDF explained, Vector space models for text
5	Word Embeddings and Semantic Meaning: Introduction to Word2Vec and GloVe, Capturing context with embeddings, Visualizing word relationships
6	Language Modeling Basics: N-grams and their limitations, Introduction to language models, Probability of word sequences
7	Neural Networks for NLP: Basics of neural networks in NLP, RNNs and LSTMs for sequence data, Limitations of traditional models
8	Transformers and the Rise of LLMs: Introduction to the Transformer model, Attention and self-attention mechanisms, How transformers changed NLP
9	Large Language Models: What are LLMs? Overview of models like GPT, BERT
10	Prompt Engineering: What is a prompt? Effective prompt design, Zero-shot, few-shot, and chain-of-thought prompting
11	Applications of LLMs: Text generation and summarization, Sentiment analysis and Q&A systems, Chatbots and virtual assistants
12	Hands-On with NLP Tools: Exploring Hugging Face and OpenAI APIs, Working with pretrained models, Using Colab for NLP experiments
13	Applications of LLMs: Chatbots, Copilots, Content generation
14	Applications of LLMs: Cod Translation, Retrieval-Augmented Generation
15	Case Study/Project/Demo
16	Presentation/Peer Review/Q&A Sessions/Demo

AI379 - Digital Image Processing

Credit Hours: 3(2+1) **Prerequisites:** None
Course Category: Artificial Intelligence Elective

Description

This course introduces fundamental concepts, techniques, and applications in digital image processing. Topics include image representation, enhancement, filtering, segmentation, color processing, morphological operations, and compression. Students will learn both theoretical principles and practical implementation using tools such as MATLAB, Python (OpenCV), and scikit-image. The course prepares students for advanced study in computer vision, machine learning, and medical imaging.

Textbook(s)

- Digital Image Processing by Rafael C. Gonzalez and Richard E. Woods (Latest Edition, Pearson)
- Fundamentals of Digital Image Processing by Anil K. Jain (Latest Edition Prentice-Hall)
- Digital Image Processing Using MATLAB by Rafael C. Gonzalez, Richard E. Woods, and Steven L. Eddins

Other supplemental materials

- Practical Python and OpenCV by Adrian Rosebrock (for applied lab sessions)
- Computer Vision: Algorithms and Applications by Richard Szeliski (Springer, Advanced Reading)

Course Learning Objectives

#	Course Learning Outcomes (CLOs)	BT/ PLO #
CLO 1	Understand fundamental concepts in image formation and perception.	C2 1,2,10
CLO 2	Apply spatial and frequency domain techniques for image enhancement and restoration.	C3 1,2,10
CLO 3	Implement image segmentation, morphology, and color processing techniques.	C4 2,3, 5,9
CLO 4	Analyze image compression methods and performance metrics.	C2 3, 8
CLO 5	Develop practical image processing solutions using software tools.	C5 4,6,8

Week wise Lecture Schedule

W #	Lecture Topics
1	Introduction to Digital Image Processing: Applications and Scope
2	Digital Image Fundamentals: Image formation, Sampling, Quantization
3	Image Enhancement in Spatial Domain: Point processing, Histogram Equalization

W #	Lecture Topics
4	Image Enhancement in Frequency Domain: Fourier Transform, Frequency filtering
5	Image Restoration: Noise models, Inverse & Wiener filtering
6	Color Image Processing: RGB, HSI models, Pseudo color techniques
7	Morphological Image Processing: Dilation, Erosion, Opening, Closing
8	Image Segmentation: Thresholding, Edge-based, Region-based segmentation
9	Mid-Term Examination and Course Review
10	Feature Extraction and Description: Texture, Shape, Color descriptors
11	Object Recognition and Classification Techniques
12	Image Compression: Lossless & Lossy methods, JPEG, Huffman coding
13	Image Registration and Fusion Techniques
14	Motion Analysis and Object Tracking
15	Student Project Demonstrations and Peer Review
16	Final Examination and Course Wrap-Up

AI471 - Reinforcement Learning

Credit Hours: 3(3+0) **Prerequisites:** AI322
Course Category: Artificial Intelligence Elective

Description

Reinforcement Learning (RL) is a foundational area in Artificial Intelligence that enables autonomous agents to learn optimal behaviors through interaction with their environment. This course provides a comprehensive introduction to RL concepts and methods, including Markov decision processes, value-based and policy-based learning, model-free algorithms like Q-learning, function approximation, and exploration strategies. Students will also explore cutting-edge developments in deep reinforcement learning and apply their knowledge through coding assignments and a final project.

Textbook(s)

Mathematical Foundations of Reinforcement Learning, Shiyu Zhao, 2025, ISBN: 978-9819739431.

Sutton, R. S., & Barto, A. G. Reinforcement Learning: An Introduction, 2nd Edition. Available free: <http://incompleteideas.net/book/the-book-2nd.html>

Other supplemental materials

Handouts and research articles may also be used by the instructor.

Course Learning Objectives

#	Course Learning Outcomes (CLOs)	BT/ PLO #
CLO 1	Explain the foundational principles of reinforcement learning, including Markov decision processes.	C4
CLO 2	Apply model-free RL methods like Q-learning and SARSA to solve decision-making tasks.	C3
CLO 3	Analyze policy iteration and value iteration techniques for policy optimization.	C6
CLO 4	Implement RL algorithms using function approximation and explore advanced methods like imitation learning.	C5

Week wise Lecture Schedule

W #	Lecture Topics
1	Introduction to Reinforcement Learning; Real-world applications (Robotics, Healthcare, Games)
2	Agent-Environment Interface; Goals and Rewards
3	Markov Processes and Markov Decision Processes (MDPs)
4	Policies, Returns, and Value Functions
5	Dynamic Programming: Policy Evaluation and Policy Iteration

W #	Lecture Topics
6	Value Iteration; Bellman Optimality Equation
7	Monte Carlo Methods for Prediction and Control
8	Temporal Difference Learning: TD(0), TD(λ)
9	Model-Free Control: Q-Learning and SARSA
10	Exploration vs. Exploitation; ϵ -greedy, Softmax Strategies
11	Function Approximation for RL: Linear Models, Tile Coding, and Gradient Descent
12	Policy Gradient Methods and Actor-Critic Algorithms
13	Imitation Learning and Learning from Demonstrations
14	Batch Reinforcement Learning and Off-Policy Learning, Introduction to Deep Reinforcement Learning; Deep Q-Networks (DQN)
15	Case Study/Project/Demo
16	Presentation/Peer Review/Q&A Sessions/Demo

AI472 - Swarm Intelligence

Credit Hours: 3(3+0) **Prerequisites:** CC351
Course Category: Artificial Intelligence Elective

Description

Swarm Intelligence is a field of Artificial Intelligence inspired by the collective behavior of decentralized, self-organizing systems such as ant colonies, bird flocks, and fish schools. This course explores both natural and artificial swarm systems, focusing on the emergent behavior that arises from simple rules followed by individual agents. Students will study classical algorithms like Ant Colony Optimization (ACO) and gain hands-on experience in designing and analyzing swarm-based systems, including applications in robotics and optimization.

Textbook(s)

- Dorigo, M., & Stuetzle, T. Ant Colony Optimization, MIT Press, 2004. ISBN: 9780262042192
 - Yang, X.-S. Nature-Inspired Optimization Algorithms, Elsevier, 2020. ISBN: 9780128196674

Other supplemental materials

- Bonabeau, E., Dorigo, M., & Theraulaz, G. Swarm Intelligence: From Natural to Artificial Systems, Oxford University Press, 1999. ISBN: 9780195131598

Course Learning Objectives

#	Course Learning Outcomes (CLOs)	BT/ PLO #
CLO 1	Explain the foundational principles of swarm intelligence and its natural inspirations.	C2
CLO 2	Apply swarm-based algorithms such as ACO to solve optimization problems.	C3
CLO 3	Design and simulate swarm robotic systems with features like task allocation and fault tolerance.	C4
CLO 4	Analyze and evaluate the performance of swarm intelligence algorithms in various domains.	C5

Week wise Lecture Schedule

W #	Lecture Topics
1	Introduction to Swarm Intelligence and key concepts (decentralization, self-organization)
2	Biological inspiration: ants, birds, fish, termites, and social insects
3	Swarm-based problem solving in natural systems
4	Travelling Salesman Problem (TSP) and its significance in optimization
5	Introduction to Ant Colony Optimization (ACO)
6	Construction graph, pheromone trails, and probability transition rules in ACO
7	Variants and enhancements of ACO

W #	Lecture Topics
8	Applications of ACO in routing, scheduling, and other combinatorial problems
9	Introduction to swarm robotics and its distinction from classical robotics
10	Robot-to-robot communication and local decision-making
11	Division of labour and task allocation strategies in swarm robotics
12	Fault detection and resilience in swarm systems
13	Case studies in swarm robotics and collective transport
14	Simulation tools and platforms for swarm robotics (e.g., Webots, ARGoS)
15	Case Study/Project/Demo
16	Presentation/Peer Review/Q&A Sessions/Demo

AI473 - Agent Based Modeling

Credit Hours: 3(3+0) **Prerequisites:** CC351
Course Category: Artificial Intelligence Elective

Description

This course introduces students to Agent-Based Modeling (ABM)—a computational modeling paradigm for simulating interactions among autonomous agents—and its applications in biological and social systems using platforms like NetLogo. It also explores the formal foundations of logic, including propositional and predicate logic, and model checking techniques essential for verifying the correctness of complex systems. Emphasis is placed on simulation, verification, and reasoning in multi-agent environments.

Textbook(s)

- Modeling Social Behavior: Mathematical and Agent-Based Models of Social Dynamics and Cultural Evolution, Paul Smaldino, 2023, ISBN: 978-0691224145

- Agent-Based Models, Nigel Gilbert, SAGE Publications, 2008

Course Learning Objectives

#	Course Learning Outcomes (CLOs)	BT/ PLO #
CLO 1	Explain agent-based modeling principles and simulate natural/social systems using tools like NetLogo.	C4 1,2,5
CLO 2	Apply concepts of complexity, emergence, and critical phenomena in the design of ABMs	C3 1,2,3,10
CLO 3	Use propositional and predicate logic to formally express and reason about software/system behavior.	C4 8,9,10
CLO 4	Evaluate and verify system properties using model checking techniques and logic-based verification tools.	C6 8,10

Week wise Lecture Schedule

W #	Lecture Topics
1	Introduction to Agent-Based Modeling (ABM); Why ABM? Use Cases
2	NetLogo Overview: Interface, Commands, and Procedures
3	Modeling Natural Systems I: Flocking, Fireflies, Bees, and Slime Mold
4	Modeling Natural Systems II: Predator–Prey Dynamics; Debugging and Validation
5	Modeling Social Systems: Schelling’s Segregation, Micromotives and Macrobehavior
6	Emergence, Self-Organization, and Complexity in ABMs
7	Cellular Automata, Critical Phenomena, and Sandpile Models

W #	Lecture Topics
8	Research Trends and Case Studies in ABM
9	Introduction to Propositional Logic: Syntax, Semantics, and Natural Deduction
10	Predicate Logic: Quantifiers, Semantics, Expressiveness, and Undecidability
11	Modal Logic and Knowledge in Multi-Agent Systems
12	Model Checking: Motivation, Linear-Time Temporal Logic (LTL) and Tools
13	Branching-Time Temporal Logic (CTL, CTL*) and Fixed-Point Semantics
14	Symbolic Model Checking, BDDs, and the Mu-Calculus
15	Case Study/Project/Demo
16	Presentation/Peer Review/Q&A Sessions/Demo

CY321 - Cyber Security**Credit Hours:** 3(3+0) **Prerequisites:** None**Course Category:** Cybersecurity Domain**Description**

In this course, the student will be able to understand the intricacies of Cybersecurity as they embark on a transformative journey through the dynamic realm of digital defense. From grasping the fundamental principles of network security, cryptography, and web application security to navigating the ever-evolving landscape of cybersecurity threats. Students will study about secure communication protocols, conducting vulnerability assessments, and formulating effective incident response strategies.

Textbook(s)

- Charles J. Brooks, Christopher Grow, Philip A. Craig, Donald Short. Cybersecurity Essentials 1st Edition, Kindle Edition Sybex, (2018), ISBN:978-1119362395

- William Stallings. Network Security Essentials: Applications and Standards 6th Edition, Pearson, (2016). ISBN:978-0134527338

- Bryan Sullivan, Vincent Liu. Web Application Security, A Beginner's Guide 1st Edition, Kindle Edition McGraw Hill (2011). ISBN:978-0071776165

Other supplemental materials

- Jason T. Luttgens, Matthew Pepe, Kevin Mandia. Incident Response & Computer Forensics, Third Edition 3rd Edition, Kindle Edition, McGraw Hill (2014). ISBN: 978-0071798686

Course Learning Objectives

#	Course Learning Outcomes (CLOs)	BT/ PLO #
CLO 1	Develop Comprehensive Cybersecurity Knowledge	C3 1,2,3, 10
CLO 2	Apply Cybersecurity Strategies and Techniques	C3 2,4,5, 10
CLO 3	Analyze Cybersecurity Challenges and Compliance	C4 3, 10

Week wise Lecture Schedule

W #	Lecture Topics
1	Cybersecurity and its importance. Types of Cybersecurity threats. Common attack vectors and methods. Cybersecurity principles and best practices
2	Introduction to Network Security. Firewalls and Intrusion Detection/Prevention Systems (IDS/IPS). Secure Network Architecture and Segmentation. Virtual Private Networks (VPNs) and Tunneling

W #	Lecture Topics
3	Cryptography. Types of Cryptographic algorithms (Symmetric and Asymmetric). Encryption and Decryption. Key Management and Public Key Infrastructure (PKI)
4	Secure Communication Protocols. HTTPS and TLS/SSL. Secure Email (PGP/GPG) Securing DNS (DNSSEC). Secure File Transfer Protocols (SFTP, SCP)
5	Web Application Security. Common Web Application Vulnerabilities (e.g., OWASP Top 10). Web Application Firewalls (WAF). Secure Coding Practices and Code Reviews. Web Application Penetration Testing
6	Operating System Security. User Management and Access Control. Patch Management and System Updates. Endpoint Protection and Antivirus
7	Malware and Antivirus. Malware types (Viruses, Worms, Trojans, Ransomware). Antivirus and Endpoint Security Solutions
8	Incident Response and Handling. Incident Response Process and Phases Incident Handling and Containment. Digital Forensics and Evidence Collection
9	Wireless Security. Wireless Network Vulnerabilities (e.g., WEP, WPA). Securing Wi-Fi Networks (WPA2, WPA3). Bluetooth and IoT Security
10	Mobile Security. Mobile Device Security. Mobile Application Security. Mobile Device Management (MDM). Mobile Threats and Vulnerabilities
11	Cloud Security. Cloud Computing and Security Challenges. Cloud Service Models (IaaS, PaaS, SaaS). Cloud Security Controls and Best Practices. Cloud Security Assessments and Audits
12	Identity and Access Management (IAM). IAM Concepts and Components. Authentication and Authorization. Single Sign-On (SSO) and Multi-Factor Authentication (MFA). Identity and Access Governance
13	Social Engineering Attacks. Phishing Techniques and Countermeasures. Ethical Hacking and Responsible Disclosure
14	Vulnerability Assessment and Scanning. Penetration Testing Methodologies. Exploitation Techniques and Post-Exploitation.
15	Overview of Cybersecurity Laws and Regulations. Cybersecurity Frameworks (e.g., NIST, ISO 27001)
16	Case studies and Presentations

CY322 - Network Security**Credit Hours:** 3(3+0) **Prerequisites:** CC261**Course Category:** Cybersecurity Domain**Description**

This course is intended to provide the students not only a broad coverage of the typical network security topics, but also a practical and real-world exposure to network security challenges faced by professionals in the workplace.

Textbook(s)

- Jie Wang, Zachary A. Kissel. Introduction to Network Security: Theory and Practice, , John Wiley & Sons, (2019) ISBN: 1118939484, 9781118939482

- Shafiullah Khan, Al-Sakib Khan Pathan. Wireless Networks and Security, Springer Berlin Heidelberg, (2013), ISBN: 978-364236168-5.

Other supplemental materials

- Nitesh Dhanjani, Justin Clarke. Network Security Tools, , O'Reilly Media, Inc, (2005), ISBN: 9780596007942.

Course Learning Objectives

#	Course Learning Outcomes (CLOs)	BT/ PLO #
CLO 1	Understand Network Security Fundamentals	C2 1, 10
CLO 2	Apply Cryptographic Techniques	C3 2, 4, 5
CLO 3	Analyze Key Distribution and User Authentication	C4 4, 6
CLO 4	Assess Network Security Protocols and Technologies	C6 3, 10

Week wise Lecture Schedule

W #	Lecture Topics
1	Introduction to Network Security Concepts: Threats, Vulnerabilities, Types of Attacks, Security Goals
2	Secured Network Architecture & OSI Layer-wise Security Overview
3	Physical Layer Security: Attacks (eavesdropping, jamming), Countermeasures (shielding, physical controls)
4	MAC Layer Security: Attacks (MAC spoofing), Countermeasures (authentication, encryption)
5	Network Layer Security: Attacks (IP spoofing, routing attacks), Countermeasures (IPsec, secure routing)
6	Transport Layer Security: Attacks (SYN flooding), Countermeasures (TLS/SSL, secure handshake)
7	Application Layer Security: Attacks (SQL injection, XSS), Countermeasures (input validation, HTTPS)

W #	Lecture Topics
8	Key Management & Digital Signatures: PKI, symmetric/asymmetric encryption
9	AAA Servers: Authentication, Authorization, and Accounting in secure network access
10	Wireless Network Security: IEEE 802.11 & 802.11i, Threats (rogue APs, wardriving), Countermeasures
11	Cloud Security (AWS & Azure): IAM, VPC, shared responsibility, network/firewall configurations
12	SDN Security: Architecture, Attack Surface, Controller/Data Plane Risks, Mitigations
13	Decentralized Network Security: P2P networks, Blockchain security applications
14	Intrusion Detection Systems (IDS): Host-based vs Network-based, Signature vs Anomaly-based IDS
15	Case Study – SNORT: Configuration, Rules, Live Attack Detection
16	Review and Integration: Layered Defense Strategies, Emerging Trends

CY331 - Secure Software Design and Development

Credit Hours: 3(3+0) **Prerequisites:** None

Course Category: Cybersecurity Domain

Description

The module aims to develop core competencies in the fields of Secure Software Concepts, Secure Software Requirements, Secure Software Design, Secure Software Implementation/Coding, and Secure Software Testing. The course details the software security activities that need to be incorporated throughout the software development lifecycle. It provides comprehensive coverage that includes the people, processes, and technology components of software, networks, and host defenses.

Textbook(s)

- Gary McGraw. Software Security: Building Security In, 1st Edition, (2006), Addison-Wesley Professional ISBN: 978-0321356703
- Loren Kohnfelder. Designing Secure Software: A Guide for Developers, 978-1718501928, (2021), ISBN: 978-1718501928

Other supplemental materials

- Official (ISC)2 Guide to the CSSLP (latest)

Course Learning Objectives

#	Course Learning Outcomes (CLOs)	BT/ PLO #
CLO 1	A good comprehension of software security standards, models, processes and best practices that need to be incorporated throughout the software development lifecycle.	C2 1, 2, 10
CLO 2	Identify insecure programming patterns and the ability to replace them with secure alternatives.	C3 2, 3
CLO 3	Know tools for software security analysis and testing, and the ability to use them in practice and understand their capabilities and limitations.	C3 4, 5, 6, 10

Week wise Lecture Schedule

W #	Lecture Topics
1	Secure software concepts: System issues, System properties, Software Project Time Management, Software Project Costing, Software Quality Assurance
2	Security Concepts in the SDLC: Risk management, Security standards (e.g., coding standards, NIST standards, Federal Information Processing Standards);
3	Best practices: (e.g., OWASP development guide, OWASP code review guide, OWASP testing guide);
4	Security methodologies: (e.g., Socratic Methodology, Operationally Critical Threat, Asset, and Vulnerability

W #	Lecture Topics
	Evaluation, STRIDE and DREAD, Open Source Security Testing Methodology Manual);
5	Security frameworks: (e.g., Zachman Framework, Control Objectives for Information and Related Technology, Sherwood Applied Business Security Architecture (SABSA));
6	Regulations- Privacy and Compliance;
7	Security Models: (e.g., BLP Confidentiality Model, Clark and Wilson Model (Access Triple Model));
8	Trusted Computing; Secure Software Requirements (Sources for Security Requirements, Types of Security Requirements);
9	Secure Software Design: (Design consideration, Information Technology Security Principles and Secure Design, Designing Secure Design Principles), Design Processes
10	Secure Software Implementation/Coding: Software Development Methodologies, Common Software Vulnerabilities and Controls
11	Secure Software Implementation/Coding: Defensive Coding Practices—Concepts and Techniques
12	Code Vulnerabilities and Avoiding Polymorphic Malware Attacks: : Buffer overflow, Format string, bug, Code vulnerabilities SQL Injection
13	Code Vulnerabilities and Avoiding Polymorphic Malware Attacks: Cross-site Scripting, Cross-site Request Forgery, Session management, Replication of vulnerabilities and exploitation;
14	Secure Software Testing: Security Testing Methodologies, Software Security Testing, Software Acceptance
15	Secure Software Testing: Legal Protection Mechanisms, Software Deployment- Operations-Maintenance and Disposal.
16	Case Studies

CY351 - Information Assurance**Credit Hours:** 3(3+0) **Prerequisites:** None**Course Category:** Cybersecurity Domain**Description**

In this course, students will be able to protect digital assets, understand risk management, and design effective information security plans. Gain practical proficiency in implementing strategies to secure data, detect anomalies, and ensure business continuity.

Textbook(s)

- Corey Schou, Steven Hernandez. Information Assurance Handbook: Effective Computer Security and Risk Management Strategies, McGraw Hill, (2015), ISBN: 78-0071821650

- Darril Gibson, Andy Igonor. Managing Risk in Information Systems (Information Systems Security & Assurance) 3rd Edition, Jones & Bartlett Learning, (2020), ISBN: 978-1284183719

Course Learning Objectives

#	Course Learning Outcomes (CLOs)	BT/ PLO #
CLO 1	Comprehend the fundamental principles and concepts of information assurance, including its importance in safeguarding information assets, managing risk, and ensuring confidentiality, integrity, and availability of data.	C1 1, 10
CLO 2	Design and develop effective information security plans, incorporating preventive, detective, and corrective measures to mitigate risks and protect against potential threats to information systems.	C3 2, 3
CLO 3	Demonstrate practical proficiency in implementing information assurance strategies, technologies, and best practices, to secure data, detect anomalies, respond to security incidents, and recover from disruptions.	C6 4, 5, 10

Week wise Lecture Schedule

W #	Lecture Topics
1	Information Security Infrastructure: The Need for Information Security, Concepts in Information Security
2	Information Security Infrastructure: Assets, Threats, Vulnerabilities, Risks, and Controls, Security Professionals and Organizations
3	Information Security Infrastructure: Information Security management System, Implementing Information Security Strategy into Current Practices, Regulations, and Plans
4	Information Security Planning: Approaches to Implementing Information Security, Organizational

W #	Lecture Topics
	Structure for Managing Information Security, Asset Management
5	Information Security Planning: Information Security Risk Management, Information Security Policy
6	Information Security Planning: Human Resource Security, Certification, Accreditation, and Assurance
7	Information Security Prevention: Information Security in System Development, Physical and Environmental Security Controls
8	Information Security Prevention: Information Security Awareness, Training, and Education
9	Information Security Prevention: Preventive Tools and Techniques
10	Information Security Prevention: Access Control
11	Information Security Detection: Information Security Monitoring Tools and Methods
12	Information Security Detection: Information Security Measurements and Metrics
13	Information Security Recovery: Information Security Incident Handling, Computer Forensics
14	Information Security Recovery Process: Business Continuity, Disaster Management
15	Information Security Recovery Process: Backup and Restoration
16	Case studies, Recent Trends

CY441 - Digital Forensics**Credit Hours:** 3(3+0) **Prerequisites:** None**Course Category:** Cybersecurity Domain**Description**

This course is an introduction to computer forensics and investigation. It provides an understanding of how to conduct investigations to correctly gather, analyze and present digital evidence to different audiences. It also outlines the tools to locate and analyze digital evidence on a variety of devices, how to keep up to date with changing technologies, and laws and regulations in digital forensics.

Textbook(s)

- John Sammons. The Basics of Digital Forensics: The Primer for Getting Started in Digital Forensics, 2nd Edition, Elsevier, (2014) ISBN: 9780128016350

- Gerard Johansen. Digital Forensics and Incident Response : Incident Response Techniques and Procedures to Respond to Modern Cyber Threats, 2nd Edition, Packt Publishing (2020) ISBN: 978-1838649005

Other supplemental materials

- Joakim Kävrestad. Guide to Digital Forensics : A Concise and Practical Introduction, Springer (2017), ISBN: 978-3319674490

Course Learning Objectives

#	Course Learning Outcomes (CLOs)	BT/ PLO #
CLO 1	To develop knowledge about forensic law, standards, regulations and ethical values	C2 1, 8, 9, 10
CLO 2	To be able to conduct digital forensics for multiple platforms and applications by various tools	C3 2, 5, 6
CLO 3	To be able to generate reports based on digital forensic tools for security systems and platforms	C3 3, 4, 6

Week wise Lecture Schedule

W #	Lecture Topics
1	An introduction to Digital Forensics; use of digital forensics; Key technical concepts;
2	Challenges in Digital Forensics ; The Difference between Computer Experts and Digital Forensics Experts;
3	Investigative Process Methodologies ; Education, Training, and Awareness;
4	Laws, Standards, and Regulations; Ethics and Professional Conduct;
5	Digital Evidence Management; Collecting evidence; Antiforensics;
6	Network forensics;
7	Mobile and Embedded Forensics;

W #	Lecture Topics
8	Cloud forensics;
9	Internet Forensics;
10	Social media forensics;
11	Investigation Methods for Collecting Digital Evidence; Digital Forensic Readiness;
12	Digital forensics tools; Discovery of Computers and Storage Media;
13	Discovery of Audio/ Video Evidence;
14	Data Visualization; Data Sources; Graphing and Charting;
15	Analyzing Data; Data Distributions; Analysis Scenarios; Data Visualization Tools.
16	Case Studies

CY371 - System and Network Administration

Credit Hours: 3(3+0) **Prerequisites:** CC261

Course Category: Cybersecurity Elective

Description

In this course on Information Assurance, students will gain essential knowledge and practical skills to protect digital assets and mitigate security risks. Explore server environments, user management, network monitoring, and more. Join us to master the critical realm of Information Assurance.

Textbook(s)

- Thomas A. Limoncelli Study guide for Practice of System and Network Administration, Cram101, 2nd Edition (2011). ISBN-10: 1428851755.

- Wale Soyinka. Linux Administration: A Beginner's Guide, Eighth Edition 8th Edition, McGraw Hill (2020), ISBN: 978-1260441703

- Dishan Francis Mastering Active Directory: Design, deploy, and protect Active Directory Domain Services for Windows Server 2022, 3rd Edition, Packt Publishing, (2022), ISBN: 978-1801070393

Other supplemental materials

Activity Directory - <https://learn.microsoft.com/en-us/azure/active-directory/hybrid/cloud-sync/tutorial-basic-ad-azure>

Linux tutorial - <https://www.javatpoint.com/linux-tutorial>

Course Learning Objectives

#	Course Learning Outcomes (CLOs)	BT/ PLO #
CLO 1	Familiarize with fundamental terminologies and technologies in System and Network Administration.	C1 1,2 10
CLO 2	Understand the role of operating system configuration for system and network administration.	C2 6
CLO 3	Recognize the different operating systems utilized for managing system and network administration tasks.	C4 2, 3,5, 10

Week wise Lecture Schedule

W #	Lecture Topics
1	Definition and Role of System Administration, Key Responsibilities and Skills of a System Administrator, Understanding System Architecture and Components, Introduction to Command-Line Interface (CLI), Server Environment (Microsoft and Linux)
2	Overview of Server Operating Systems (Windows Server and Linux Server): Different Server Roles and Functions, Server Hardware and Virtualization Concepts, Introduction to Hypervisors (e.g., VMware,

W #	Lecture Topics
	Hyper-V), Comparative Analysis of Most Demanded Operating Systems
3	Detailed Comparison of Windows Server and Linux Server Features: Licensing and Cost Considerations, Performance and Security Aspects of Both Systems, Linux Installation and Verification
4	Choosing a Linux Distribution: Installation Methods (Graphical and Command-Line),
5	User Account Management in Linux and Windows: Group Management and User Group Associations, User Authentication and Password Policies, Services and Management
6	Understanding System Services and Daemons: Starting, Stopping, and Restarting Services, Configuration Management Tools (e.g., systemd), Software Management in Linux
7	Package Management Systems (APT, YUM/DNF): Installing, Updating, and Removing Software Packages, Handling Dependencies and Repositories, Managing Network Services and Network Establishment in VM
8	Configuring Network Interfaces in Linux and Windows: Network File Sharing (SMB, NFS) Introduction to Virtual Networking (bridged, NAT, host-only), Managing Permissions in Linux
9	File System Permissions (chmod, chown, chgrp): Setting File and Directory Access Control Lists (ACLs), Understanding Security Contexts (SELinux, AppArmor), Monitoring of Network Resources
10	Network Monitoring Tools (e.g., netstat, nmap): Resource Monitoring (CPU, Memory, Disk Usage), Setting Up Alerts and Notifications, Securing Network Traffic using Nagios XI
11	Introduction to Nagios XI for Network Monitoring: Configuring Nagios Plugins for Services and Hosts, Implementing SSL/TLS for Secure Communication, Logs Handling and Logs Management
12	Understanding System Logs (syslog, event logs): Log Rotation and Archiving, Log Analysis and Troubleshooting, Installation of Remote Desktop Server
13	Remote Desktop Protocols (RDP, VNC): Setting Up Remote Desktop Services (Windows Remote Desktop Services, VNC Server), Installation and Configuration of Web Server on CentOS
14	Installing Apache Web Server: Configuring Virtual Hosts, Introduction to SSL Certificates and HTTPS, Installation and Configuration of FTP on CentOS
15	Setting Up FTP Server (vsftpd): User Access and Authentication, FTP Security Considerations Installation of Active Directory
16	Introduction to Active Directory (Windows)

CY372 - Vulnerability Assessment & Reverse Engineering

Credit Hours: 3(3+0) **Prerequisites:** None

Course Category: Cybersecurity Elective

Description

The course aims to develop core competencies in the field of vulnerability assessment covering software, networks and Web applications. It also covers reverse engineering techniques to analyze software, exploit targets, and defend against security threats like malware and viruses.

Textbook(s)

- Reversing: Secrets of Reverse Engineering, Eldad Eilam, 2005, ISBN: 978-0764574812
- Finding and Fixing Vulnerabilities in Information Systems: The Vulnerability Assessment and Mitigation Methodology by Philip S. Anton
- The Art of Software Security Assessment: Identifying and Preventing Software Vulnerabilities by Mark Dowd

Other supplemental materials

- Reversing: Secrets of Reverse Engineering by Eldad Eilam (latest edition)

Course Learning Objectives

#	Course Learning Outcomes (CLOs)	BT/ PLO #
CLO 1	Basic Understanding of Hacking and Ethical Hacking.	C2 1,2,10
CLO 2	Apply techniques for vulnerability assessment and penetration testing.	C3 2,3,4
CLO 3	Understand Software vulnerabilities, Network vulnerabilities, Types of Malware and its Analysis.	C3 1,8,9 ,10

Week wise Lecture Schedule

W #	Lecture Topics
1	Understanding the need for security assessments;
2	Classifying vulnerabilities; Software vulnerabilities; Network vulnerabilities;
3	Vulnerability assessment versus penetration testing; Vulnerability Assessment Tools;
4	Vulnerability management Regulatory compliance; Calculating ROIs;
5	Application review process; Pre-assessment;
6	Code navigation; Code- auditing tactics;
7	Memory corruption
8	Understanding issues in programming languages;
9	Steps in Reverse engineering,
10	Common tools used for Reverse engineering;

W #	Lecture Topics
11	Binary Obfuscation techniques;
12	Understanding core assembly concepts to perform malicious code analysis, Identifying key assembly logic structures with a disassembler
13	Malware analysis Types of malware analysis, Malware Taxonomy;
14	Static analysis; Dynamic analysis;
15	Malware Inspection; Malware analysis tools; Sandboxing and virtualization;
16	Case Studies and Student Presentations

CY373 - Hardware Security**Credit Hours:** 3(3+0) **Prerequisites:** None**Course Category:** Cybersecurity Elective**Description**

With the proliferation of embedded devices, IoT systems, and cyber-physical infrastructure, hardware has become a critical attack surface. Hardware Security explores techniques to protect digital systems from physical attacks, reverse engineering, side-channel attacks, and IP theft. This course covers the principles, threat models, and modern defenses in hardware security, with emphasis on secure hardware design, trusted execution environments, and cryptographic hardware primitives.

Textbook(s)

- Tehranipoor, M., & Wang, C. Introduction to Hardware Security and Trust, Springer, 2011.

- Xiao, K., Tehranipoor, M. Hardware Security: A Hands-on Learning Approach, Springer, 2019.

Other supplemental materials

- Zhang, Y. (Ed.) Hardware Security: Design, Threats, and Safeguards, CRC Press, 2016.

Course Learning Objectives

#	Course Learning Outcomes (CLOs)	BT/ PLO #
CLO 1	Explain threats, vulnerabilities, and security principles at the hardware level.	C5
CLO 2	Identify and analyze hardware attack vectors, including physical and side-channel.	C2
CLO 3	Design and evaluate countermeasures such as PUFs, logic locking, and obfuscation.	C4
CLO 4	Understand the role of secure hardware in trusted systems and cryptographic devices.	C2

Week wise Lecture Schedule

W #	Lecture Topics
1	Introduction to Hardware Security: Scope, Applications, and Threat Models
2	Review of Digital Hardware Design and Basic Cryptographic Primitives
3	Hardware Attack Taxonomy: Physical, Semi-invasive, Invasive, and Remote Attacks
4	Reverse Engineering and Counterfeit Detection
5	Side-Channel Attacks (SCA): Power Analysis and Timing Attacks
6	Fault Injection Attacks: Clock Glitching, Laser, and EM Injection
7	Countermeasures Against SCA and Fault Attacks
8	Logic Locking and Obfuscation Techniques

W #	Lecture Topics
9	Hardware Trojans: Types, Detection, and Prevention
10	Physically Unclonable Functions (PUFs): Types and Applications
11	Secure Boot and Trusted Execution Environments (TEEs)
12	Design-for-Security (DfS) and Security-Aware EDA Tools
13	Secure FPGA Design and Bitstream Protection
14	Trusted Platform Modules (TPMs) and Hardware Root of Trust
15	Hardware Security in IoT and Cyber-Physical Systems
16	Project Presentations, Case Studies, and Future Trends in Hardware Security

CY374 - Malware Analysis**Credit Hours:** 3(3+0) **Prerequisites:** None**Course Category:** Cybersecurity Elective**Description**

This course provides foundational and advanced techniques in the field of malware analysis. Students will explore various types of malwares, methods for analyzing malicious software through static and dynamic approaches, disassembly techniques, and hands-on debugging. Emphasis will be placed on understanding malware behavior, using professional tools like IDA Pro and x64dbg, and preparing students for careers in reverse engineering and cybersecurity.

Textbook(s)

Monnappa K A. Learning Malware Analysis: Explore the concepts, tools, and techniques to analyze and investigate Windows malware, Packt Publishing (2018), ISBN: 978-1788392501

Other supplemental materials

Michael Sikorski . Practical Malware Analysis: The Hands-On Guide to Dissecting Malicious Software, No Starch Press, (2012), ISBN:978-1593272906

Course Learning Objectives

#	Course Learning Outcomes (CLOs)	BT/ PLO #
CLO 1	Grasp fundamental concepts of Malware, distinguish types, and understand their impact on computer systems.	C1 1, 10
CLO 2	Perform proficient Static and Dynamic Malware Analysis using tools like IDA Pro, interpreting assembly code and identifying malware behavior.	C3 2,3,4, 5
CLO 3	Master advanced techniques such as disassembling Windows API, using IDA scripting and plugins, and patching binaries for defense and analysis purposes.	C6 1,2, 4, 5, 6,10

Week wise Lecture Schedule

W #	Lecture Topics
1	Basic concepts about Malware: Types, history, classifications. Malware Analysis, Types Of Malware Analysis
2	Static Analysis: Determining the File Type, Fingerprinting the Malware
3	Static Analysis: Multiple Anti-Virus Scanning, Extracting Strings, Determining File Obfuscation,
4	Static Analysis: Inspecting PE Header Information, Comparing and Classifying the Malware
5	Dynamic Analysis: System And Network Monitoring, Dynamic Analysis (Monitoring) Tools
6	Dynamic Analysis: Dynamic Analysis Steps, Analyzing a Malware Executable

W #	Lecture Topics
7	Dynamic Analysis: Dynamic-Link Library (DLL) Analysis
8	Assembly Language and Disassembly Primer: Computer Basics, CPU Registers, Data Transfer Instructions
9	Assembly Language and Disassembly Primer: Arithmetic Operations, Bitwise Operations, Branching And Conditionals, Loops
10	Assembly Language and Disassembly Primer: Functions, Arrays And Strings, Structures, x64 Architecture
11	Disassembly Using IDA: Code Analysis Tools, Static Code Analysis (Disassembly) Using IDA
12	Disassembly Using IDA: Disassembling Windows API, Patching Binary Using IDA, IDA Scripting and Plugins
13	Debugging Malicious Binaries: General Debugging Concepts
14	Debugging Malicious Binaries: Debugging a Binary Using x64dbg
15	Debugging Malicious Binaries: Debugging a Binary Using IDA
16	Casse Studies and Students Presentations

CY375 - Wireless and Mobile Security

Credit Hours: 3(3+0) **Prerequisites:** CC261

Course Category: Cybersecurity Elective

Description

The main objective of this course is to let the students learn the art of Cybersecurity and Network Protection. Throughout the journey, we will delve into various topics, including Wireless Networks, Mobile Device Security, Application Level Security, Ubiquitous Computing, IoT Security, Data Center Operations, and Cloud Computing. By mastering encryption techniques, understanding security challenges, and implementing robust measures, students will become skilled defenders in the ever-evolving cyber landscape. Join us now to strengthen your cybersecurity prowess and make a significant impact in the world of digital security.

Textbook(s)

Pallapa Venkataram, Satish Babu. Wireless and Mobile Network Security, 1st Edition, McGraw Hill (2010), ISBN: 978-0070700246

Jim Doherty. Wireless and Mobile Device Security, Jones and Bartlett Publishers, Inc (2021) ISBN: 978-1284211726

Other supplemental materials

Frank Adelstein, K.S.Gupta. Fundamentals of Mobile and Pervasive Computing. 1st Edition, Tata McGraw Hill (2005), ISBN: 978-0070603646

Course Learning Objectives

#	Course Learning Outcomes (CLOs)	BT/ PLO #
CLO 1	Understand the fundamentals of Wireless Networks and Mobile Device Security, including wireless communication components, architectures, and mobile device security features.	C2 1,2,10
CLO 2	Identify and evaluate security risks associated with Wireless Networks, Mobile Apps, Ubiquitous Computing, and IoT, along with the implementation of security countermeasures.	C4 2,3,4
CLO 3	Comprehend various encryption protocols for securing Wireless Networks, Mobile Data, Cloud Computing, and IoT devices, as well as key management for ensuring data confidentiality and integrity.	C2 1,4,10
CLO 4	Analyze the security challenges and solutions in Mobile Device Management (MDM), Bring Your Own Device (BYOD) policies, Application Level Security in Mobile Ad Hoc Networks (MANETs), Data Center Operations, and Cloud Computing to develop effective security recommendations	C4 3,4, 10

Week wise Lecture Schedule

W #	Lecture Topics
1	Introduction to Wireless Networks: Basics of Wireless Communication, Wireless Network Architectures and Components, Wireless Standards (Wi-Fi, Bluetooth, etc.)
2	Wireless Network Security Risks: Common Security Threats in Wireless Networks, Wireless Eavesdropping and Data Interception
3	Wireless Encryption Protocols: WEP, WPA, WPA2, and WPA3 Security Protocols, Wi-Fi Protected Setup (WPS), Vulnerabilities
4	Mobile Device Security Fundamentals: Overview of Mobile Device Architecture, Mobile Operating System Security (Android, iOS)
5	Mobile App Security: Secure Mobile App Development Best Practices, Code Signing and App Distribution Security Mobile Device Management (MDM) and BYOD: Implementing MDM Solutions for Enhanced Security, Security Challenges of Bring Your Own Device (BYOD) Policies
6	Mobile Data Protection: Data Encryption on Mobile Devices, Remote Wiping and Tracking Lost Devices
7	Mobile Authentication and Biometrics, Multi-factor Authentication for Mobile Devices, Biometric Security on Smartphones
8	Wireless Intrusion Detection and Prevention: Deploying Wireless IDS/IPS Solutions, Detecting Rogue Access Points and Attacks
9	Application Level Security in MANETs: MANETs, Some applications of MANETs, MANET Features, Security Challenges in MANETs,
10	Application Level Security in MANETs: Security Attacks on MANETs, External Threats for MANET applications, Internal threats for MANET Applications, Some of the Security Solutions.
11	Ubiquitous Computing, Need for Novel Security Schemes for UC, Security Challenges for UC, and Security Attacks on UC networks, Some of the security solutions for UC.
12	IoT Security challenges, Standards and countermeasures
13	Data Center Operations - Security challenge, implement. Five Principal Characteristics of Cloud Computing
14	Data center Security Recommendations. Encryption for Confidentiality and Integrity, Encrypting data at rest, Key Management Lifecycle
15	Cloud Encryption Standards
16	Case Studies and Recent Trends

CY376 - Web App Development and Security

Credit Hours: 3(3+0) **Prerequisites:** None

Course Category: Cybersecurity Elective

Description

In this course, students will gain expertise in web application development, web app security, vulnerability comprehension, and the application of security principles.

Textbook(s)

- Andrew Hoffman. Web Application Security: Exploitation and Countermeasures for Modern Web Applications, O'Reilly Media (2020), ISBN: 978-1492053118

- Bryan Sullivan, Vincent Liu, Web Application Security, A Beginner's Guide 1st Edition, McGraw Hill (2011), ISBN: 978-0071776165

Other supplemental materials

- Matthew Leonard. Full Stack Web Development Complete Course: Complete Beginner to Expert Guide to Learn Everything About Full Stack Web Development Using CSS, HTML, PHP, Independently published (2023), ISBN: 979-8387438608

Course Learning Objectives

#	Course Learning Outcomes (CLOs)	BT/ PLO #
CLO 1	Acquire comprehensive expertise in web application development, encompassing front-end and back-end technologies, frameworks, and deployment options.	C1 1,2 4,5, 6,10
CLO 2	Gain comprehensive understanding of web application security, covering architecture, HTTP security headers, global standards (WASC, OWASP, NIST, CWE, SANS), and fundamentals for a secure environment (CIA, policies, secure SDLC).	C2 1,2, 8,10
CLO 3	Comprehend diverse web application vulnerabilities, including common attack categories and prevalent risks. Implement effective countermeasures to enhance application security.	C2 1,3, 5
CLO 4	Apply security testing, Secured Development Approach, and Secure Design Principles for web applications.	C3 5,6, 10

Week wise Lecture Schedule

W #	Lecture Topics
1	Introduction to Web application development: Basic concepts, tools and technologies, HTML, CSS
2	Front-end Web Development: Advanced CSS, JavaScript
3	Back-end Web Development: Server-side languages, Introduction to PHP, Node.js, Database Connectivity
4	Web Development Frameworks and Deployment: Front-end Frameworks (e.g., React, Angular, Vue.js),

W #	Lecture Topics
	Back-end Frameworks (e.g., Django, Ruby on Rails), Deployment Options (e.g., Heroku, AWS)
5	Introduction to Web app security: Case studies and statistics. Introduction to web applications. Basics of web application architecture. Application security risks. Attack vectors. Threat agents.
6	HTTP protocol: HTTP response headers, HTTP versus HTTPS, HTTP Strict Transport Security (HSTS), X-Frame-Options, X-XSS-Protection, X-Content-Type-Options. Content-Security-Policy, Referrer-Policy, Expect-CT.
7	Global Organizations, Standards and Frameworks: The Web Application Security Consortium (WASC), The Open Web Application Security Project (OWASP), The National Institute of Standards and Technology (NIST), The Common Weakness Enumeration (CWE) category system. The SysAdm, Audit, Network, Security (SANS) Institute.
8	Security fundamentals for secured environment: CIA: Confidentiality, integrity, availability. Policies and standards, Acquiring secure software, Secure architecture, Physical security. Introduction to secure SDLC.
9	Common Attack categories: Insecure interaction between components, Risky resource management, Porous defenses.
10	OWASP Top 10 Web Application Vulnerabilities: Injection, Broken authentication and session management. Sensitive data exposure, XML external entity (XXE), Broken access control, Security misconfiguration.
11	OWASP Top 10 Web Application Vulnerabilities: Cross-site scripting (XSS), Insecure deserialization, Using components with known vulnerabilities, Insufficient logging & monitoring, Definitions, explanations and examples. Countermeasures.
12	Other Common Vulnerabilities: Clickjacking: Cross-Site Request Forgery (CSRF). Server Side Request Forgery (SSRF). Definitions, explanations and examples. Countermeasures.
13	Testing and Monitoring: Static application security testing. Dynamic application security testing. Interactive application security testing. Runtime application self-protection. Monitoring tools.
14	Secured Development Approach: The secure SDLC. Threat modelling. Source code review, Common dangerous programming practices. Common development mistakes.
15	Secure Design Principles and Best Practices: Defense in depth, Fail safe, Least privilege, Separation of duties, Economy of mechanism, Complete mediation
16	Secure Design Principles and Best Practices: Open design, Least common mechanism, Psychological acceptability. Weakest link, Leveraging existing components.

CY471 - Penetration Testing

Credit Hours: 3(2+1) **Prerequisites:** None

Course Category: Cybersecurity Elective

Description

This course provides in-depth knowledge and practical skills in offensive security techniques used by ethical hackers and red teams. It covers advanced penetration testing methodologies involving both digital and physical attack surfaces. Students will explore social engineering, custom exploit creation, advanced web app attacks, and report writing in alignment with professional ethics. The course assumes prior knowledge of networking, security fundamentals, and basic scripting.

Textbook(s)

Vijay Kumar Velu, Mastering Kali Linux for Advanced Penetration Testing (4th Edition), Packt, 2022. ISBN: 9781801819770.

Other supplemental materials

Dafydd Stuttard, Marcus Pinto, *The Web Application Hacker's Handbook: Finding and Exploiting Security Flaws*, Wiley 2011, ISBN: 9781118026472 ISBN: 9781118026472

Course Learning Objectives

#	Course Learning Outcomes (CLOs)	BT/ PLO #
CLO 1	Demonstrate the ability to plan and execute penetration tests using Kali Linux tools.	C3
CLO 2	Perform advanced attacks including multi-stage, non-tool-based, and Layer 2 attacks.	C4
CLO 3	Evaluate and report security risks ethically, complying with professional standards.	C5
CLO 4	Build and deliver custom exploits and social engineering campaigns.	C6

Week wise Lecture Schedule

W #	Lecture Topics
1	Introduction to Offensive Security & Kali Linux: Ethical frameworks and legal considerations
2	Penetration Testing Lifecycle, Rules of Engagement, Scoping
3	Advanced Information Gathering & Enumeration Techniques
4	Vulnerability Scanning & Analysis
5	Exploit Development: Basics, Buffer Overflows, Payload Crafting
6	Advanced Web Application Attacks (XSS, SQLi, Command Injection)
7	Multi-Stage and Pivoting Attacks: Internal network compromise

W #	Lecture Topics
8	Layer 2 Attacks: ARP Spoofing, DHCP Starvation, VLAN Hopping
9	Wireless Attacks and Physical Intrusion Techniques
10	Social Engineering Attacks: Phishing, Pretexting, Payload Delivery
11	Antivirus and IDS Evasion, Encoding, Obfuscation
12	Insider Attacks, Privilege Escalation, Persistence
13	Red Team Operations: Attack simulation, stealth, OPSEC
14	Custom Exploit Generation: Shellcode, Exploit Packaging
15	Reporting: Professional Documentation, Risk Rating, Recommendations
16	Case Studies and Students Presentations

CY472 - Ethical Hacking**Credit Hours:** 3(3+0) **Prerequisites:** None**Course Category:** Cybersecurity Elective**Description**

Student will learn to explore the exciting domain of Ethical Hacking and develop essential skills to protect networks and systems responsibly. This dynamic course introduces the principles of ethical hacking, legal considerations, and effective methods for gathering critical target information. Dive into system hacking, secure wireless networks, and defend against potential threats. Gain hands-on experience and tackle real-world scenarios, empowering yourself to become a trusted guardian of digital security in an ever-changing technological landscape. Enroll now to unlock your potential as an ethical hacker!

Textbook(s)

- Kimberly Graves. Certified Ethical Hacker, SYBEX, (2014), ISBN: 978-0470525203

- Dale Meredith, Christopher Rees. Certified Ethical Hacker (CEH), Packt Publishing, (2022), ISBN: 978-1801813099

Other supplemental materials

Online Tutorial - <https://www.guru99.com/ethical-hacking-tutorials.html>

Course Learning Objectives

#	Course Learning Outcomes (CLOs)	BT/ PLO #
CLO 1	Exploring the importance of ethical behavior and complying with laws to maintain a responsible approach to hacking.	C1 1,8,9, 10
CLO 2	Learn to gather target information, footprint, and execute social engineering tactics ethically and responsibly.	C3 2, 5, 6
CLO 3	Acquire skills to crack passwords, escalate privileges, and hide files securely and ethically.	C4 5,6, 10
CLO 4	Explore various wireless hacking techniques and discover methods to protect and secure wireless networks from potential attacks.	C5 1,2,3, 6,8,10

Week wise Lecture Schedule

W #	Lecture Topics
1	Introduction to Ethical Hacking, Ethics, and Legality: Defining Ethical Hacking, Keeping It Legal, U.S. State Laws
2	Gathering Target Information: Reconnaissance, Footprinting, and Social Engineering
3	Gathering Network and Host Information: Scanning, Enumeration
4	Gathering Network and Host Information: Enumeration
5	System Hacking: Password Cracking

W #	Lecture Topics
6	System Hacking: Escalating
7	System Hacking: Privileges, and Hiding Files
8	Trojans, Backdoors, Viruses, and Worms
9	Gathering Data from Networks: Sniffers
10	Denial of Service and Session Hijacking
11	Google, Web Servers, Web Application Vulnerabilities, and Web-Based Password Cracking Techniques
12	Attacking Applications: SQL Injection
13	Attacking Applications: Buffer Overflows
14	Wireless Network Hacking: Wi-Fi and Ethernet, Authentication and Cracking Techniques,
15	Wireless Network Hacking: Using Wireless Sniffers to Locate SSIDs, MAC Filters and MAC Spoofing
16	Wireless Network Hacking: Rogue Access Points Evil Twin or AP Masquerading, Wireless Hacking Techniques 251 Securing Wireless Networks

CY473 - Cyber Laws and Crimes**Credit Hours:** 3(3+0) **Prerequisites:** None**Course Category:** Cybersecurity Elective**Description**

In this course, students will explore cyber crimes, their categories, and typical examples in Pakistan and the United States. They will grasp the legal aspects and regulations related to cyber crimes and digital evidence. Additionally, they will comprehend privacy and data protection principles in the digital age and understand cyber crime laws in both countries. Moreover, students will familiarize themselves with emerging cyber technologies' legal implications and develop awareness of cyberterrorism's impact on cyberspace security.

Textbook(s)

- Catherine D. Marcum. Cyber Crimes Third Edition, ASPEN Publishing, (2022), ISBN 978-1543839579
- Benny Forer, Privacy & Cyber Law: A Textbook, Independent Publisher (2023) ISBN:9798395732569
- Marjie Britz. Computer Forensics and Cyber Crime: An Introduction 3rd Edition, Pearson (2013), ISBN: 978-0132677714

Other supplemental materials

- Cyber laws in Pakistan:
<https://lms.rehnumai.punjab.gov.pk/course/view.php?id=46>
- Cybercrime Laws in Pakistan:
[https://www.graana.com/blog/cybercrime-law-in-pakistan/#:~:text=Prevention%20of%20Electronic%20Crimes%20Act%202016&text=It%20provides%20a%20comprehensive%20framework,DoS%20assault%20\(DOS%20Attack\)](https://www.graana.com/blog/cybercrime-law-in-pakistan/#:~:text=Prevention%20of%20Electronic%20Crimes%20Act%202016&text=It%20provides%20a%20comprehensive%20framework,DoS%20assault%20(DOS%20Attack))

Course Learning Objectives

#	Course Learning Outcomes (CLOs)	BT/ PLO #
CLO 1	Understand cyber crimes, their categories, and typical examples in Pakistan and the United States. Grasp legal aspects and regulations related to cyber crimes and digital evidence.	C2 1, 2, 10
CLO 2	Comprehend privacy and data protection principles in the digital age.	C2 1, 9, 10
CLO 3	Understand the cyber crime law passed by the Majlis-e-Shoora (Parliament) and cybercrime laws of the United States.	C2 1, 3, 9, 10
CLO 4	Familiarize with emerging cyber technologies and their legal implications. Develop awareness of cyberterrorism and its impact on cyberspace security.	C2 1, 8, 9, 10

Week wise Lecture Schedule

W #	Lecture Topics
1	Introduction to Cyber Crimes: Creation of the Internet and Cybercrime, The Dark Web and Cybercrime,

W #	Lecture Topics
	Transnational Cybercrimes, Key Legal Concepts and Terminology
2	Cyber Crime Categories: Types of Cyber Crimes: Unauthorized access to information system, data or critical infrastructure system and its data, Unauthorized copying or transmission of data or critical infrastructure information system, Interference with an information system, data or critical infrastructure system
3	Cyber Crime Categories: Cyber Terrorism, Hate Speech, Recruitment, Funding, and planning of terrorism, Electronic Forgery, Electronic Fraud, Malicious Coding
4	Cyber Crime Categories: Unauthorized Issuance of SIM Card, Making, obtaining and/or supplying device for use in offence, Spamming, Spoofing / Spreading of Fake News, Tampering of Communication Equipment,
5	Cyber Crime Categories: Unauthorized Interception, Offences against the Modesty and/or Dignity of a Person/People, Child Pornography, Cyber Stalking
6	Cyber Laws and Regulations: International Cyber Law Frameworks, National Cyber Laws and Regulations, Jurisdiction and Challenges in Cyber Law Enforcement
7	Legal Aspects of Digital Evidence: Digital Forensics and Its Role in Cyber Crime Investigations. Admissibility and Challenges of Digital Evidence in Court
8	Privacy and Data Protection: Data Protection Laws and Regulations, Privacy Rights and Issues in the Digital Age
9	Cyber Law and Emerging Technologies: Legal Issues with AI, IoT, Blockchain, and Cloud Computing, Regulatory Responses to New Technologies
10	Cyberterrorism: Technology and Terrorism in the Physical World, Recruitment, Education, Fundraising, The Internet and Counterterrorism
11	Typical Examples of Cyber-Crimes in Pakistan: Creating/Using Fake Social Media Accounts. Hacking your phone and/or Computer, Writing Comments online of a threatening nature. Using the internet to defraud people, Spreading photos of people without their consent
12	Typical Examples of Cyber-Crimes in Pakistan: Engaging in Fraud Online, Hijacking a Whats-app Account, Using keystroke software to record passwords, Setting up bugging devices illegally. Public Defaming based on lies, Creating a website to cheat the General Public, Fake bank calls
13	Cyber Crime Law passed by the majlis.e.shoora (parliament)
14	cybercrime laws of the united states
15	Recent Topics
16	Case Studies

DS331 - Data Science**Credit Hours:** 3(3+0) **Prerequisites:** None**Course Category:** Data Science Domain**Description**

Data science is a rapidly growing field that is transforming the way we live and work. Data scientists use their skills to collect, analyze, and interpret data to solve real-world problems. Data science is an interdisciplinary field that draws on knowledge from a variety of areas, including statistics, computer science, mathematics, and business. Data scientists need to be able to collect, clean, and analyze data, as well as communicate their findings to stakeholders. The demand for data scientists is high, and the field is expected to grow in the coming years. If you are interested in a career in data science, there are many online and in-person courses available to help you get started. Here are some of the key concepts in data science: Data collection: The process of gathering data from a variety of sources, such as surveys, social media, and sensors. Data cleaning: The process of removing errors and inconsistencies from data. Data analysis: The process of using statistical and machine learning techniques to extract insights from data. Data visualization: The process of communicating the results of data analysis using charts, graphs, and other visual representations. Communication and presentation: The ability to communicate your findings to a variety of audiences.

Textbook(s)

- Joel Grus. Data Science from Scratch: First Principles with Python, 2nd Edition, O'Reilly Media, (2019), ISBN: 978-1492041139

- Wes McKinney. Python for Data Analysis, 3rd Edition, O'Reilly Media, (2022) ISBN: 978-1098104030

Other supplemental materials

- Trevor Hastie, Robert Tibshirani, and Jerome Friedman. The Elements of Statistical Learning, 2nd Edition, Springer, (2016), ISBN: 978-0387848570

Course Learning Objectives

#	Course Learning Outcomes (CLOs)	BT/ PLO #
CLO 1	Understand the basic concepts of data science, statistics, and probability and their application in understanding behavior of data.	CS 1,2,3,4
CLO 2	Apply basic tools for performing exploratory data analysis and visualization.	C3 1,2,3,4
CLO 3	Understand basic predictive modeling and data analysis methods.	CS 1,3

Learn Python for performing different data science steps.

CLO 4	Learn Python for performing different data science steps.	C3 2,4,5,6
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Week wise Lecture Schedule

W #	Lecture Topics
1	Introduction to data science
2	Data preprocessing
3	Statistical inferences and probability
4	Machine learning
5	Big data
6	Data visualization
7	Ethics in data science
8	Data distribution analysis
9	Correlation and covariance analysis
10	Feature selection and dimensionality reduction
11	Supervised learning (classification, regression)
12	Unsupervised learning (clustering, dimensionality reduction)
13	Model evaluation and validation techniques
14	Time series analysis and forecasting techniques
15	Predictive modeling and feature engineering
16	Evaluation metrics for predictive models

DS332 - Data Warehousing & Business Intelligence

Credit Hours: 3(3+0) **Prerequisites:** None

Course Category: Data Science Domain

Description

Gives an overview about importance & significance of Data Warehousing (DWH) and Business Intelligence (BI). Discusses the main concepts and solutions for DWH and BI. The key concepts underpinning the logical design, physical design and implementation of data warehouses are appraised. Data collection, data extraction, cleansing, transformation and loading methods are considered along with query optimization techniques. Differentiation between OLAP & OLTP. Data Warehousing supports information processing by providing a solid platform of integrated, historical, and consistent data for performing enterprise- wide data analysis.

Textbook(s)

- W. H. Inmon, "Building the Data Warehouse", Wiley-India Edition 2020.

- Ralph Kimball, "The Data Warehouse Toolkit – Practical Techniques for Building Dimensional Data Warehouse," John Wiley & Sons, Inc.

Other supplemental materials

- Matteo Golfarelli, Stefano Rizzi, "Data Warehouse Design
- Modern Principles and Methodologies", McGraw Hill Publisher

Course Learning Objectives

#	Course Learning Outcomes (CLOs)	BT/ PLO #
1	CLO Demonstrate an appreciation of the role that Data Warehouses and Business Intelligence play in enhancing the decision-making process	C2
2	CLO Demonstrate an understanding of the fundamental concepts of the Star and the Snowflake Schema; learn how to design the schema of a DW based on these two models.	C2
3	CLO Understand the architecture of DW Systems and be able to specify the advantages and potential problem areas	C3
4	CLO Use Analytic SQL to aggregate, analyze and report, and model data.	C3

Week wise Lecture Schedule

W #	Lecture Topics
1	Introduction to Data Warehouse and Business Intelligence
2	Necessities and Essentials of Business Intelligence

W #	Lecture Topics
3	Data Warehouse Life Cycle and Basic Architecture
4	Data Warehouse Architecture in SQL Server
5	Logical Model of Data Warehousing
6	Indexes in Data Warehousing
7	Physical Model and Storage Design
8	Optimization Techniques in Data Warehousing
9	OLAP Operations and Concepts
10	OLAP Queries and Query Optimization
11	Building the Data Warehouse
12	Data Visualization and Reporting Using SSAS and Tableau
13	Data Visualization and Reporting Based on OLAP Cubes
14	Reports and Dashboard Management in Power BI
15	Dashboard Enrichment Techniques and Custom Visuals
16	Overview and Comparison of Business Intelligence Tools

DS333 - Data Mining**Credit Hours:** 3(3+0) **Prerequisites:** None**Course Category:** Data Science Domain**Description**

Data Mining combines AI, statistics, and databases to uncover hidden patterns in large datasets. This course aims to develop students' ability to analyze and extract knowledge from data. It covers data preprocessing, summary statistics, frequent itemset generation, association and correlation measures, and core algorithms in classification, prediction, and clustering. Students will also apply programming and problem-solving skills to develop data mining applications.

Textbook(s)

- Jiawei Han & Micheline Kamber, Jian Pei. Data Mining: Concepts and Techniques, Latest Edition.

- Pang-Ning Tan, Michael Steinbach, and Vipin Kumar (2005). Introduction to Data Mining.

Other supplemental materials

- Charu C. Aggarwal (2015). Data Mining: The Textbook

- D. Hand, H. Mannila, P. Smyth (2001). Principles of Data Mining. MIT Press.

Course Learning Objectives

#	Course Learning Outcomes (CLOs)	BT/ PLO #
CLO 1	Apply preprocessing techniques on any given raw data.	C3 1,3,10
CLO 2	Select and apply proper data mining algorithm to discover interesting patterns	C3 1,3,4
CLO 3	Analyze and extract patterns to solve problems and point out how to deploy solution	C4 1,3, 4,5
CLO 4	Evaluate systematically supervised, semi supervised and unsupervised models and algorithms with respect to their accuracy	C4 3,4,5, 10

Week wise Lecture Schedule

W #	Lecture Topics
1	Introduction to Data Mining, Basic Concepts, and Applications
2	Data Preprocessing Techniques and Summary Statistics
3	Association Rule Mining: Apriori Algorithm
4	Association Rule Mining: Frequent Pattern Trees (FP-Growth)

W #	Lecture Topics
5	Introduction to Classification and Types
6	Supervised Classification: Decision Trees
7	Supervised Classification: Naïve Bayes and K-Nearest Neighbors
8	Supervised Classification: Support Vector Machines (SVM)
9	Introduction to Unsupervised Classification: Concepts and Use Cases
10	Unsupervised Classification: K-Means, K-Median
11	Unsupervised Classification: Hierarchical & Divisive Clustering
12	Unsupervised Classification: Kohonen Self-Organizing Maps
13	Outlier and Anomaly Detection Techniques
14	Web and Social Network Mining
15	Data Mining Trends and Research Frontiers
16	Implementation of Key Concepts Using Python: Review and Final Integration

DS351 - Advanced Statistics**Credit Hours:** 3(2+1) **Prerequisites:** QR-2**Course Category:** Data Science Domain**Description**

Statistical methods are used for analysis of different datasets for forecasting the values, predicting the unknowns, relating the variables for getting deeper insights and relating data differences with real world complexities. Data Science extracts knowledge from data on the basis of hidden patterns which can be made explicit by incorporating the statistical algorithms in it. This course is designed to prepare students on statistical techniques with a purview of artificial intelligence and data science.

Textbook(s)

- Probability and Statistics for Computer Scientists, 2nd Edition, Michael Baron.

Other supplemental materials

- Probability for Computer Scientists, online Edition, David Forsyth
 - Discovering Statistics using SPSS for Windows, Andy Field

Course Learning Objectives

#	Course Learning Outcomes (CLOs)	BT/ PLO #
CLO 1	Describe what part of statistics is meant for data scientist and what the applications of statistics in data science are.	C1 1,2, 3, 10
CLO 2	Apply Statistical techniques in real life problems.	C3 1,2, 6,10
CLO 3	Analyze, Correlate, Forecast data by using different statistical techniques.	C2 1,2, 3
CLO 4	Apply basic data science statistical techniques by using SPSS on real world datasets.	C3 1,2, 3

Week wise Lecture Schedule

W #	Lecture Topics
1	Introduction to Statistics, Use of Statistics in Data Science, Experimental Design,
2	Statistical Techniques for Forecasting, Interpolation/ Extrapolation, Introduction to Probability,
3	Conditional Probability, Prior and Posterior Probability, Random number generation (RNG),
4	Techniques for RNG, Correlation analysis,
5	Chi Square Dependency tests, Diversity Index,
6	Data Distributions Multivariate Distributions
7	Error estimation, Confidence Intervals, Linear Transformations,

W #	Lecture Topics
8	Gradient Descent and Coordinate Descent, Likelihood inference,
9	Revision of linear regression and likelihood inference
10	Fitting algorithms for nonlinear models and related diagnostics
11	Generalized linear model; exponential families; variance and link functions
12	Proportion and binary responses; logistic regression,
13	Count data and Poisson responses; log-linear models
14	Overdispersion and quasi-likelihood; estimating functions,
15	Mixed models, random effects, generalized additive models and penalized regression;
16	Introduction to SPSS, Probability/ Correlation analysis/ Dependency tests/ Regression in SPSS.

DS361 - Data Visualization**Credit Hours:** 3(2+1) **Prerequisites:** None**Course Category:** Data Science Domain**Description**

Data Visualization is a process of obtaining detailed insights hidden in the data. It is a necessary component in the pipeline of any data science project. This course teaches skills specifically in terms of how to effectively present the data and findings. Further, this course provides hands on skills using R for data exploration and visualization.

Textbook(s)

- Exploratory Data Analysis with R” by Roger D. Peng
 - Kieran Healy Data Visualization: A Practical Introduction, 2018, Princeton University Press

Other supplemental materials

Claus O. Wilke, Fundamentals of Data Visualization, 2019, O’Reilly Media

Course Learning Objectives

#	Course Learning Outcomes (CLOs)	BT/ PLO #
CLO 1	Provides knowledge about importance, necessity, and justification of performing exploratory data analysis and visualization	C2 1,2,3
CLO 2	Introduce various type of charts along with their alternatives solution to show same data from versatile aspects.	C2 3,4,10
CLO 3	Improving the competency of the students to analyze different problems and select the most appropriate solution.	C3 1,10
CLO 4	Use of R, various recent tools, and technologies to develop hands-on skills for exploratory data analysis and visualization.	C3 1,4,5,10

Week wise Lecture Schedule

W #	Lecture Topics
1	Introduction to EDA and Visualization – Importance of EDA – Overview of visualization principles
2	Building Blocks of Visualization – Data types, variables – Grammar of graphics (ggplot2 style)
3	Types of Exploratory Graphs – Univariate, bivariate, multivariate plots – Best practices
4	Single and Multi-Dimensional Summaries – Mean, median, mode, SD – Cross-tabulation, pair plots

W #	Lecture Topics
5	Five-Number Summary and Box Plots – Min, Q1, Median, Q3, Max – Interquartile range and outliers
6	Histograms, Bar Plots, and Others – Usage, pitfalls – Frequency vs. relative frequency
7	Distributions and Their Visualization – Normal, skewed, heavy-tailed – Outlier detection
8	Probability Mass Functions (PMF) – Visualizing discrete distributions – Real-world examples
9	Cumulative Distribution Functions (CDF) – ECDF plots – Percentiles, quartiles, randomness
10	Modeling Distributions – Exponential, Normal, Lognormal, Pareto – Applications
11	Probability Density Functions (PDF) – Continuous distribution curves – Kernel density estimation
12	Relationships Between Variables – Scatter plots – Covariance, correlation
13	Estimation and Hypothesis Testing – Confidence intervals – Visual inference with plots
14	Clustering & Grouping – K-Means and Hierarchical clustering – Visual cluster interpretation
15	Time Series and Survival Analysis – Trends, seasonality – Kaplan-Meier curves, log-rank tests
16	Implementation & Capstone – End-to-end EDA in R – Group project or case study demo

DS371 - Business Process Analysis**Credit Hours:** 3(3+0) **Prerequisites:** None**Course Category:** Data Science Elective**Description**

This course focuses on the role of Business Process Management (BPM) within the field of Data Science, emphasizing how data-driven insights can optimize both manufacturing and service processes. Students will learn to use modeling and charting tools to represent and analyze business workflows, apply Lean process improvement techniques, and evaluate real-world cases of process re-engineering and business process outsourcing (BPO). The course bridges process understanding with data analytics to support informed decision-making and operational efficiency.

Textbook(s)

Business Process Improvement; The Breakthrough Strategy for Total Quality, Productivity, and Competitiveness, H. J. Harrington

Business Intelligence: A Managerial Approach by Turban, Sharda, Delen, King, 2nd Edition, Prentice Hall (2011). ISBN: 13-978-0-136-10066-9

Course Learning Objectives

#	Course Learning Outcomes (CLOs)	BT/ PLO #
CLO 1	Explain key concepts of business process management, improvement strategies, and their role in data science.	C5 1,3,10
CLO 2	Model and analyze business processes using standard tools to identify inefficiencies and improvement areas.	C3 1, 2, 3, 4
CLO 3	Apply data-driven techniques and business intelligence tools to monitor, evaluate, and optimize processes.	C3 3, 4, 5
CLO 4	Design and present process improvement solutions using analytics, dashboards, and visual reports.	C4 3, 4, 5, 10

Week wise Lecture Schedule

W #	Lecture Topics
1	Introduction to Business Processes and Business Process Analytics (BPA); Role in Data Science
2	Understanding Process Quality, Productivity, and Competitiveness (Harrington)
3	Fundamentals of Business Process Improvement; Total Quality Management Concepts
4	Process Modeling Techniques: Flowcharts, SIPOC, Swimlane Diagrams
5	Business Intelligence (BI) Concepts and Architecture (Turban)

W #	Lecture Topics
6	Business Process Metrics and Performance Measurement (KPIs, Cycle Time, Cost, Efficiency)
7	Process Analysis Using Data; Identifying Bottlenecks, Redundancies, and Delays
8	Business Process Reengineering (BPR): Concepts, Methodologies, and Case Studies (Harrington)
9	Business Process Outsourcing (BPO) and Shared Services; Strategic Considerations
10	BI Tools and Platforms: Overview of OLAP, ETL, Data Warehousing (Turban)
11	Lean and Six Sigma Techniques for Process Improvement; DMAIC Cycle
12	Analytical Techniques: Descriptive, Predictive, and Prescriptive Analytics in Business Processes
13	Data Visualization and Dashboards for Process Monitoring (Power BI, Tableau)
14	Case Studies: Process Analytics Applications in Manufacturing and Services
15	Workshop: Process Improvement Using Analytics (Hands-on lab with process data)
16	Capstone Presentations / Final Project Review: Business Process Analytics Recommendations and Insights

DS372 - Platforms & Architectures for Data Science

Credit Hours: 3(3+0) **Prerequisites:** None

Course Category: Data Science Elective

Description

This course introduces the essential platforms, architectures, and tools that support modern data science workflows, from data ingestion and storage to model monitoring and production deployment. Students will explore data messaging systems (e.g., Kafka, Kinesis), feature store implementations, and tools for data versioning and workflow management. The course emphasizes working with cloud-based platforms, particularly Microsoft Azure (HDInsight, Synapse Analytics, Data Lake), and tools for monitoring, orchestration, and deployment of data science solutions at scale.

Textbook(s)

Data Science on the Google Cloud Platform books
Valliappa Lakshmanan, 2022

Other supplemental materials

IBM Cloud Pak for Data Hemanth Manda, Sriram Srinivasan ,
Deepak Ranarao · 2021

Course Learning Objectives

#	Course Learning Outcomes (CLOs)	BT/ PLO #
CLO 1	Describe key components of data science architectures including data messaging, storage, and workflow tools.	C2
CLO 2	Analyze and evaluate different platforms and cloud services used in scalable data science environments.	C4
CLO 3	Apply tools like DVC, MLflow, and Spark to manage, deploy, and monitor data science models and pipelines.	C3
CLO 4	Design and implement a complete data science workflow using messaging, feature stores, and Azure services.	C3

Week wise Lecture Schedule

W #	Lecture Topics
1	Introduction to Data Science Platforms and Architectures
2	The ETL Pipeline and Messaging Systems: Concepts and Architecture
3	Apache Kafka and RabbitMQ: Distributed Messaging for Real-Time Data
4	Amazon Kinesis and TIBCO Spotfire: Event Stream Processing
5	Data Storage Challenges: Volume, Variety, Velocity, and Latency
6	Introduction to Feature Stores and Their Role in ML Pipelines

W #	Lecture Topics
7	Hands-on with Feast, Tecton, or Custom Feature Stores with Spark
8	Data Versioning and Experiment Tracking: DVC and MLflow
9	Workflow Orchestration and Production Pipelines
10	Dashboards and Visualization for Monitoring Production Models
11	Monitoring Tools: Performance Metrics, Logging, and Alerting
12	Meta Data Stores and Pipeline Metadata Management
13	Overview of Azure Data Science Stack
14	Azure HDInsight (Spark, Hive), Azure Synapse Analytics, Data Lake
15	End-to-End Workflow: Ingest to Model Deployment on Azure
16	Final Project: Design and Demonstration of a Scalable DS Platform

DS373 - Text Mining

Credit Hours: 3(3+0) **Prerequisites:** None

Course Category: Data Science Elective

Description

This course introduces the theory and practice of Text Mining, focusing on transforming unstructured text into meaningful data for analysis. Students will learn core NLP techniques, methods for document representation, and key algorithms for categorization, clustering, topic modeling, and sentiment analysis. Advanced topics include neural approaches, sequence-to-sequence models, and text visualization. Real-world applications include document classification, summarization, dialogue systems, and machine translation using popular NLP toolkits.

Textbook(s)

Mining Text Data. Charu C. Aggarwal and ChengXiang Zhai, Springer, 2012.

Data Science and Predictive Analytics: Biomedical and Health Applications using R, Ivo D. Dinov, 2023 (2nd ed.), ISBN: 978-3031174834

Other supplemental materials

Text Mining Handbook: Advanced approaches in analyzing unstructured Data, Cambridge University Press

Course Learning Objectives

#	Course Learning Outcomes (CLOs)	BT/ PLO #
CLO 1	Explain the fundamental techniques of NLP and their role in transforming unstructured text into structured data.	C2
CLO 2	Apply supervised and unsupervised algorithms for text categorization, clustering, and topic modeling.	C3
CLO 3	Evaluate text summarization, sentiment analysis, and machine translation methods for real-world applications.	C5
CLO 4	Use NLP toolkits and visualization tools to design and implement end-to-end text mining solutions.	C4

Week wise Lecture Schedule

W #	Lecture Topics
1	Introduction to Text Mining and Applications in Data Science
2	Natural Language Processing (NLP): Overview and Basic Techniques
3	NLP Techniques I: Tokenization, POS Tagging, Chunking, Named Entity Recognition
4	NLP Techniques II: Syntax, Parsing, Public NLP Toolkits (NLTK, spaCy, Stanford CoreNLP)
5	Document Representation: Bag of Words, TF-IDF, Word Embeddings

W #	Lecture Topics
6	Text Categorization: Naïve Bayes, k-NN, and Logistic Regression
7	Text Categorization: SVM and Decision Trees
8	Introduction to Neural Networks for NLP (Word2Vec, RNNs, Transformers)
9	Text Clustering: Hierarchical and K-Means Clustering
10	Topic Modeling I: pLSI and LDA – Theory and Implementation
11	Topic Modeling II: Applications in Annotation, Filtering, Structure Modeling
12	Document Summarization: Extraction-based Methods
13	Sentiment Analysis: Polarity, Review Mining, Aspect Identification
14	Sequence-to-Sequence Models: Machine Translation and Dialogue Systems
15	Text Visualization: Word Clouds, Dimensionality Reduction (t-SNE), LDAViz
16	Capstone Project Presentations: Text Mining Solution on Real-World Dataset

DS374 - Big Data Analytics**Credit Hours:** 3(3+0) **Prerequisites:** CC221**Course Category:** Data Science Elective**Description**

The course objective is to develop understanding about the core concept of Big Data, why Big Data requires a different programming paradigm and mindset, and what are the various programming approaches used, what type of data can be processed.

Textbook(s)

Rajkumar Buyya, Rodrigo N. Calheiros, Amir Vahid Dastjerdi, Big Data: Principles and Paradigms, Morgan Kaufmann, 2016. ISBN: 9780128053942.

Jules S. Damji, Brooke Wenig, Tathagata Das, Denny Lee. *Learning Spark: Lightning-Fast Big Data Analysis*, O'Reilly Media, 2020 (2nd Edition). ISBN: 9781492050049.

Course Learning Objectives

#	Course Learning Outcomes (CLOs)	BT/ PLO #
CLO 1	Understand the fundamental concepts of Big Data and its programming paradigm.	C2
CLO 2	Hadoop/MapReduce Programming, Framework, and Ecosystem	C3
CLO 3	Apache Spark Programming	C3

Week wise Lecture Schedule

W #	Lecture Topics
1	Introduction and Overview of Big Data Systems, Applications and Use Cases
2	Platforms for Big Data, Batch vs. Stream Processing
3	Hadoop as a Platform, Hadoop Ecosystem Overview
4	Hadoop Distributed File System (HDFS), HDFS Architecture
5	MapReduce Framework, Map and Reduce Functions
6	Resource Management in Hadoop, Introduction to YARN
7	Deep dive into YARN, Cluster Scheduling
8	Apache Scala Basics, Syntax, Data Types, Functions
9	Apache Scala Advanced Topics, Collections, Functional Programming
10	Introduction to Apache Spark, Spark vs. MapReduce
11	Resilient Distributed Datasets (RDD), RDD Operations
12	Apache Spark SQL, DataFrames, SQL Queries on Big Data

W #	Lecture Topics
13	Data Analytics on Hadoop and Spark, Real-time and Batch Examples
14	Machine Learning on Hadoop and Spark, MLlib Introduction
15	Spark Streaming, Real-time Data Pipelines
16	Case Study/Project/Demo

DS471 - Artificial Neural Networks & Deep Learning

Credit Hours: 3(3+0) **Prerequisites:** None

Course Category: Data Science Elective

Description

This course will introduce Artificial Neural Networks and Deep Learning. ANN's basic architecture and how they mimic the human brain using simple mathematical models. Many of the important concepts and techniques around brain computing and the major types of ANN will also be introduced. Emphasis is made on the mathematical models, understanding learning laws, selecting activation functions and how to train the networks to solve classification problems. Deep neural networks have achieved state of the art performance on several computer vision and speech recognition benchmarks. This course will further build on the fundamentals of Neural networks and artificial intelligence and will introduce advanced topics in neural networks, convolutional and recurrent network structures, deep unsupervised and reinforcement learning.

Textbook(s)

Neural Network Design, Latest Edition, Martin T. Hagan, Howard, B. Demuth, Mark Hudson Beale and Orlando De Jesus, Publisher: Martin Hagan, ISBN-10: 0971732116

Other supplemental materials

An Introduction to Neural Networks, James A Anderson, Publisher: A Bradford Book (March 16, 1995), ISBN-10: 0262011441

Course Learning Objectives

#	Course Learning Outcomes (CLOs)	BT/ PLO #
CLO 1	Understand the fundamentals of neural networks in AI	C2
CLO 2	Explain how simple ANNs can be designed	C2
CLO 3	Apply ANN for classification Problems	C3
CLO 3	Apply deep learning algorithms to real-world problems	C3
CLO 4	Analyze results from deep learning to select appropriate solutions	C4

Week wise Lecture Schedule

W #	Lecture Topics
1	Introduction and history of neural networks, Applications in AI

W #	Lecture Topics
2	Basic architecture of neural networks, Perceptron, Adaline, Minimum Error Learning
3	Machine learning theory, Training vs. test sets, Evaluation metrics
4	Gradient descent (Delta) rule, Hebbian learning, Neo-Hebbian learning
5	Differential Hebbian Learning, Drive Reinforcement Theory
6	Kohonen Self-Organizing Maps, Associative memory, Bi-directional associative memory (BAM)
7	Energy surfaces, Boltzmann machines, Theory of generalization
8	Backpropagation algorithm, Multi-layer perceptrons
9	Feedforward networks, Deep convolutional networks
10	Computational complexity of neural networks, Optimization techniques
11	Autoencoders, Sparse coding, Unsupervised deep learning
12	Deep belief networks, Restricted Boltzmann Machines
13	Deep recurrent neural networks, BPTT, LSTM, GRU
14	GPU programming for deep learning, CuDNN basics
15	Case Study/Project/Demo
16	Presentation/Peer Review/Q&A Sessions/Demo

DS472 - Data Ethics, Governance and Laws

Credit Hours: 3(3+0) **Prerequisites:** None

Course Category: Data Science Elective

Description

In the era of data-driven decision-making, understanding the ethical, legal, and governance aspects of data is crucial. This course explores the ethical considerations, governance frameworks, and legal principles that guide the responsible handling and use of data. Students will gain insights into the complexities of data ethics, governance strategies, and the legal landscape surrounding data in various industries.

Textbook(s)

- Data and Goliath: The Hidden Battles to Collect Your Data and Control Your World Reprint Edition by Bruce Schneier (2019), ISBN: 978-0393352177.

Other supplemental materials

Ethics of Big Data: Balancing Risk and Innovation 1st Edition, by Kord Davis (2012), ISBN: 978-1449311797
Data Governance: How to Design, Deploy and Sustain an Effective Data Governance, by John Ladley (2012), ISBN: 978-0124158290

Course Learning Objectives

#	Course Learning Outcomes (CLOs)	BT/ PLO #
CLO 1	Define the concepts of data ethics, governance, and laws, emphasizing their significance in the data landscape.	C2 1,2,3
CLO 2	Address ethical considerations in data handling, covering privacy, consent, and issues related to bias, fairness, transparency, and accountability in data.	C3 1,2, 10
CLO 3	Explore data governance frameworks, focusing on the principles of data governance, data stewardship, ownership, and effective implementation.	C4 3, 4
CLO 4	Provide an overview of the regulatory landscape, including data protection laws such as GDPR and CCPA, industry-specific regulations, and global data governance standards.	C2 1,2,10
CLO 5	Examine data compliance and risk management strategies, emphasizing risk assessment in data management and mitigation of legal and ethical risks.	C4 3,4,5, 10
CLO 6	Discuss ethics in data science and AI, emphasizing responsible AI practices, ethical considerations in algorithmic decision-making, and addressing bias and discrimination.	C2 1,2,3
CLO 7	Consider corporate and social responsibility in the context of data, covering corporate	C2

ethics, social implications of data use, and ethical business practices in the data industry. 1,2,3

Week wise Lecture Schedule

W #	Lecture Topics
1	Introduction to Data Ethics, Governance, and Laws: Defining Data Ethics, Importance of Data
2	Governance, Legal Frameworks in the Data Landscape
3	Ethical Considerations in Data Handling: Privacy and Consent,
4	Bias and Fairness in Data, Transparency and Accountability
5	Data Governance Frameworks: Principles of Data Governance, Data Stewardship and Ownership, Implementing Effective Data Governance
6	Regulatory Landscape: Overview of Data Protection Laws (e.g., GDPR, CCPA),
7	Industry-Specific Regulations
8	Global Data Governance Standards
9	Data Compliance and Risk Management: Compliance Strategies, Risk Assessment in Data Management
10	Mitigating Legal and Ethical Risks.
11	Ethics in Data Science and AI: Responsible AI Practices, Ethical Considerations in Algorithmic Decision-Making,
12	Addressing Bias and Discrimination.
13	Corporate and Social Responsibility: Corporate Ethics and Data, Social Implications of Data Use
14	Ethical Business Practices in the Data Industry.
15	Data Security and Cybersecurity Laws: Legal Frameworks for Cybersecurity,
16	Incident Response and Reporting, International Collaboration on Cybersecurity.

DS473 - Topics in Data Science and AI**Credit Hours:** 3(3+0) **Prerequisites:** None**Course Category:** Data Science Elective**Description**

This course delves into specialized areas within the domains of Data Science and Artificial Intelligence (AI). It explores advanced topics, cutting-edge methodologies, and emerging trends that are shaping the future of these fields. Participants will gain insights into specialized applications, recent advancements, and innovative techniques driving developments in Data Science and AI.

Textbook(s)

- Deep Learning By Ian Goodfellow, Yoshua Bengio and Aaron Courville, (2019), ISBN: 9780262035613.
- Natural Language Processing in Action by Hobson Lane, Hannes Hapke, Cole Howard, (2019), ISBN: 978-1617294631
- Computer Vision: Algorithms and Applications (Texts in Computer Science) Latest Edition by Richard Szeliski, ISBN: 978-1848829343

Course Learning Objectives

#	Course Learning Outcomes (CLOs)	BT/ PLO #
CLO 1	Apply advanced machine learning techniques, including deep learning architectures, neural networks, reinforcement learning, transfer learning, and generative adversarial networks (GANs).	C3 1,2,3
CLO 2	Explore natural language processing (NLP) and understanding, covering text mining, sentiment analysis, information extraction, named entity recognition (NER), language translation, and contextual embeddings.	C4 1,2,3,6
CLO 3	Engage in computer vision and image processing, addressing object detection, image segmentation, generation, style transfer, facial recognition, and the use of deep convolutional networks (CNNs).	C3 1,2,6 ,7
CLO 4	Implement scalable machine learning on big data platforms, encompassing stream processing, real-time analytics, parallel and distributed computing, and data warehousing.	C5 6,7,9
CLO 5	Investigate reinforcement learning and AI in robotics, including applications in autonomous agents, intelligent systems, and human-AI collaboration.	C5 1,2,3

Week wise Lecture Schedule

W #	Lecture Topics
1	Advanced Machine Learning Techniques: Deep Learning Architectures and Neural Networks, Reinforcement Learning and its Applications,
2	Transfer Learning and Domain Adaptation, Generative Adversarial Networks (GANs) and Variational Autoencoders (VAEs), Advanced Optimization Algorithms
3	Natural Language Processing and Understanding: Text Mining and Sentiment Analysis, Information Extraction and Named Entity Recognition (NER),
4	Language Translation and Multilingual Models, Contextual Embeddings and Transformers (e.g., BERT, GPT).
5	Computer Vision and Image Processing: Object Detection and Image Segmentation,
6	Image Generation and Style Transfer,
7	Facial Recognition and Biometric Systems
8	Deep Convolutional Networks (CNNs) for Vision Tasks. Big Data Analytics and Scalable AI.
9	Scalable Machine Learning on Big Data Platforms: Stream Processing and Real-time Analytics
10	Parallel and Distributed Computing for AI Applications
11	Data Warehousing and Architectures for Big Data.
12	Reinforcement Learning and AI in Robotics: Reinforcement Learning in Robotics
13	Autonomous Agents and Intelligent Systems
14	Applications of AI in Robotics and Automation, Human-AI Collaboration in Robotics.
15	Case Study/Project/Demo
16	Presentation/Peer Review/Q&A Sessions/Demo

CS101 – Applications of Information and Communication Technologies

Credit Hours: 3(2+1) **Prerequisites:** None

Course Category: General Education Course

Description

This course is designed to provide students with an exploration of the practical applications of Information and Communication Technologies (ICT) and software tools in various domains. Students will gain hands-on experience with a range of software applications, learning how to leverage ICT to solve daily life problems, enhance productivity and innovate in different fields. Through individual and interactive exercises and discussions, students will develop proficiency in utilizing software for communication, creativity, and more.

Textbook(s)

- "Discovering Computers" by Vermaat, Shaffer, and Freund

"GO! with Microsoft Office" Series by Gaskin, Vargas, and McLellan.

"Exploring Microsoft Office" Series by Grauer and Poatsy.

"Computing Essentials" by Morley and Parker.

"Technology in Action" by Evans, Martin, and Poatsy.

Course Learning Objectives

#	Course Learning Outcomes (CLOs)	BT/ PLO #
CLO 1	Explain the fundamental concepts, components, and scope of Information and Communication Technologies (ICT).	C2
CLO 2	Identify uses of various ICT platforms and tools for different purposes.	C1
CLO 3	Apply ICT platforms and tools for different purposes to address basic needs in different domains of daily, academic, and professional life.	C3
CLO 4	Understand the ethical and legal considerations in use of ICT platforms and tools.	C2

Week wise Lecture Schedule

W #	Lecture Topics
1	Introduction to Information and Communication Technologies:• Components of Information and Communication Technologies (basics of hardware, software, ICT platforms, networks, local and cloud data storage etc.).
2	• Scope of Information and Communication Technologies (use of ICT in education, business, governance, healthcare, digital media and entertainment, etc.).
3	• Emerging technologies and future trends.

W #	Lecture Topics
4	Basic ICT Productivity Tools:• Effective use of popular search engines (e.g., Google, Bing, etc.) to explore World Wide Web.• Formal communication tools and etiquettes (Gmail, Microsoft Outlook, etc.).
5	• Microsoft Office Suites (Word, Excel, PowerPoint).• Google Workspace (Google Docs, Sheets, Slides).
6	• Dropbox (Cloud storage and file sharing), Google Drive (Cloud storage with Google Docs integration) and Microsoft OneDrive (Cloud storage with Microsoft Office integration).
7	• Evernote (Note-taking and organization applications) and OneNote (Microsoft's digital notebook for capturing and organizing ideas).• Video conferencing (Google Meet, Microsoft Teams, Zoom, etc.).• Social media applications (LinkedIn, Facebook, Instagram, etc.).
8	ICT in Education:• Working with learning management systems (Moodle, Canvas, Google Classrooms, etc.).
9	• Sources of online education courses (Coursera, edX, Udemy, Khan Academy, etc.).• Interactive multimedia and virtual classrooms.
10	ICT in Health and Well-being:• Health and fitness tracking devices and applications (Google Fit, Samsung Health, Apple Health, Xiaomi Mi Band, Runkeeper, etc.).
11	• Telemedicine and online health consultations (OLADOC, Schat Kahani, Marham, etc.).
12	ICT in Personal Finance and Shopping:• Online banking and financial management tools (JazzCash, Easypaisa, Zong PayMax, ILINK and MNET, Keenu Wallet, etc.).
13	• E-commerce platforms (Daraz.pk, Telemart, Shophive, etc.).
14	Digital Citizenship and Online Etiquette:• Digital identity and online reputation.• Netiquette and respectful online communication.• Cyberbullying and online harassment.
15	Ethical Considerations in Use of ICT Platforms and Tools:• Intellectual property and copyright issues.• Ensuring originality in content creation by avoiding plagiarism and unauthorized use of information sources.
16	• Content accuracy and integrity (ensuring that the content shared through ICT platforms is free from misinformation, fake news, and manipulation).